HOCHSCHULE
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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

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1 Basics

1.1 Quantities, Units and Ohm's Law

This chapter collects the tools that every later chapter uses: the basic quantities and their units, Ohm's law, Kirchhoff's laws, the voltage divider, equivalent sources, the LED with its series resistor and the RC circuit. Most of it is known from the introductory course on electrical engineering. Here it is repeated with a view to the lab: at the end of the semester you will build a measurement chain "From Sensor to Digital Value". A resistive sensor sits in a bridge, an instrumentation amplifier scales the signal to $0\,\mathrm{V} \ldots 4\,\mathrm{V}$, a low-pass filter limits the bandwidth, a sample and hold circuit freezes the value and a 3-bit flash converter with a resistor ladder turns it into a number that is shown on LEDs. Every block of this chain is built from the elements of this chapter.
We also introduce the circuit simulator Falstad (CircuitJS). Every circuit of the lecture exists as a Falstad file, so you can change values and watch the effect immediately.

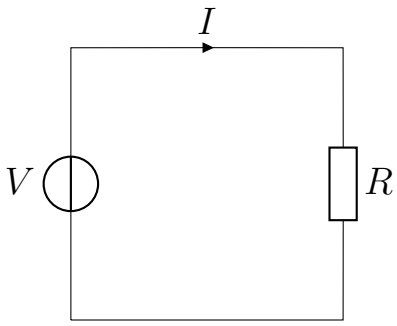
Quantities and units

quantity	symbol	unit	definition
voltage	V	volt, V	energy per charge, J/C
current	I	ampere, A	charge per time, C/s
resistance	R	ohm, Ω	$R = V/I$
power	P	watt, W	$P = V \cdot I$
capacitance	C	farad, F	$C = Q/V$
time, frequency	t, f	s, Hz	$f = 1/T$

- prefixes: p (10^{-12}), n (10^{-9}), µ (10^{-6}), m (10^{-3}), k (10^{3}), M (10^{6})
- useful combination: V / kΩ = mA, mA · kΩ = V

Calculating with kΩ, mA and V avoids most errors with powers of ten: 5 V across 2.2 kΩ gives 2.27 mA without any exponent. Voltages are always measured *between two points*. If only one point is named, the second one is ground (0 V). In the diagrams voltages are drawn as arrows, currents as arrows on the wires; the direction of an arrow defines the sign, not the physics. A negative result simply means that the real direction is opposite to the arrow.

Ohm's law



Ohm's law

$$V = R \cdot I \qquad I = \frac{V}{R} \qquad R = \frac{V}{I}$$

- a resistor is **linear**: double voltage ⇒ double current
- the conductance $G = 1/R$ is measured in siemens, S

Ohm's law describes resistors, but many other components are nonlinear: for a diode or an LED the current grows exponentially with the voltage, so there is no single value R. For such components we use a characteristic curve or a simplified model, for example a constant forward voltage V_F.

Power and power rating

Power in a resistor

$$P = V \cdot I = I^2\,R = \frac{V^2}{R}$$

- the power is turned into heat; every resistor has a **power rating**
  - SMD 0603: 0.1 W, SMD 1206: 0.25 W, through-hole (0207): 0.25 W … 0.6 W
- design rule: stay below about $50\,\%$ of the rating
- small signal circuits (kΩ, mA) dissipate only mW
- example: $100\,\Omega$ at 5 V: $P = 25\ \text{V}^2/100\,\Omega = 250$ mW, too much for a 0.25 W resistor

The power rating is specified at an ambient temperature of typically $70\,^\circ$C; above that it must be derated. A resistor at its full rating becomes hot (well above $100\,^\circ$C), its value drifts and its life is shortened. Therefore we keep a safety factor of two. In measurement circuits the dissipated power matters for another reason: a sensor that heats itself measures its own temperature. This *self-heating* is the reason why the sensor current in the capstone task is limited to 1 mA.

Standard values: the E-series

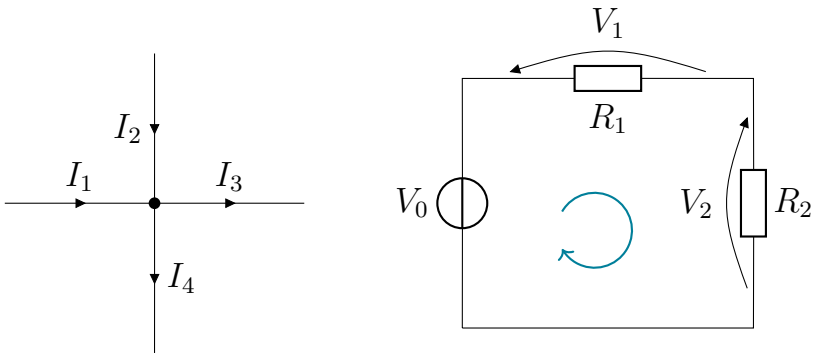
- resistors are not available in every value, but in logarithmic series
- E12 ($10\,\%$ tolerance): 1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
- E24 ($5\,\%$) adds: 1.1 1.3 1.6 2.0 2.4 3.0 3.6 4.3 5.1 6.2 7.5 9.1
- E96 ($1\,\%$): 96 values per decade, standard for metal film resistors today
- each value exists in every decade: $4.7\,\Omega$, $47\,\Omega$, …, 4.7 MΩ

The neighbouring values of a series differ by a constant factor ($10^{1/12} \approx 1.21$ for E12), so that the tolerance bands of neighbouring values just touch. Today most resistors have a tolerance of $1\,\%$ even if one uses E12 or E24 values; the coarser series are simply easier to remember and to keep in stock. A design normally ends with choosing the next standard value and recalculating the result with it. Capacitors are usually available in E6 (1.0, 1.5, 2.2, 3.3, 4.7, 6.8).

1.2 Kirchhoff's Laws and Networks

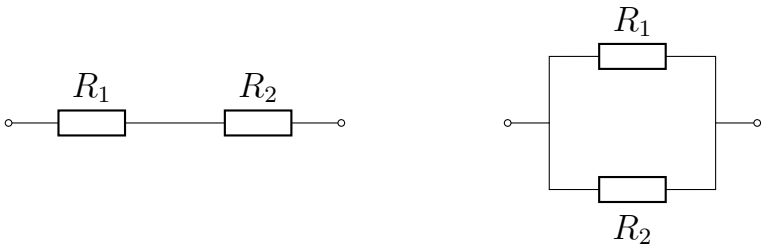
Two conservation laws are enough to analyse any network of resistors and sources: charge is not lost at a node, and energy is not gained on a closed path. Together with Ohm's law they give as many equations as there are unknown currents. For the small networks of this lecture we mostly combine resistors in series and in parallel step by step.

Kirchhoff's current law (KCL) and voltage law (KVL)



- KCL: at every node the sum of incoming currents equals the sum of outgoing currents: $I_1 + I_2 = I_3 + I_4$
- KVL: around every closed loop the sum of voltages is zero: $V_0 = V_1 + V_2$

Series and parallel connection

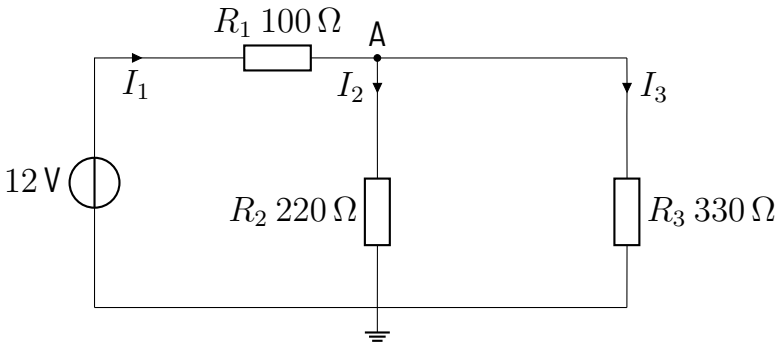


Series and parallel

series: $R = R_1 + R_2$ parallel: $R = R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

- series: same current, the voltages add up
- parallel: same voltage, the currents add up
- parallel resistance is always smaller than the smallest resistor; $R \parallel R = R/2$

Example: series-parallel network



- $R_2 \parallel R_3 = 132\,\Omega$, total $R = 232\,\Omega$, $I_1 = 12\,\text{V}/232\,\Omega = 51.7\,\text{mA}$
- $V_A = I_1 \cdot 132\,\Omega = 6.83\,\text{V}$; $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ (KCL: $I_1 = I_2 + I_3$)

▶ Falstad: `examples/basics/network.txt` The procedure is always the same: combine the parallel resistors, add the series resistor, calculate the total current, then go back and calculate the node voltage and the branch currents. KCL at node A is a cheap check of the result. Exercise 1.1 looks at the power of each resistor.

Exercise 1.1: Power in a series-parallel network

Use the network above ($12\,\text{V}$, $R_1 = 100\,\Omega$, $R_2 = 220\,\Omega$, $R_3 = 330\,\Omega$). All resistors are $0.25\,\text{W}$ types.

a) Calculate the power dissipated in each resistor and the power delivered by the source.

b) Which resistor is overloaded? Which rating would you choose instead?

c) Check the currents and powers in Falstad.

Solution 1.1

a) $P_1 = I_1^2 R_1 = (51.7\,\text{mA})^2 \cdot 100\,\Omega = 268\,\text{mW}$, $P_2 = V_A^2/R_2 = (6.83\,\text{V})^2/220\,\Omega = 212\,\text{mW}$, $P_3 = (6.83\,\text{V})^2/330\,\Omega = 141\,\text{mW}$. Source: $P = 12\,\text{V} \cdot 51.7\,\text{mA} = 621\,\text{mW} = P_1 + P_2 + P_3$.

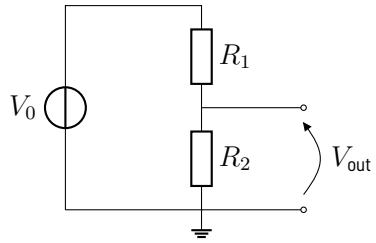
b) R_1 exceeds $0.25\,\text{W}$; R_2 is at $85\,\%$ of its rating, which is too hot as well. With the $50\,\%$ rule R_1 needs $0.6\,\text{W}$ and R_2 $0.5\,\text{W}$ (better: $1\,\text{W}$ for R_1).

c) ▶ Falstad: `examples/basics/network.txt`: `VA` $= 6.83\,\text{V}$; hovering over the resistors shows $I_1 = 51.7\,\text{mA}$, $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ and the power of each resistor.

1.3 The Voltage Divider

The voltage divider is the most frequently used circuit in this lecture. It sets operating points, scales signals, forms the reference ladder of a flash converter and, with a sensor in one branch, it is half of a measurement bridge. Its weak point is the load: as soon as current is drawn from the tap, the output voltage drops.

Unloaded divider



- the same current flows through R_1 and R_2: $I = V_0/(R_1 + R_2)$
- the voltages divide in the ratio of the resistances

Voltage divider

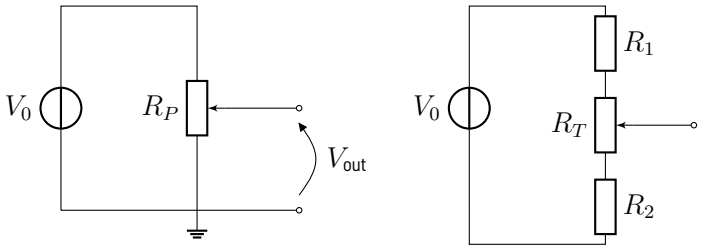
$$V_{\text{out}} = V_0 \frac{R_2}{R_1 + R_2}$$

Loaded divider

- a load R_L appears in parallel to R_2: $V_{\text{out}} = V_0 \dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$
- seen from the load the divider is a real source with $V_{\text{th}} = V_0\, R_2/(R_1 + R_2)$ and $R_{\text{th}} = R_1 \parallel R_2$
- rule of thumb: $R_L \geq 10\,(R_1 \parallel R_2)$ keeps the error below $10\,\%$
- for an error below $1\,\%$: $R_L \geq 100\,(R_1 \parallel R_2)$, or buffer the tap with an op-amp

▶ Falstad: `examples/basics/divider_loaded.txt` Choosing the resistance level of a divider is a trade-off. Small resistors make the divider stiff against a load, but they draw a large cross current from the supply and dissipate power. Large resistors save current, but every load, every leakage current and the input current of the next stage disturbs the output. In the flash ADC of the capstone task the reference ladder uses $8 \times 1\,\text{k}\Omega$ at $4\,\text{V}$ ($0.5\,\text{mA}$); it is loaded only by comparator inputs, which draw practically no current.

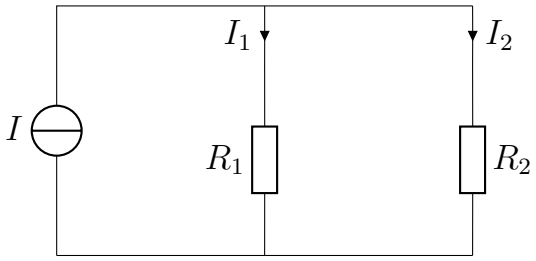
Potentiometer: adjustable divider



- the wiper divides the track: $V_{\text{out}} = V_0 \cdot x$, $x = 0 \ldots 1$ (linear taper)
- trimmer between two fixed resistors: fine adjustment over a small range
- the wiper must not carry a large current; load it lightly

A potentiometer is a resistive track with a sliding contact. Panel potentiometers have a knob, trimmers are adjusted once with a screwdriver. In the capstone task a trimmer sets the gain of the instrumentation amplifier so that the full temperature range gives exactly $0\,\text{V} \ldots 4.00\,\text{V}$. Putting the trimmer between two fixed resistors limits the adjustment range and makes the adjustment much finer: a $1\,\text{k}\Omega$ trimmer between two $4.7\,\text{k}\Omega$ resistors at $10\,\text{V}$ covers only $4.5\,\text{V} \ldots 5.5\,\text{V}$. In Falstad a potentiometer is found under *Draw* → *Passive Components* → *Potentiometer*; it gets a slider automatically.

Current divider



Current divider

$$I_1 = I\,\frac{R_2}{R_1 + R_2} \qquad I_2 = I\,\frac{R_1}{R_1 + R_2}$$

- the **smaller** resistor takes the **larger** current
- example: $10\,\text{mA}$ into $1\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega$: $I_1 = 8.25\,\text{mA}$, $I_2 = 1.75\,\text{mA}$

▶ Falstad: `examples/basics/current_divider.txt`

Exercise 1.2: Loaded voltage divider

A divider with $R_1 = R_2 = 10\,\text{k}\Omega$ is supplied with $V_0 = 10\,\text{V}$.

a) Calculate V_{out} without load.

b) Calculate V_{out} with a load $R_L = 10\,\text{k}\Omega$.

c) Check both results in Falstad.

Solution 1.2

a) $V_{\text{out}} = 10\,\text{V} \cdot \frac{10}{10+10} = 5\,\text{V}$.

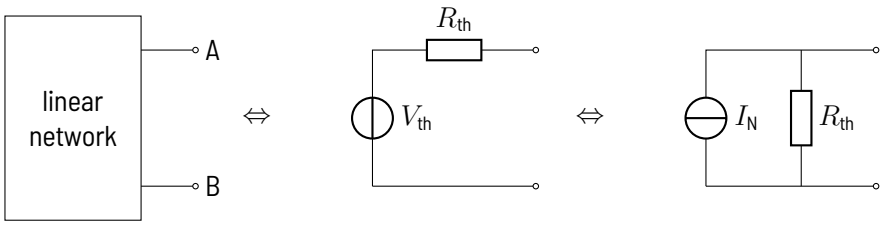
b) $R_2 \parallel R_L = 5\,\text{k}\Omega$, so $V_{\text{out}} = 10\,\text{V} \cdot \frac{5}{10+5} = 3.33\,\text{V}$. The load costs one third of the output voltage, because R_L is only twice the source resistance $R_{\text{th}} = 5\,\text{k}\Omega$.

c) ▶ Falstad: `examples/basics/divider_loaded.txt`
the labels show `Vout_open` $= 5.00\,\text{V}$ and `Vout_load` $= 3.33\,\text{V}$ (move the mouse over the label).

1.4 Equivalent Sources and Superposition

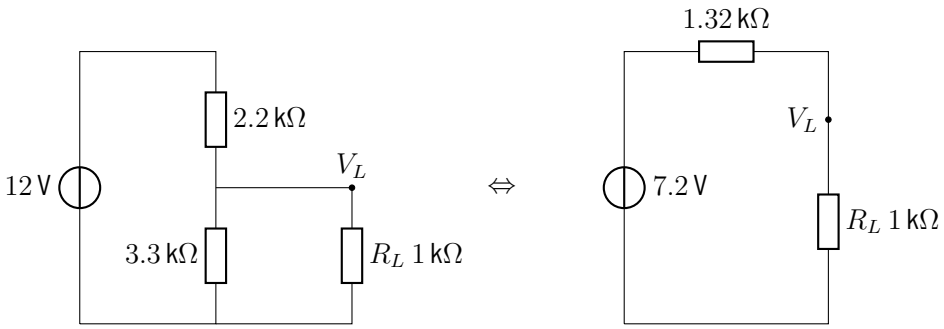
Every linear network of sources and resistors behaves at two terminals exactly like one ideal voltage source with one series resistor (Thevenin) or like one ideal current source with one parallel resistor (Norton). This is the most useful simplification in circuit design: a bridge half, a sensor with its supply or a divider are replaced by one source and one resistor, and the effect of the next stage (the load) can be calculated in one line. The superposition principle is the second tool for linear networks with several sources.

Thevenin and Norton equivalent



- open-circuit voltage: $V_{th} = V_{oc}$ (nothing connected to A–B)
- short-circuit current: $I_N = I_{sc}$ (A and B shorted)
- internal resistance $R_{th} = V_{oc}/I_{sc}$, or: set all sources to zero (voltage source $\rightarrow$ wire, current source $\rightarrow$ open) and calculate the resistance between A and B

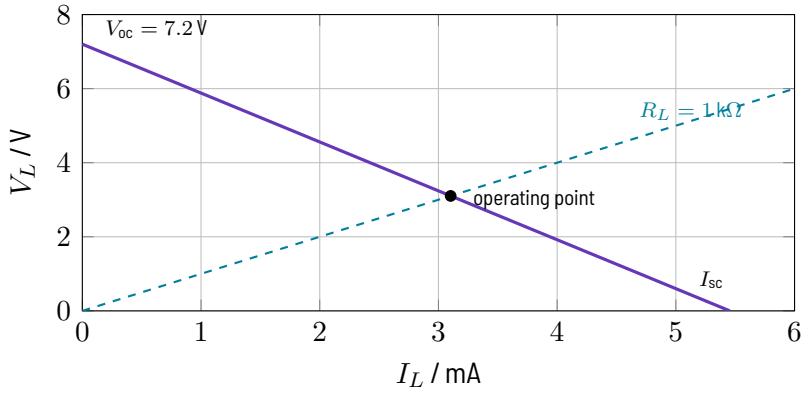
Example: Thevenin equivalent of a divider



- $V_{oc} = 12\,\text{V} \cdot 3.3/5.5 = 7.2\,\text{V}, \quad I_{sc} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$
- $R_{th} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega = 2.2\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega$
- with R_L: $V_L = 7.2\,\text{V} \cdot 1/(1 + 1.32) = 3.10\,\text{V}$ in both circuits

▶ Falstad: `examples/basics/thevenin.txt`

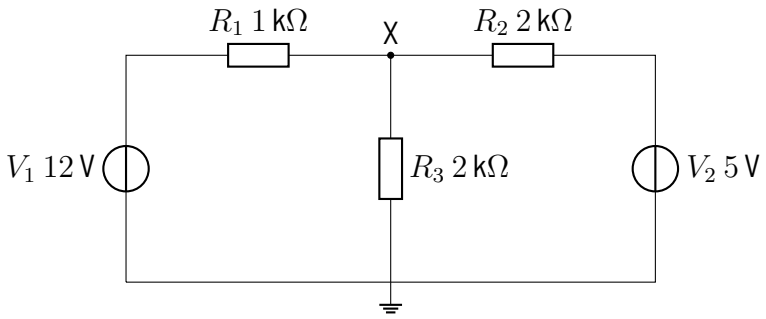
Load line of a real source



- the source can deliver every point on the line $V_L = V_{th} - R_{th}\, I_L$
- the load forces $V_L = R_L\, I_L$; the intersection is the operating point

The load line is the graphical form of the Thevenin equivalent. It connects the open-circuit voltage on the voltage axis with the short-circuit current on the current axis. Its slope is $-R_{\text{th}}$: the flatter the line, the stiffer the source. The method also works for nonlinear loads (an LED, a Zener diode): draw the characteristic of the load into the same diagram and read the intersection. The next chapter uses it for real voltage sources.

Superposition



- in a linear network the effect of all sources is the sum of the effects of each source alone
- the other sources are set to zero: voltage source → wire, current source → open
- V_1 alone: $V_{X1} = 12\,\text{V} \cdot \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} = 6\,\text{V}$; V_2 alone: $V_{X2} = 5\,\text{V} \cdot \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} = 1.25\,\text{V}$
- both: $V_X = 7.25\,\text{V}$ ▶ Falstad: `examples/basics/superposition.txt`

Superposition is valid only for linear circuits: it fails as soon as a diode switches or an op-amp clips. It is also not valid for power, because power is a quadratic function of current. In the lecture it is used for amplifiers with several inputs, for example the summing amplifier or an op-amp circuit with an offset voltage at one input.

Exercise 1.3: Thevenin equivalent from open-circuit voltage and short-circuit current

The divider $12\,\text{V}$, $R_1 = 2.2\,\text{k}\Omega$ (top), $R_2 = 3.3\,\text{k}\Omega$ (bottom) feeds a load $R_L = 1\,\text{k}\Omega$.

a) Determine V_{oc} and I_{sc} at the tap and from them R_{th}.

b) Calculate V_L with the equivalent circuit.

c) Which load would reduce the open-circuit voltage by only $5\,\%$?

d) Compare the original circuit and the equivalent in Falstad.

Solution 1.3

a) $V_{\text{oc}} = 12\,\text{V} \cdot 3.3/(2.2+3.3) = 7.2\,\text{V}$. With the tap shorted no current flows in R_2: $I_{\text{sc}} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$. $R_{\text{th}} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega$.

b) $V_L = 7.2\,\text{V} \cdot \frac{1}{1+1.32} = 3.10\,\text{V}$.

c) $V_L/V_{\text{oc}} = R_L/(R_L + R_{\text{th}}) = 0.95 \Rightarrow R_L = 19\,R_{\text{th}} = 25.1\,\text{k}\Omega$ (E12: $27\,\text{k}\Omega$).

d) ▶ Falstad: `examples/basics/thevenin.txt`: `VL_orig` = `VL_equiv` = $3.10\,\text{V}$. Replace R_L by a wire and hover over it: the short-circuit current is $5.45\,\text{mA}$ in both circuits.

Exercise 1.4: Superposition

Use the circuit of the superposition slide.

a) Calculate V_X with superposition.

b) Check the result with KCL at node X.

c) How large is the current through R_2, and in which direction does it flow?

d) Check the three cases in Falstad.

Solution 1.4

a) V_1 alone: $R_2 \parallel R_3 = 1\,\text{k}\Omega$, $V_{X1} = 12\,\text{V} \cdot 1/2 = 6\,\text{V}$. V_2 alone: $R_1 \parallel R_3 = 0.667\,\text{k}\Omega$, $V_{X2} = 5\,\text{V} \cdot 0.667/2.667 = 1.25\,\text{V}$. $V_X = 7.25\,\text{V}$.

b) $\frac{12-7.25}{1} = 4.75\,\text{mA}$ flows in through R_1; $\frac{7.25}{2} = 3.625\,\text{mA}$ flows out through R_3 and $\frac{7.25-5}{2} = 1.125\,\text{mA}$ through R_2: $4.75 = 3.625 + 1.125$ ✓

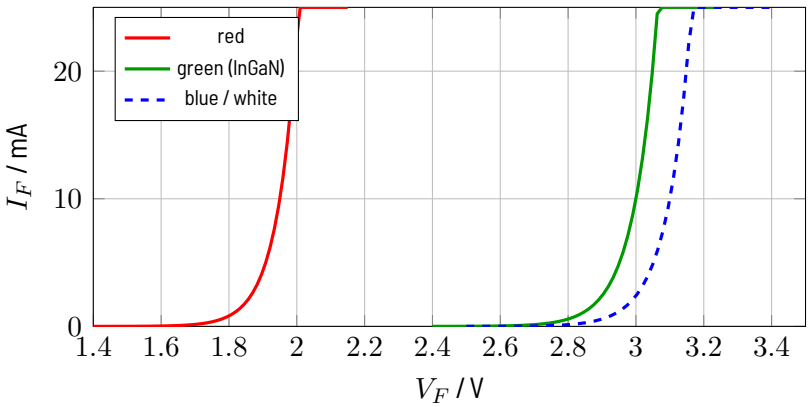
c) $1.125\,\text{mA}$ from X into the $5\,\text{V}$ source: V_2 is charged (absorbs power).

d) ▶ Falstad: `examples/basics/superposition.txt`: `VX1` $= 6.00\,\text{V}$, `VX2` $= 1.25\,\text{V}$, `VX` $= 7.25\,\text{V}$.

1.5 The LED with Series Resistor

The light emitting diode is the first nonlinear component of the lecture and the first task of the lab. It is also the last element of the capstone chain: the digital value of the flash converter is shown on LEDs, "exactly the LED and resistor from Task 1". An LED must never be connected directly to a voltage source. Its current rises exponentially with the voltage, so a few tenths of a volt make the difference between dark and destroyed. The series resistor turns the voltage source into a (rough) current source and sets the current.

LED characteristic



- below a threshold almost no current, above it the current rises steeply
- simple model: constant forward voltage V_F at the operating current

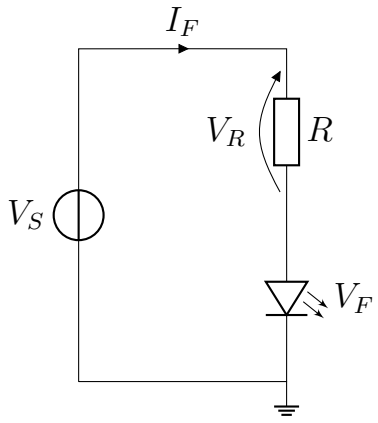
Forward voltage and current

colour	material	V_F at 10 mA … 20 mA
infrared	GaAs	1.2 V … 1.5 V
red	AlGaInP	1.8 V … 2.1 V
yellow, orange	AlGaInP	2.0 V … 2.2 V
green (classic, yellow-green)	GaP	2.1 V … 2.4 V
green (true green), blue, white	InGaN	2.8 V … 3.4 V

- standard 3 mm/5 mm LEDs: $I_F \leq 20$ mA; modern LEDs are clearly visible at 2 mA … 5 mA
- reverse voltage only about 5 V: an LED is not a rectifier

The forward voltage grows with the photon energy, i.e. it is higher for shorter wavelengths: red needs less than 2 V, blue and white (a blue LED with a phosphor) need about 3 V. This matters for a 3.3 V supply, where a blue LED leaves only a few hundred millivolts for the series resistor and the current depends strongly on the spread of V_F. The forward voltage also decreases by about 2 mV/K, so an LED without a resistor heats up, draws more current and heats up further.

Dimensioning the series resistor



LED series resistor

$$R = \frac{V_S - V_F}{I_F} \qquad P_R = (V_S - V_F) \cdot I_F$$

- example: $V_S = 5\,\text{V}$, red LED $V_F = 2.0\,\text{V}$, $I_F = 10\,\text{mA}$: $R = 300\,\Omega \Rightarrow$ E12: $330\,\Omega$, $I_F = 9.1\,\text{mA}$, $P_R = 27\,\text{mW}$

Design recipe

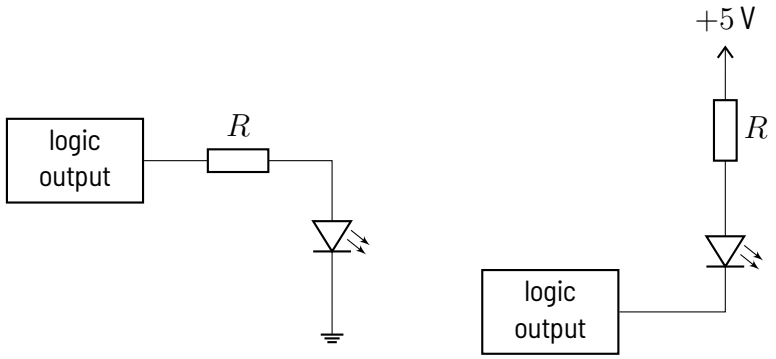
Design card: LED with series resistor

1. take V_F from the datasheet at the desired current (or the table: red $2.0\,\text{V}$, blue/white $3.0\,\text{V}$)
2. choose I_F: $2\,\text{mA}$ … $10\,\text{mA}$ for an indicator, check the maximum current of the driver
3. $R = (V_S - V_F)/I_F$, round **up** to the next E12/E24 value
4. recalculate I_F and check $P_R = I_F^2 R$ against the rating

- rounding up keeps the current below the target, which protects LED and driver
- the larger $V_S - V_F$, the less I_F depends on the spread of V_F

The last point explains why the resistor works as a current source: the current is $I_F = (V_S - V_F)/R$. If V_F changes by $0.1\,\text{V}$, the current changes by $0.1\,\text{V}/R$, which is $3\,\%$ at $V_S = 5\,\text{V}$ but $8\,\%$ at $3.3\,\text{V}$ for a red LED. For a blue LED at $3.3\,\text{V}$ the scheme hardly works any more; a real current source (chapter Sources) is better.

Driving an LED from a logic output



source: high = on sink: low = on

- a logic output is a switch to V_S or to ground with an output resistance of some $10\,\Omega$
- 74HC at $5\,\text{V}$: $\pm 4\,\text{mA}$ with guaranteed levels, absolute maximum $25\,\text{mA}$ per pin
- microcontrollers: some mA per pin and a limit for the whole port
- $330\,\Omega$ at $5\,\text{V}$ gives about $9\,\text{mA}$, $1\,\text{k}\Omega$ about $3\,\text{mA}$

▶ Falstad: `examples/basics/led_resistor.txt` A CMOS output stage consists of a p-channel transistor to the supply and an n-channel transistor to ground. Each acts like a resistor of some $10\,\Omega$ to $50\,\Omega$ when it is switched on. With an LED current of a few milliampere the output voltage therefore differs by only $0.1\,\text{V}$ … $0.3\,\text{V}$ from the rail, and the simple formula still applies. The LED can be connected to ground

(the output *sources* the current, the LED is on at high level) or to the supply (the output *sinks* the current, the LED is on at low level). Older TTL outputs can sink much more current than they can source; that is why many datasheets and old circuits connect LEDs to the supply. In the capstone task the three bits of the encoder drive three LEDs to ground. The Falstad example uses ideal logic outputs, so the full $5\,\mathrm{V}$ is available.

Exercise 1.5: LED on a $5\,\mathrm{V}$ logic output

A red LED ($V_F = 2.0\,\mathrm{V}$ at $10\,\mathrm{mA}$) is to be driven by a $5\,\mathrm{V}$ logic output.

a) Calculate the series resistor for $10\,\mathrm{mA}$ and choose an E12 value.

b) Calculate the actual current and the power in the resistor.

c) The same LED is to be driven from a $3.3\,\mathrm{V}$ microcontroller pin with at most $4\,\mathrm{mA}$. Choose the resistor.

d) Falstad uses an LED model with $V_F \approx 1.8\,\mathrm{V}$. Which current does the simulation show with $330\,\Omega$?

Solution 1.5

a) $R = (5\,\mathrm{V} - 2.0\,\mathrm{V})/10\,\mathrm{mA} = 300\,\Omega$; rounded up to E12: $330\,\Omega$.

b) $I_F = 3.0\,\mathrm{V}/330\,\Omega = 9.1\,\mathrm{mA}$, $P_R = 3.0\,\mathrm{V} \cdot 9.1\,\mathrm{mA} = 27\,\mathrm{mW}$: any resistor will do, even SMD 0402.

c) $R \geq (3.3\,\mathrm{V} - 2.0\,\mathrm{V})/4\,\mathrm{mA} = 325\,\Omega$, again $330\,\Omega$ ($I_F = 3.9\,\mathrm{mA}$; V_F is a bit lower at this current, so the real current is slightly higher).

d) ▶ Falstad: `examples/basics/led_resistor.txt`: $V_{LED} = 1.78\,\mathrm{V}$, so $I_F = (5 - 1.78)\,\mathrm{V}/330\,\Omega = 9.76\,\mathrm{mA}$ (hover over the resistor). Use the slider to see how the current changes with R.

1.6 Capacitor and RC Circuits

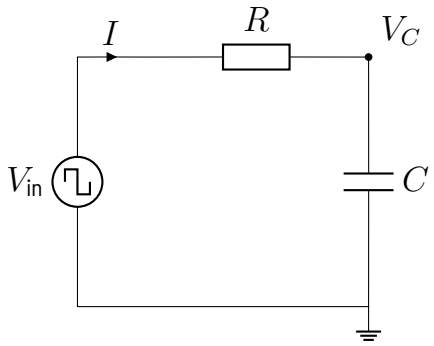
Capacitors store charge. In this lecture they appear as decoupling capacitors at every supply pin, in filters, in the hold capacitor of the sample and hold circuit and as the reason for every delay. The RC circuit is the simplest dynamic circuit; its exponential response and its time constant $\tau = RC$ appear again and again.

The capacitor

- charge proportional to voltage: $Q = C \cdot V$
- current = change of charge: $I = C \dfrac{dV}{dt}$
  - the voltage across a capacitor cannot jump (that would need an infinite current)
  - DC: once charged, no current flows (open circuit)
- energy: $E = \frac{1}{2} C V^2$
- types: ceramic (pF to 100 µF; C0G stable, X7R voltage dependent), film (precise, for filters and hold capacitors), electrolytic (large, polarized)

The relation $I = C\,dV/dt$ is the capacitor's counterpart of Ohm's law. A constant current charges a capacitor linearly: 1 mA into 1 µF gives 1 V/ms. This is used in integrators and ramp generators. Ceramic capacitors of type X7R or Y5V lose a large part of their capacitance with DC voltage and temperature; for filters and the hold capacitor of a sample and hold circuit use C0G (NP0) ceramic or film capacitors.

Charging through a resistor

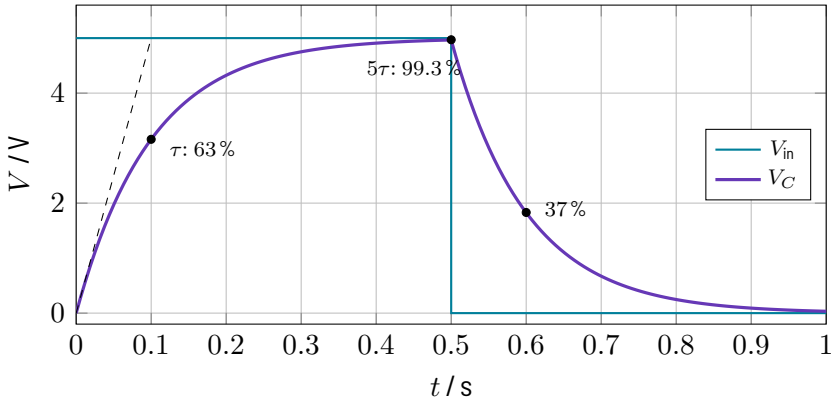


RC charging and discharging

$V_C(t) = V_0\left(1 - e^{-t/\tau}\right) \qquad V_C(t) = V_0\, e^{-t/\tau} \qquad \tau = R\,C$

- step of V_0 at the input; the current $I = (V_{in} - V_C)/R$ decreases as V_C rises
- time to reach a voltage V_1: $t = \tau \ln \frac{V_0}{V_0 - V_1}$

The time constant

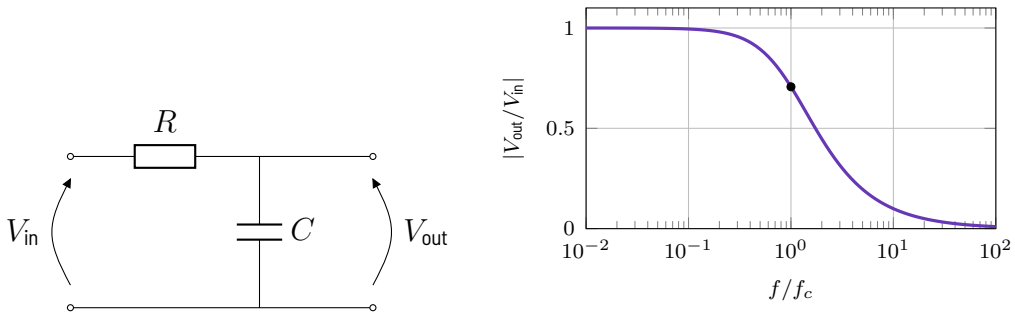


- after τ: 63 %, after 3τ: 95 %, after 5τ: 99.3 % ("settled")

- the tangent at $t = 0$ reaches the final value after exactly τ
- example $R = 10\,\text{k}\Omega$, $C = 10\,\mu\text{F}$: $\tau = 0.1\,\text{s}$ ▶ Falstad: `examples/basics/rc_charging.txt`

How long one has to wait depends on the required accuracy. Settling to within $1\,\%$ takes $4.6\,\tau$, to within half an LSB of an n-bit converter $\tau \ln(2^{n+1})$: $2.8\,\tau$ for the 3-bit converter of the capstone task, but $9\,\tau$ for a 12-bit converter. This is the reason why the acquisition time of a sample and hold circuit is specified together with an accuracy.

Preview: the RC low-pass



Cut-off frequency

$$f_c = \frac{1}{2\pi RC} \qquad \left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{1 + (f/f_c)^2}}$$

- at f_c the amplitude drops to $1/\sqrt{2} = 0.707$ (-3 dB), phase $-45°$
- example: $1.6\,\text{k}\Omega$, $100\,\text{nF}$: $f_c = 995\,\text{Hz}$ ▶ Falstad: `examples/basics/rc_lowpass.txt`

For slow signals the capacitor follows the input; for fast signals it cannot charge in time and the output amplitude decreases. Above f_c the amplitude falls by a factor of 10 per decade of frequency (-20 dB per decade). In the capstone task an anti-aliasing low-pass with $f_c \approx 2.5\,\text{Hz}$ precedes the sample and hold circuit that samples at $5\,\text{Hz}$; with $C = 10\,\mu\text{F}$ this needs $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot 10\,\mu\text{F}) = 6.4\,\text{k}\Omega$, E24: $6.2\,\text{k}\Omega$ ($f_c = 2.57\,\text{Hz}$). The chapter on active filters treats filters in detail. The RC low-pass in the Falstad example is driven with $1\,\text{kHz}$, almost exactly f_c: the output amplitude is $0.705\,\text{V}$ for an input amplitude of $1\,\text{V}$, and the output lags by about a quarter period.

Exercise 1.6: RC charging

A capacitor $C = 10\,\mu\text{F}$ is charged through $R = 10\,\text{k}\Omega$ from a square wave that switches between $0\,\text{V}$ and $5\,\text{V}$ at $1\,\text{Hz}$.

a) Calculate τ and V_C after τ and after $0.5\,\text{s}$.

b) After which time does V_C reach $4\,\text{V}$?

c) To which voltage does C discharge $0.1\,\text{s}$ after the falling edge?

d) Read the values from the Falstad scope.

Solution 1.6

a) $\tau = 10\,\text{k}\Omega \cdot 10\,\mu\text{F} = 0.1\,\text{s}$. $V_C(\tau) = 5\,\text{V}\,(1 - e^{-1}) = 3.16\,\text{V}$; $V_C(0.5\,\text{s}) = V_C(5\tau) = 5\,\text{V}\,(1 - e^{-5}) = 4.97\,\text{V}$.

b) $t = \tau \ln \frac{5}{5-4} = 0.1\,\text{s} \cdot \ln 5 = 0.161\,\text{s}$.

c) The discharge starts at $4.97\,\text{V}$: $V_C = 4.97\,\text{V} \cdot e^{-1} = 1.83\,\text{V}$.

d) ▶ Falstad: `examples/basics/rc_charging.txt`: the scope shows $3.16\,\text{V}$ at $0.1\,\text{s}$, $4\,\text{V}$ at $0.161\,\text{s}$, $4.97\,\text{V}$ at the falling edge and $1.83\,\text{V}$ $0.1\,\text{s}$ later.

1.7 Working with Falstad

Falstad's circuit simulator (CircuitJS) runs in the browser and needs no installation [11]. It simulates the circuit continuously and shows the voltages as colours and the currents as moving dots, which makes it ideal for getting a feeling for a circuit. It is not a replacement for SPICE: its models are simple (the op-amp has no bandwidth limit by default, transistors have no capacitances) and it cannot print exact plots. For the circuits of this lecture it is accurate enough, and every example has been checked against the hand calculation.

Opening and building circuits

- https://www.falstad.com/circuit/circuitjs.html
- open an example: click its Falstad box in the script, or *File* → *Import From Text* and paste the content of `examples/basics/*.txt`
- new circuit: *Circuits* → *Blank Circuit*
- place elements: *Draw* menu or right click; frequent ones have shortcuts (`r` resistor, `c` capacitor, `w` wire, `g` ground)
- drag to place, double click to edit values (`4.7k`, `100n`, `10u` are accepted)
- save your work: *File* → *Export As Text* or *Export As Link*

Pitfalls when drawing

- wires connect **only at their end points**: a wire passing over a terminal is not connected; split the wire there
- unconnected terminals are shown as red dots
- every circuit needs a ground (or a source that defines the potential)
- *labeled nodes* (*Draw* → *Wiring* → *Labeled Node*) with the same name are connected without a wire, and their name can be used for scopes
- "singular matrix": usually a floating node or two ideal voltage sources in parallel

Most errors in self-drawn circuits come from the first rule. A wire that runs over the end of a resistor looks connected, but it is not. Check the dots: a junction of three or more wires is drawn with a large dot. Moving an element with the mouse shows which wires follow it.

Measuring

- hover the mouse over an element: the bottom right corner shows voltage, current and power
- hover over a labeled node: its voltage to ground
- voltage colours: green positive, red negative, grey zero; yellow dots: current
- scope: right click on an element → *View in New Scope*; right click on a scope for settings (scale, several traces, *Max Scale*)
- sliders: *Edit* dialog of an element, check *Show slider*; the examples have sliders for the values worth changing
- simulation speed: slider on the right; the time step is in *Options* → *Other Options*

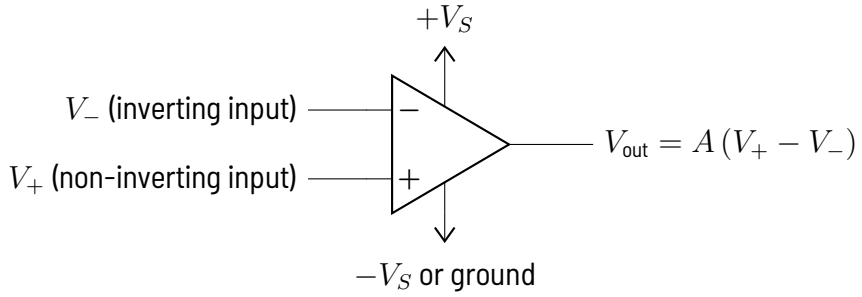
The examples of this lecture are generated by a Python program that also simulates them and asserts the numbers quoted in this script. When you open an example, compare the displayed values with the solution of the exercise. If they differ, check the time: circuits with capacitors need a few time constants before they reach their final state.
Example: open `led_resistor`. Hover over the label `VLED` to read the forward voltage (1.78 V), hover over the resistor to read the current (9.76 mA), and move the slider to change the resistor. The two LEDs on the right are driven by inverters and blink in turn: one is connected as a current source, one as a current sink.

2 Op-Amp Basics

2.1 The Op-Amp in a Nutshell

The operational amplifier (op-amp) appears in almost every chapter of this lecture: as a buffer behind a voltage reference, as the control element of a voltage or current source, as the amplifier behind a sensor bridge, in active filters and in the sample-and-hold circuit. This short chapter gives the minimum that the following chapters need: what an op-amp does, why it is always used with negative feedback, and the two golden rules that make the analysis of op-amp circuits easy. With these rules we look at the most useful op-amp circuit of all, the unity-gain buffer (voltage follower), and at the non-inverting amplifier.
Chapter 6 comes back to op-amps in detail: inverting, summing, difference and instrumentation amplifiers, and the limits of real op-amps. Good introductions are [1] (chapter 4) and [5].

Symbol and model



- an amplifier for the **difference** of two voltages, with a huge gain $A = 10^5 \ldots 10^7$
- inputs draw (almost) no current; the output can deliver some 10 mA
- the output stays inside the supply range; supply pins are often not drawn

The open-loop gain A is so large that a difference of a few hundred microvolts between the inputs is enough to drive the output to one of the supply rails. Without feedback the op-amp only tells which input is higher: it is a comparator (used in the flash ADC, chapter 9). To get a well-defined linear circuit, a part of the output voltage is fed back to the inverting input.

Negative feedback

- a part of V_{out} is fed back to the **inverting** input
- V_{out} rises $\Rightarrow$ V_- rises $\Rightarrow$ $V_+ - V_-$ shrinks $\Rightarrow$ V_{out} is pulled back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$ (50 µV for $V_{\text{out}} = 5$ V, $A = 10^5$)

Rule 2.1 Golden rules for an op-amp with negative feedback:

1. The output does whatever is necessary to make the voltage difference between the inputs zero: $V_+ = V_-$.
2. The inputs draw no current: $I_+ = I_- = 0$.

Rule 1 holds only while the feedback loop works: the output must not be stuck at a supply rail, and the feedback must go to the inverting input. Rule 2 is a property of the op-amp itself; real inputs draw picoamperes (CMOS) to some 10 nA (bipolar).

Real op-amps on a single supply

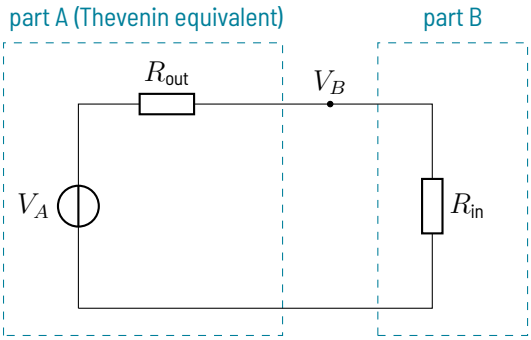
- most circuits in the lab run from a single supply (5 V or 12 V) and ground
- **input common-mode range**: the input voltages must stay inside it (often the supply minus 1.5 V)
- **output swing**: a rail-to-rail output reaches the rails within some 10 mV, older types (LM358) only within about 1.5 V of the positive rail
- **output current**: typically 10 mA … 40 mA, short-circuit protected
- examples: MCP6002, TLV9002 (CMOS, rail-to-rail, 5 V), LM358 (bipolar, up to 32 V)

The data sheet tells which voltages are allowed at the inputs and which output voltages are possible at a given load current. These two ranges are the first thing to check in every op-amp circuit, before any formula: the golden rules only hold inside them.

2.2 The Unity-Gain Buffer

A circuit is built from parts: a sensor divider, a filter, a reference, an amplifier, a converter input. Each part is designed for itself, with the formulas of chapter 1. When two parts are connected, the second part loads the first one, and the formulas are no longer valid: the voltage divider delivers less voltage, the filter has another cut-off frequency. The unity-gain buffer (voltage follower) separates the two parts. It copies the voltage from its input to its output, takes no current from the first part and delivers the current for the second part from the supply.

Every connection is a voltage divider



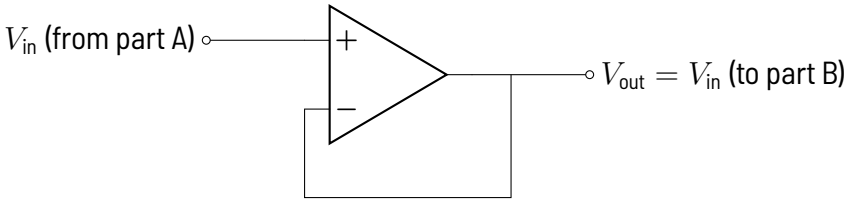
Loading

$$V_B = V_A \frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{out}}} \qquad \text{error} \approx \frac{R_{\text{out}}}{R_{\text{in}}} \text{ for } R_{\text{in}} \gg R_{\text{out}}$$

- rule of thumb: $R_{\text{in}} \geq 100\, R_{\text{out}}$ for an error below $1\,\%$

Part A is replaced by its Thevenin equivalent (chapter 1): a voltage divider of R_1 and R_2 has $R_{\text{out}} = R_1 \parallel R_2$, an RC low-pass has the output impedance $R \parallel (1/j\omega C)$. Part B is represented by its input resistance. If the condition $R_{\text{in}} \gg R_{\text{out}}$ cannot be met, a buffer is needed.

The voltage follower



- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$ (gain 1)
- rule 2: input current ≈ 0, input resistance GΩ …TΩ: part A is **not loaded**
- output resistance with feedback $r_o/(1+A)$: $50\,\Omega/10^5 = 0.5\,\text{m}\Omega$: part B sees an **ideal source**

Unity-gain buffer

$$V_{\text{out}} = V_{\text{in}} \qquad R_{\text{in}} \to \infty \qquad R_{\text{out}} \to 0$$

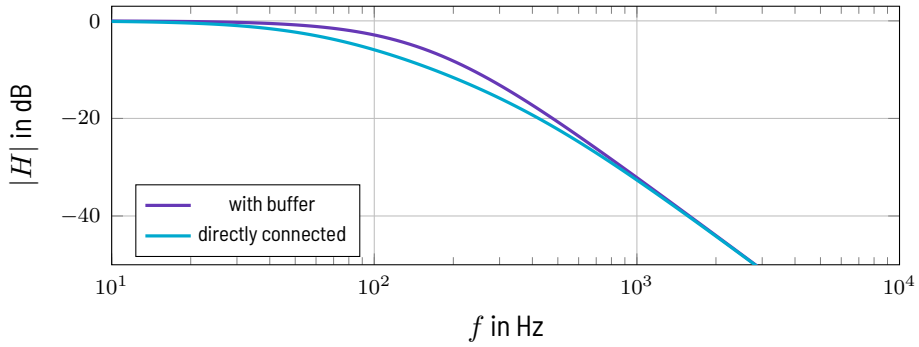
Example: potentiometer as a set-point

- potentiometer $10\,\text{k}\Omega$ at $5\,\text{V}$, wiper in the middle: $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$
- load $1\,\text{k}\Omega$ directly at the wiper: $V_B = 2.5\,\text{V} \cdot \frac{1}{1+2.5} = 0.714\,\text{V}$
- load behind a follower: $2.50\,\text{V}$; the op-amp delivers $2.5\,\text{mA}$
- the wiper itself still has $2.50\,\text{V}$: the follower does not load it

▶ Falstad: `examples/opampbasics/buffer_pot.txt`

The op-amp in the Falstad example runs from $5\,\text{V}$ and ground, like a CMOS rail-to-rail op-amp such as the MCP6002. With the slider of the directly connected load you can see how large R_L must be until the wiper voltage is close to $2.5\,\text{V}$. Exercise 2.1 works through this example.

Example: two RC low-passes in cascade



- each stage $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 159\,\text{Hz}$
- with buffer: $H = H_1 \cdot H_2 = \frac{1}{(1+j\omega RC)^2}$, at f_c: $|H| = 1/2$ (-6 dB)
- directly: $H = \frac{1}{1+3j\omega RC-(\omega RC)^2}$, at f_c: $|H| = 1/3$ (-9.5 dB)

▶ Falstad: `examples/opampbasics/buffer_rc_cascade.txt`

Without the buffer the second stage draws current from the capacitor of the first stage. The transfer function is no longer the product of the two single transfer functions, the cut-off frequency moves down and the corner becomes softer. The Falstad example drives both versions with a sine of 1 V amplitude at f_c: the scope shows 0.333 V for the direct and 0.500 V for the buffered cascade. With the buffer each stage can be designed on its own, which is the idea behind active filters (chapter 7).

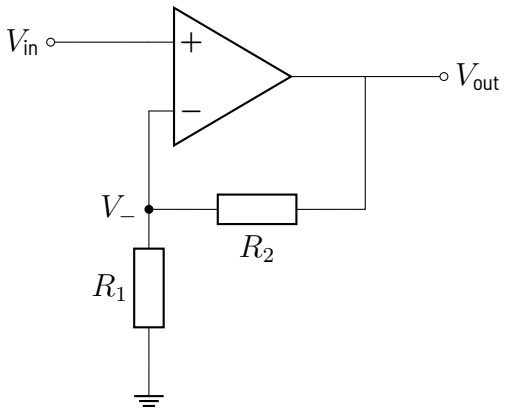
Where buffers are used – and their limits

- in this lecture: behind a voltage reference (chapter 4), at bridge taps (chapter 5), between filter stages (chapter 7), behind the hold capacitor (chapter 8)
- limits
  - output current (some 10 mA) and output swing
  - input common-mode range: e.g. LM358 inputs only up to $V_S - 1.5$ V
  - capacitive loads (long cables): the follower may oscillate; add $50\,\Omega$ … $100\,\Omega$ in series with the output
  - offset voltage: $V_{\text{out}} = V_{\text{in}} + V_{\text{OS}}$ (µV …mV)

2.3 The Non-Inverting Amplifier

If only a part of the output voltage is fed back, the follower becomes an amplifier. It keeps the good input property of the follower: the source is not loaded. The next chapter uses it to scale a reference voltage into an exact output voltage.

Non-inverting amplifier



- R_1, R_2 divide the output: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$
- $R_2 = 0$ (or $R_1 \to \infty$): the follower with $G = 1$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$, output resistance ≈ 0

The gain depends only on the ratio of two resistors, not on the op-amp. With $1\,\%$ resistors the gain is accurate to about $2\,\%$; with $0.1\,\%$ resistors to $0.2\,\%$. Chapter 6 derives the inverting amplifier and shows how a finite open-loop gain reduces the accuracy.

Exercise 2.1: Buffering a potentiometer

A $10\,\text{k}\Omega$ potentiometer between $5\,\text{V}$ and ground sets a voltage. The wiper is in the middle position. The voltage is used by a following stage with an input resistance of $R_L = 1\,\text{k}\Omega$.

a) Draw the Thevenin equivalent of the wiper. In which wiper position is R_{out} largest?

b) Calculate the voltage at R_L without a buffer and the error in percent.

c) How large must R_L be for an error below $1\,\%$ without a buffer?

d) Now a voltage follower (op-amp at $5\,\text{V}$, rail-to-rail) is placed between wiper and R_L. Calculate the voltage at R_L, the current from the wiper and the output current of the op-amp.

e) The wiper is turned to the upper end ($5\,\text{V}$). What does an LM358 at $5\,\text{V}$ supply deliver, and why?

f) Check (b) and (d) with Falstad.

Solution 2.1

a) $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$. For a wiper position x (0 ...1): $R_{\text{out}} = x(1-x)\,10\,\text{k}\Omega$, largest in the middle ($R/4$).

b) $V_B = 2.5\,\text{V} \cdot 1\,\text{k}\Omega/(1\,\text{k}\Omega + 2.5\,\text{k}\Omega) = 0.714\,\text{V}$, error $-71\,\%$.

c) $R_L \geq 100 \cdot 2.5\,\text{k}\Omega = 250\,\text{k}\Omega$.

d) $V_B = 2.50\,\text{V}$; wiper current ≈ 0 (picoamperes for a CMOS op-amp); op-amp output current $2.5\,\text{V}/1\,\text{k}\Omega = 2.5\,\text{mA}$, which comes from the $5\,\text{V}$ supply.

e) The input common-mode range of the LM358 ends at about $V_S - 1.5\,\text{V} = 3.5\,\text{V}$, and its output does not reach the positive rail either: the output stays at about $3.5\,\text{V}$. For signals up to the supply a rail-to-rail op-amp is needed.

f) ▶ Falstad: `examples/opampbasics/buffer_pot.txt`

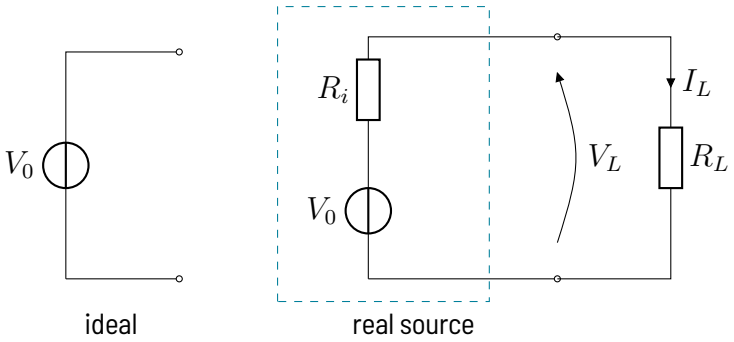
`V_direct` $= 0.714\,\mathrm{V}$, `V_wiper` $=$ `V_buffered` $= 2.50\,\mathrm{V}$.

3 Voltage and Current Sources

3.1 Ideal and Real Voltage Sources

Every circuit needs energy, and every measurement needs a quantity that is known exactly: a supply voltage, a reference voltage or an excitation current for a sensor. This chapter shows how real sources differ from the ideal ones of circuit theory, and how simple circuits with Zener diodes, transistors and op-amps produce stable voltages and currents. In the capstone task the bridge needs a stable supply, the reference ladder of the flash converter needs a stable 4 V, and the resistive sensor must not carry more than 1 mA. The chapter on voltage references continues with precise references (bandgap references); here we treat the basic circuits.

Ideal and real voltage source



- ideal: the voltage is V_0 for every current
- real: the terminal voltage drops with the load current

Terminal voltage of a real source

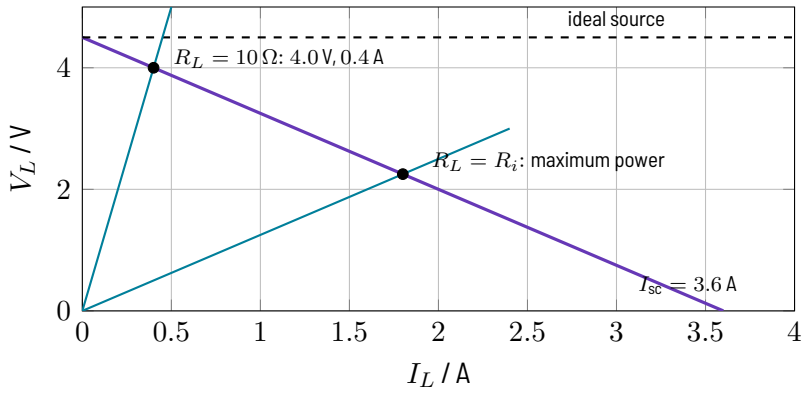
$$V_L = V_0 - R_i\, I_L = V_0\, \frac{R_L}{R_i + R_L}$$

Examples of internal resistances

- AA alkaline cell: $V_0 = 1.5\,\text{V}$, $R_i \approx 0.1\,\Omega \ldots 0.3\,\Omega$ (rises as the cell discharges)
- 9 V block battery: $R_i \approx 1\,\Omega \ldots 2\,\Omega$
- laboratory power supply in voltage mode: some mΩ, plus the leads
- voltage divider: $R_i = R_1 \parallel R_2$ (chapter Basics)
- logic output: $20\,\Omega \ldots 50\,\Omega$
- op-amp output with feedback: well below $1\,\Omega$ at low frequencies

The internal resistance is not a resistor one could find inside the source; it is a model for everything that makes the voltage drop under load: the chemistry of a battery, the control loop of a power supply, the output transistors of a logic gate. A real source is therefore described by two numbers, V_0 and R_i, which can be measured: the open-circuit voltage directly, the internal resistance from a second measurement with a known load, $R_i = (V_0 - V_L)/I_L$. A short circuit to measure I_{sc} directly is not a good idea for batteries and power supplies.

Load line and operating point

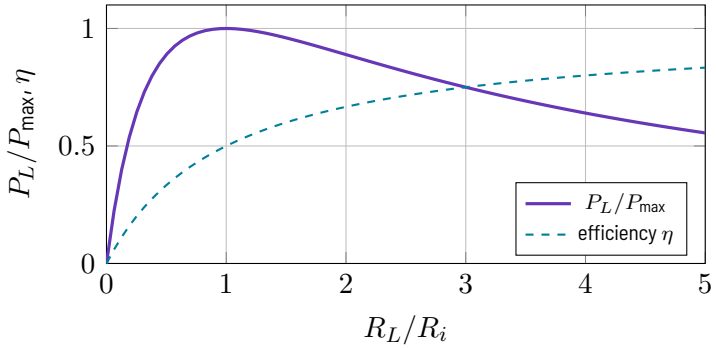


- example: $V_0 = 4.5\,\text{V}$, $R_i = 1.25\,\Omega$; load lines of $R_L = 10\,\Omega$ and $R_L = R_i$

- the operating point is the intersection of source line and load line

▶ Falstad: `examples/sources/real_source.txt`

Power matching



- load power $P_L = V_0^2\, R_L/(R_i + R_L)^2$ is maximum for $R_L = R_i$: $P_{\max} = V_0^2/(4R_i)$
- then half of the power is lost in the source: efficiency $\eta = 50\,\%$
- power supplies and signal sources for measurement: $R_L \gg R_i$ (voltage matching)

Power matching is important in radio frequency circuits and for antennas, where the cable impedance must be matched, and for weak energy sources such as thermoelectric generators. For supplies and in measurement circuits it is the wrong goal: we want the voltage to stay constant, so the load resistance must be large compared with the internal resistance. The same rule appeared for the loaded voltage divider.

Exercise 3.1: Measuring a real voltage source

A battery pack shows $4.5\,\text{V}$ without load and $4.0\,\text{V}$ with a load of $10\,\Omega$.

a) Calculate the load current and the internal resistance.

b) Calculate the short-circuit current.

c) Which load takes the maximum power from the battery? How large is this power and the efficiency?

d) Check (a) in Falstad and use the slider to find the maximum load power.

Solution 3.1

a) $I_L = 4.0\,\text{V}/10\,\Omega = 0.4\,\text{A}$, $R_i = (4.5\,\text{V} - 4.0\,\text{V})/0.4\,\text{A} = 1.25\,\Omega$.

b) $I_{\text{sc}} = V_0/R_i = 4.5\,\text{V}/1.25\,\Omega = 3.6\,\text{A}$.

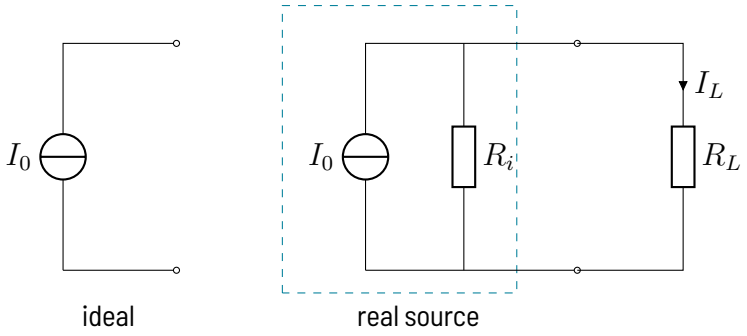
c) $R_L = R_i = 1.25\,\Omega$: $P_{\max} = (4.5\,\text{V})^2/(4 \cdot 1.25\,\Omega) = 4.05\,\text{W}$, $V_L = 2.25\,\text{V}$, $\eta = 50\,\%$; the battery heats up with the same $4.05\,\text{W}$.

d) ▶ Falstad: `examples/sources/real_source.txt`: `V_ideal` $= 4.50\,\text{V}$, `V_term` $= 4.00\,\text{V}$, $I_L = 0.400\,\text{A}$ (hover over R_L). With the slider at $R_L = 1.25\,\Omega$ the resistor shows $4.05\,\text{W}$.

3.2 Current Sources and Source Transformation

A current source delivers a given current whatever voltage the load needs. In the lab it is less familiar than a voltage source, but it is everywhere inside integrated circuits: it biases transistor stages, it charges capacitors linearly, and it excites resistive sensors, because the voltage across the sensor is then directly proportional to its resistance.

Ideal and real current source

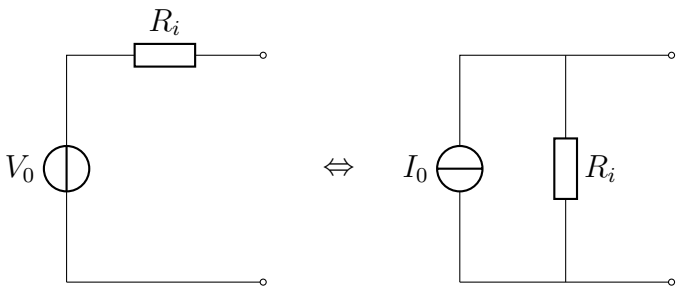


Load current of a real current source

$$I_L = I_0 - \frac{V_L}{R_i} = I_0 \frac{R_i}{R_i + R_L}$$

- a good current source has a **large** internal resistance ($R_i \gg R_L$)
- an ideal current source must never be left open: its voltage would rise without limit
- real current sources work only in a limited voltage range: the **compliance range**

Source transformation



- both have the same open-circuit voltage V_0 and the same short-circuit current $I_0 = V_0/R_i$
- therefore they behave identically for **every** load (Thevenin ↔ Norton)
- inside, they are different: the power lost in R_i is not the same
- example: $10\,\text{V}$ with $10\,\text{k}\Omega \Leftrightarrow 1\,\text{mA}$ with $10\,\text{k}\Omega$; with $R_L = 4.7\,\text{k}\Omega$ both give $3.20\,\text{V}$

▶ Falstad: `examples/sources/source_transform.txt` Whether a source is called a voltage source or a current source is a question of the load. A source with $R_i = 10\,\text{k}\Omega$ is a fairly good voltage source for a $1\,\text{M}\Omega$ load and a fairly good current source for a $100\,\Omega$ load. Source transformation is also a practical tool for calculations: series connections are easy with voltage sources, parallel connections with current sources.

Exercise 3.2: Source transformation

A source consists of $10\,\text{V}$ in series with $10\,\text{k}\Omega$. It feeds $R_L = 4.7\,\text{k}\Omega$.

a) Transform it into a current source with a parallel resistor.

b) Calculate the load voltage with both models.

c) Calculate the power dissipated in the internal resistance for both models. What do you conclude?

d) Check (b) in Falstad.

Solution 3.2

a) $I_0 = 10\,\text{V}/10\,\text{k}\Omega = 1\,\text{mA}$ parallel to $10\,\text{k}\Omega$.

b) Voltage source: $V_L = 10\,\text{V} \cdot 4.7/14.7 = 3.20\,\text{V}$. Current source: $V_L = 1\,\text{mA} \cdot (10\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega) = 1\,\text{mA} \cdot 3.20\,\text{k}\Omega = 3.20\,\text{V}$.

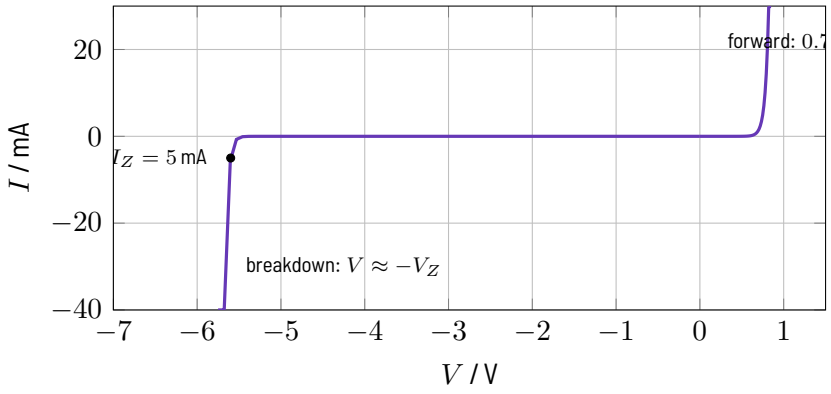
c) Voltage source: $I = 0.680\,\text{mA}$ flows through R_i, $P = 4.63\,\text{mW}$. Current source: R_i sees $3.20\,\text{V}$, $P = 1.02\,\text{mW}$. The models are equivalent only at the terminals, not inside.

d) ▶ Falstad: `examples/sources/source_transform.txt`: `VL_thevenin` = `VL_norton` = $3.20\,\text{V}$.

3.3 The Zener Diode Shunt Regulator

A Zener diode is a diode that is operated in reverse breakdown. Above the breakdown voltage the reverse current rises steeply, so the voltage stays almost constant over a wide current range. Together with a series resistor this gives the simplest voltage regulator. It is good enough for small currents and as a reference for the transistor regulator of the next section.

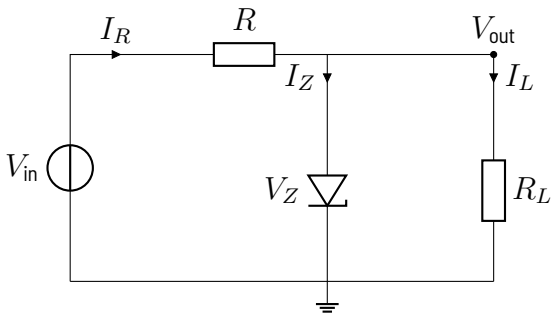
Zener diode characteristic



- breakdown voltages $2.4\,\text{V}$ … $75\,\text{V}$ in E24 steps (BZX55, BZX79: $0.5\,\text{W}$; 1N47xx: $1.3\,\text{W}$)
- datasheet: V_Z at a test current (often $5\,\text{mA}$), dynamic resistance $r_Z = \Delta V / \Delta I$

Below about $5\,\text{V}$ the breakdown is caused by the Zener effect (tunnelling), above by avalanche multiplication. Diodes around $5\,\text{V}$ … $6\,\text{V}$ have the smallest temperature coefficient and the smallest dynamic resistance, which is why $5.6\,\text{V}$ and $6.2\,\text{V}$ are popular reference values. Low voltage Zener diodes have a soft knee: a $2.7\,\text{V}$ type has a dynamic resistance of some $100\,\Omega$. The Falstad model is calibrated so that the current is exactly $5\,\text{mA}$ at the nominal V_Z, and it is much stiffer than a real diode (a few ohms instead of typically $10\,\Omega$ … $40\,\Omega$).

Shunt regulator



- the resistor takes the voltage difference: $I_R = (V_{\text{in}} - V_Z)/R$
- the Zener takes what the load does not need: $I_Z = I_R - I_L$
- the Zener must keep a minimum current $I_{Z,\text{min}}$ (about $5\,\text{mA}$, or $10\,\%$ of $I_{Z,\text{max}}$)

Designing a shunt regulator

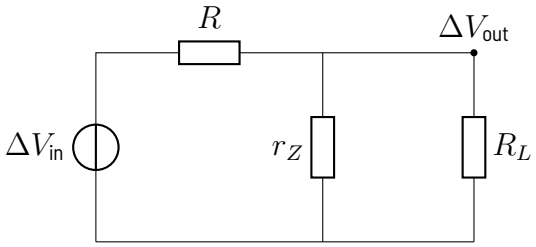
Design card: Zener shunt regulator

1. worst case for regulation: lowest V_{in}, largest load current $\Rightarrow\ R \le \dfrac{V_{\text{in,min}} - V_Z}{I_{L,\text{max}} + I_{Z,\text{min}}}$, choose the next **smaller** E-value
2. worst case for the Zener: highest V_{in}, no load $\Rightarrow\ I_{Z,\text{max}} = \dfrac{V_{\text{in,max}} - V_Z}{R}$, $P_Z = V_Z\, I_{Z,\text{max}}$
3. power of the resistor: $P_R = (V_{\text{in,max}} - V_Z)^2 / R$

- example: $12\,\text{V}$, $5.6\,\text{V}$, $I_L = 0\,\text{mA} \ldots 20\,\text{mA}$: $R \le 6.4\,\text{V}/25\,\text{mA} = 256\,\Omega \Rightarrow 220\,\Omega$
- the efficiency is poor: the full current flows all the time, even without load

▶ Falstad: `examples/sources/zener_shunt.txt`

Regulation and ripple



- for small changes the Zener is a resistor r_Z: small-signal equivalent circuit
- line regulation: $\dfrac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \dfrac{r_Z \parallel R_L}{R + r_Z \parallel R_L} \approx \dfrac{r_Z}{R}$
- load regulation: output resistance $R_{\text{out}} = R \parallel r_Z \approx r_Z$
- example $r_Z = 20\,\Omega$, $R = 220\,\Omega$: $1\,\text{V}$ ripple at the input $\rightarrow 83\,\text{mV}$ at the output

The Zener regulator suppresses input changes only by the ratio r_Z/R, typically a factor of 10. A larger resistor improves this, but requires a higher input voltage. Better regulation is obtained when the Zener diode is fed from a current source, or when a feedback loop with an amplifier is used (series regulator, voltage reference). The output resistance r_Z means that the output voltage changes with the load current: for $20\,\text{mA}$ and $r_Z = 20\,\Omega$ by $0.4\,\text{V}$.

Exercise 3.3: Design of a Zener shunt regulator

A load needs $5.6\,\text{V}$ and draws $0\,\text{mA}$ … $20\,\text{mA}$. The input voltage is $12\,\text{V}$. Use a $5.6\,\text{V}$ Zener diode with $I_{Z,\text{min}} = 5\,\text{mA}$.

a) Calculate the series resistor and choose an E12 value.

b) Calculate the maximum Zener current and the power of Zener diode and resistor. Which ratings do you choose?

c) Does the circuit still regulate at full load when the input drops to $11\,\text{V}$?

d) Read the output voltage without load and with $R_L = 270\,\Omega$ in Falstad and calculate I_Z.

Solution 3.3

a) $R \le (12\,\text{V} - 5.6\,\text{V})/(20\,\text{mA} + 5\,\text{mA}) = 256\,\Omega \Rightarrow 220\,\Omega$.

b) Without load: $I_{Z,\text{max}} = 6.4\,\text{V}/220\,\Omega = 29.1\,\text{mA}$, $P_Z = 5.6\,\text{V} \cdot 29.1\,\text{mA} = 163\,\text{mW} \Rightarrow 0.5\,\text{W}$ Zener (BZX79C5V6). $P_R = (6.4\,\text{V})^2/220\,\Omega = 186\,\text{mW} \Rightarrow 0.5\,\text{W}$ resistor.

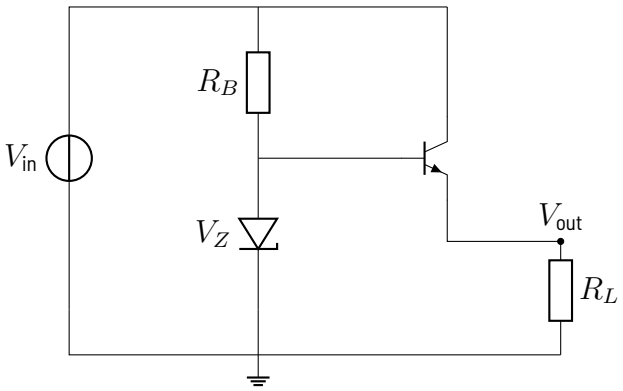
c) $I_R = (11\,\text{V} - 5.6\,\text{V})/220\,\Omega = 24.5\,\text{mA}$, $I_Z = 4.5\,\text{mA}$: just below $I_{Z,\text{min}}$; the voltage drops slightly, but the circuit still works. For a safe margin choose $200\,\Omega$ (E24).

d) ▶ Falstad: `examples/sources/zener_shunt.txt`
`Vout_open` $= 5.65\,\text{V}$ ($I_Z = I_R = 28.9\,\text{mA}$), `Vout_load` $= 5.61\,\text{V}$,
$I_L = 5.61\,\text{V}/270\,\Omega = 20.8\,\text{mA}$, $I_R = (12 - 5.61)\,\text{V}/220\,\Omega = 29.0\,\text{mA}$, $I_Z = 8.3\,\text{mA}$.

3.4 Series Regulators

The shunt regulator wastes current and cannot deliver large load currents. A series regulator puts a transistor between input and output. The transistor carries only the load current, and the Zener diode only has to deliver the small base current. Integrated linear regulators add an error amplifier, a bandgap reference and protection circuits; they are the standard solution for supplies up to about 1 A.

Emitter follower with Zener diode



Emitter follower regulator

$$V_{\text{out}} = V_Z - V_{BE} \approx V_Z - 0.65\,\text{V} \qquad I_B = \frac{I_L}{\beta + 1}$$

- the Zener sees only I_B: load changes hardly affect V_Z

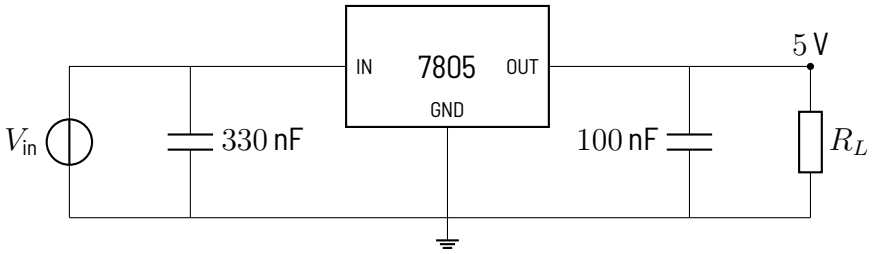
▶ Falstad: `examples/sources/series_regulator.txt`

Properties of the emitter follower regulator

- output resistance $\approx r_e + (R_B \parallel r_Z)/(\beta + 1)$ with $r_e = V_T/I_E$ ($0.25\,\Omega$ at $100\,\text{mA}$): much smaller than with the Zener alone
- V_{BE} depends on current and temperature ($-2\,\text{mV/K}$): V_{out} is only accurate to about $0.1\,\text{V}$
- transistor power: $P_Q = (V_{\text{in}} - V_{\text{out}}) \cdot I_L$ (heat sink!)
- design of R_B: $\dfrac{V_{\text{in,min}} - V_Z}{R_B} \geq I_{B,\text{max}} + I_{Z,\text{min}}$
- no short-circuit protection: a short at the output destroys the transistor

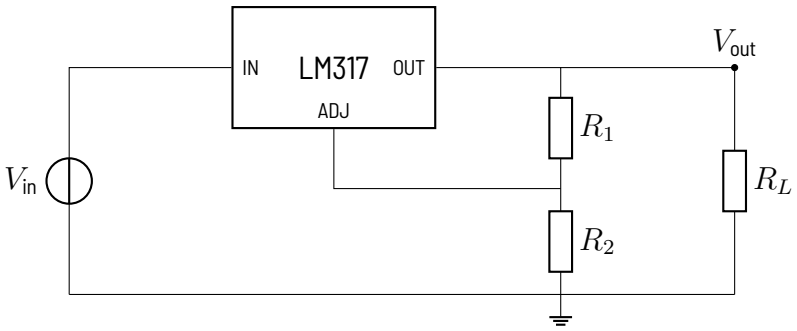
The Falstad example shows the effect: with $470\,\Omega$ (about $10\,\text{mA}$) the output is $4.95\,\text{V}$, with $47\,\Omega$ ($104\,\text{mA}$) it is $4.89\,\text{V}$. The Zener voltage itself changes by only $4\,\text{mV}$; most of the change comes from V_{BE}, which grows by $V_T \ln 10 = 60\,\text{mV}$ for ten times the current. An op-amp that compares the output with the reference and drives the transistor removes this error: that is the principle of the integrated regulator (section 3.5).

Integrated fixed regulator 78xx



- 7805, 7812, ...: fixed output voltage, $1\,\text{A}$ (TO-220), $0.1\,\text{A}$ (78L05); 79xx for negative voltages
- dropout about $2\,\text{V}$: $V_{\text{in}} \geq V_{\text{out}} + 2\,\text{V}$; LDO regulators: $0.1\,\text{V} \ldots 0.5\,\text{V}$
- internal current limit and thermal shutdown; capacitors close to the pins for stability

Adjustable regulator LM317



LM317

$$V_{out} = 1.25\,\text{V}\left(1 + \frac{R_2}{R_1}\right) + I_{ADJ}\,R_2 \qquad I_{ADJ} \approx 50\,\mu\text{A}$$

- the regulator keeps $1.25\,\text{V}$ between OUT and ADJ; $R_1 = 240\,\Omega$ sets $5.2\,\text{mA}$, enough for the minimum load

The LM317 has no ground pin. Its internal reference sits between OUT and ADJ, and all its quiescent current flows out of the output. The current through R_1 ($1.25\,\text{V}/R_1$) also flows through R_2; together with the small I_{ADJ} this gives the formula. The data sheet requires a minimum load current of about $10\,\text{mA}$ (typically $3.5\,\text{mA}$); the divider takes care of it. Without R_2 and with the load in place of R_2, the LM317 becomes a current source: $I = 1.25\,\text{V}/R_1$. Falstad has no model for the LM317; its principle (reference, error amplifier, series transistor) is the topic of the chapter on voltage references.

Exercise 3.4: Emitter follower regulator

A Zener diode $5.6\,\text{V}$ with $R_B = 1\,\text{k}\Omega$ drives the base of an npn transistor ($\beta = 100$) from $V_{in} = 12\,\text{V}$. The load is $R_L = 47\,\Omega$.

a) Estimate V_{out} and the load current with $V_{BE} = 0.7\,\text{V}$ (high current).

b) Calculate base current and Zener current.

c) Calculate the power of the transistor. Is a TO-92 transistor ($0.6\,\text{W}$) sufficient?

d) Which series resistor would a Zener shunt regulator need for the same load? Compare the idle current.

e) Compare with Falstad for $R_L = 47\,\Omega$ and $470\,\Omega$.

Solution 3.4

a) $V_{out} \approx 5.6\,\text{V} - 0.7\,\text{V} = 4.9\,\text{V}$, $I_L = 4.9\,\text{V}/47\,\Omega = 104\,\text{mA}$.

b) $I_B = 104\,\text{mA}/101 = 1.03\,\text{mA}$; $I_{R_B} = 6.4\,\text{V}/1\,\text{k}\Omega = 6.4\,\text{mA}$, $I_Z = 5.4\,\text{mA}$.

c) $P_Q = (12\,\text{V} - 4.9\,\text{V}) \cdot 104\,\text{mA} = 0.74\,\text{W}$: too much for TO-92; use a BD139 (TO-126) with a small heat sink.

d) $R \le 6.4\,\text{V}/(104\,\text{mA} + 5\,\text{mA}) = 59\,\Omega \Rightarrow 56\,\Omega$. Without load the Zener takes $114\,\text{mA}$ ($0.64\,\text{W}$); the series regulator only draws $6.4\,\text{mA}$.

e) ▶ Falstad: `examples/sources/series_regulator.txt`: $R_L = 47\,\Omega$: $V_{out} = 4.89\,\text{V}$, $I_L = 104\,\text{mA}$, $V_Z = 5.60\,\text{V}$; $R_L = 470\,\Omega$: $V_{out} = 4.95\,\text{V}$, $V_Z = 5.61\,\text{V}$.

Exercise 3.5: Adjustable regulator LM317

a) Design an LM317 regulator for $5\,\text{V}$ with $R_1 = 240\,\Omega$. Choose R_2 from E24 and calculate the actual output voltage (with $I_{\text{ADJ}} = 50\,\mu\text{A}$).

b) Which input voltage is needed at least (dropout $2\,\text{V}$)? What is the power of the LM317 at $12\,\text{V}$ input and $300\,\text{mA}$?

c) The LM317 is used as a current source for an LED string with $20\,\text{mA}$. Calculate the resistor.

Solution 3.5

a) $R_2 = R_1\,(5/1.25 - 1) = 3\,R_1 = 720\,\Omega \Rightarrow$ E24: $750\,\Omega$. $V_{\text{out}} = 1.25\,\text{V}\,(1 + 750/240) + 50\,\mu\text{A} \cdot 750\,\Omega = 5.16\,\text{V} + 0.04\,\text{V} = 5.20\,\text{V}$. (With E96: $R_2 = 715\,\Omega$ gives $4.97\,\text{V}$.)

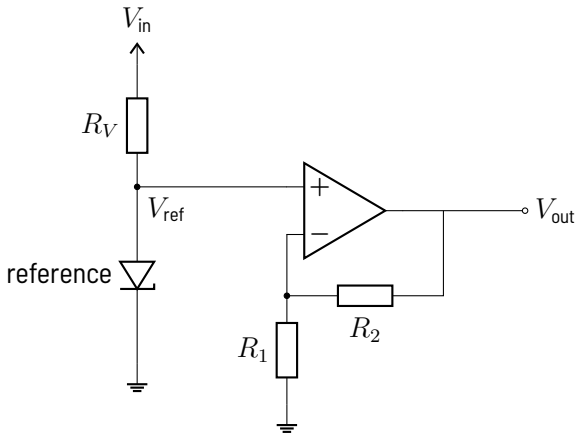
b) $V_{\text{in}} \geq 7.2\,\text{V}$. $P = (12\,\text{V} - 5.2\,\text{V}) \cdot 0.3\,\text{A} = 2.0\,\text{W}$: TO-220 with heat sink.

c) $R = 1.25\,\text{V}/20\,\text{mA} = 62.5\,\Omega \Rightarrow$ E24: $62\,\Omega$ ($20.2\,\text{mA}$); the LM317 then needs $V_{\text{in}} \geq V_{\text{LEDs}} + 1.25\,\text{V} + 2\,\text{V}$.

3.5 Op-Amp Voltage Sources

The emitter follower regulator loses about $0.1\,\text{V}$ between light and full load, because V_{BE} changes with the current. An op-amp removes this error. It compares the output voltage, divided by R_1 and R_2, with a reference voltage and drives the output until both are equal (golden rule 1, chapter 2). The reference only sees the op-amp input and is not loaded at all. The output voltage is set by a resistor ratio, and the output resistance is a few milliohms. For load currents above some $10\,\text{mA}$ a transistor inside the feedback loop boosts the output current. Reference, error amplifier and series transistor are exactly the blocks of every integrated linear regulator.

Reference and non-inverting amplifier



Op-amp voltage source

$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_2}{R_1}\right) \qquad R_{\text{out}} \approx \frac{r_o}{1 + A\beta},\ \beta = \frac{R_1}{R_1 + R_2}$$

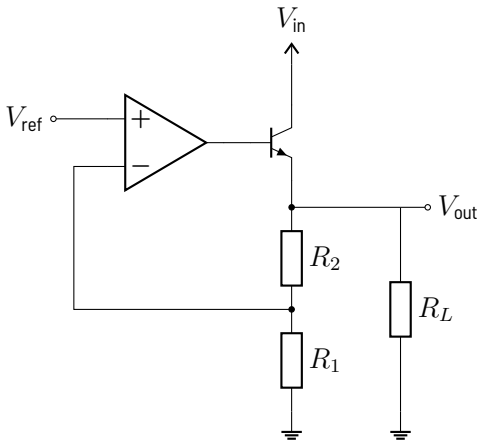
- the reference is not loaded; V_{out} is as accurate as V_{ref} and the ratio R_2/R_1

Output resistance and output current

- example: $r_o = 50\,\Omega$, $A = 10^5$, $R_1 = R_2$: $R_{\text{out}} = 50\,\Omega/(1 + 0.5 \cdot 10^5) = 1\,\text{m}\Omega$
  - 10 mA more load current changes V_{out} by only 10 µV
- but: the op-amp itself delivers only some 10 mA
- and: V_{out} must stay inside the output swing of the op-amp

The feedback makes the output resistance smaller by the loop gain $A\beta$, with the feedback factor β. This holds at low frequencies; at higher frequencies A falls (chapter 6) and the output resistance rises. A capacitor of some 100 nF at the load takes over at high frequencies. A large capacitor directly at the op-amp output can make it oscillate; data sheets state the allowed capacitive load.

More current: transistor inside the loop



- npn emitter follower as **booster**: the op-amp delivers only $I_B = I_L/(\beta + 1)$
- V_{BE} lies inside the loop: the op-amp output moves to $V_{\text{out}} + V_{BE}$, V_{out} stays exact

- transistor power $(V_{\text{in}} - V_{\text{out}}) \cdot I_L$; the op-amp output must reach $V_{\text{out}} + V_{BE}$

Example: $5.00\,\text{V}$ from $12\,\text{V}$, up to $100\,\text{mA}$

- reference LM4040-2.5 ($2.500\,\text{V}$, $0.1\,\%$) with $R_V = 4.7\,\text{k}\Omega$: $2.0\,\text{mA}$
- $G = 2$: $R_1 = R_2 = 10\,\text{k}\Omega$ ($0.1\,\%$) $\Rightarrow V_{\text{out}} = 5.00\,\text{V}$
- booster BD139 ($\beta \approx 100$): op-amp output $5.7\,\text{V}$ and $1\,\text{mA}$ at $100\,\text{mA}$ load

	$R_L = 470\,\Omega$	$R_L = 47\,\Omega$
emitter follower with Zener (section 3.4)	$4.95\,\text{V}$	$4.89\,\text{V}$
op-amp and booster	$5.000\,\text{V}$	$5.000\,\text{V}$

▶ Falstad: `examples/sources/opamp_voltage_source.txt`

In the Falstad example the reference is an ideal $2.5\,\text{V}$ source, which stands for the LM4040 with its series resistor. Both cells show `Vout_10mA` = `Vout_100mA` = $5.000\,\text{V}$, over the whole range of the R_L slider ($20\,\Omega$ to $1\,\text{k}\Omega$). The transistor dissipates $(12\,\text{V} - 5\,\text{V}) \cdot 106\,\text{mA} = 0.74\,\text{W}$ at $47\,\Omega$, the same as in the emitter follower regulator. An integrated regulator adds what this circuit lacks: a current limit and a thermal shutdown.

Exercise 3.6: Op-amp voltage source for $3.3\,\text{V}$

A $3.3\,\text{V}$ supply for a sensor board is built from a $2.5\,\text{V}$ reference, an op-amp and an npn booster ($\beta = 100$, $V_{BE} = 0.75\,\text{V}$). The input voltage is $12\,\text{V}$, the load current up to $200\,\text{mA}$.

a) Choose R_2 for $R_1 = 10\,\text{k}\Omega$ from E24 and from E96. Calculate the output voltages.

b) Which voltage and current must the op-amp output deliver at full load?

c) Calculate the power of the transistor. Compare with the power delivered to the load.

d) In the Falstad example, turn the R_L slider from $1\,\text{k}\Omega$ to $20\,\Omega$. What happens to the output voltage, and what would limit the output current in a real circuit?

Solution 3.6

a) $R_2 = R_1\,(3.3/2.5 - 1) = 3.2\,\text{k}\Omega$. E24: $3.3\,\text{k}\Omega$, $V_{\text{out}} = 2.5\,\text{V} \cdot 1.33 = 3.325\,\text{V}$ ($0.8\,\%$); E96: $3.24\,\text{k}\Omega$, $V_{\text{out}} = 3.310\,\text{V}$ ($0.3\,\%$).

b) $V_{\text{out}} + V_{BE} \approx 4.06\,\text{V}$; $I_B = 200\,\text{mA}/101 = 1.98\,\text{mA}$ (the divider takes only $0.25\,\text{mA}$ more from the emitter).

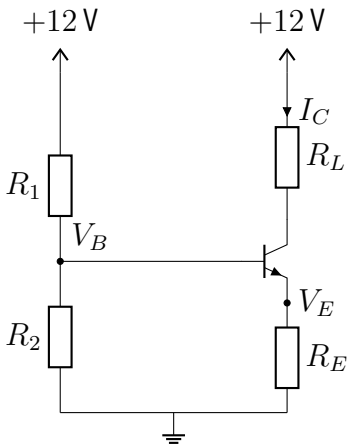
c) $P_Q = (12\,\text{V} - 3.31\,\text{V}) \cdot 0.2\,\text{A} = 1.74\,\text{W}$, the load gets only $0.66\,\text{W}$: efficiency $28\,\%$. A heat sink is needed; a switching regulator would be the better choice for this current.

d) ▶ Falstad: `examples/sources/opamp_voltage_source.txt`: `Vout_100mA` stays at $5.000\,\text{V}$ over the whole slider range ($250\,\text{mA}$ at $20\,\Omega$). The simulation has no current limit; in reality the transistor power ($1.75\,\text{W}$), its safe operating area and the op-amp output current (I_L/β) limit the current.

3.6 Transistor Current Sources

A bipolar transistor in its active region is a current source by nature: its collector current depends on the base–emitter voltage, but hardly on the collector voltage. An emitter resistor turns this into a current source with a well-defined current. Inside integrated circuits the current mirror is the most important building block; there is hardly an op-amp or a comparator without several of them.

BJT current source



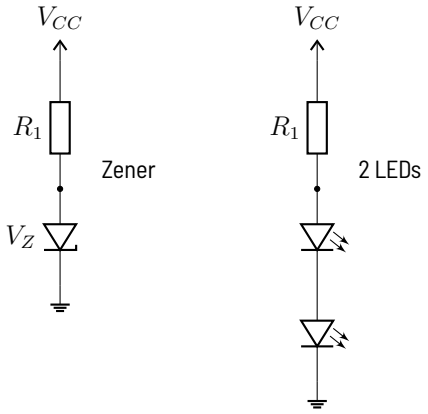
BJT current source

$$I_C \approx I_E = \frac{V_B - V_{BE}}{R_E} \approx \frac{V_B - 0.65\,\text{V}}{R_E}$$

- the emitter follows the base: $V_E = V_B - V_{BE}$ is fixed, so the emitter current is fixed
- the load sits in the collector and may vary within the compliance range

Compliance range and base voltage

- the transistor must stay active: $V_{CE} \geq$ about $0.3\,\text{V}$, i.e. $V_C \geq V_E + 0.3\,\text{V}$
- compliance: $R_{L,\text{max}} = \dfrac{V_{CC} - V_E - 0.3\,\text{V}}{I_C}$; a larger load saturates the transistor and the current drops
- stiff base voltage: divider current $\geq 10\,I_B$; base current error $I_B \cdot (R_1 \parallel R_2)$
- temperature: V_{BE} drops by $2\,\text{mV/K}$; a large V_E (1 V and more) makes the current stable



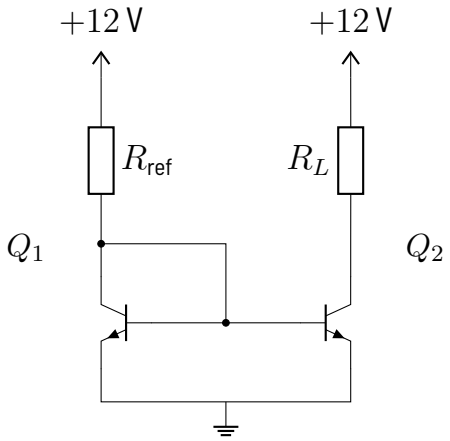
The base voltage can come from a divider, from a Zener diode or from forward-biased diodes or LEDs. The diode solutions make V_B independent of the supply voltage. Two silicon diodes ($V_B \approx 1.3\,\text{V}$) have the additional advantage that their temperature coefficient cancels that of V_{BE}: the voltage across R_E is one diode drop and stays constant. A red LED gives $V_B \approx 1.8\,\text{V}$ and $V_E \approx 1.15\,\text{V}$, again with partial temperature compensation.

Example: $1\,\text{mA}$ current source

- $V_{CC} = 12\,\text{V}$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$: $V_B = 2.16\,\text{V}$ (unloaded)
- $R_E = 1.5\,\text{k}\Omega$: $I_E = (2.16\,\text{V} - 0.65\,\text{V})/1.5\,\text{k}\Omega = 1.0\,\text{mA}$
- loads $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$: the same current; $15\,\text{k}\Omega$ is outside the compliance range

▶ Falstad: `examples/sources/bjt_current_source.txt` The simulation gives $1.023\,\text{mA}$ for both the $1\,\text{k}\Omega$ and the $4.7\,\text{k}\Omega$ load. The difference to the hand calculation comes from V_{BE}: the Falstad transistor model has $V_{BE} = 0.60\,\text{V}$ at $1\,\text{mA}$, real small signal transistors (BC547) are closer to $0.65\,\text{V}$. With $15\,\text{k}\Omega$ the collector would have to go down to $12\,\text{V} - 15\,\text{V} < 0$; the transistor saturates and the current falls to $0.71\,\text{mA}$. The Falstad model has no Early effect, so in the active region the current does not depend on the load at all. A real transistor has an output resistance of some $100\,\text{k}\Omega$, increased by the emitter resistor to several $\text{M}\Omega$.

Current mirror



- Q_1 is connected as a diode: it sets V_{BE} for the reference current
- Q_2 has the same V_{BE} and therefore the same collector current: $I_{\text{out}} = I_{\text{ref}}\,\beta/(\beta + 2)$
- $I_{\text{ref}} = (V_{CC} - V_{BE})/R_{\text{ref}}$; with $11\,\text{k}\Omega$: $1.04\,\text{mA} \rightarrow 1.02\,\text{mA}$

▶ Falstad: `examples/sources/current_mirror.txt` The current mirror works only if both transistors are identical and at the same temperature, so it is used in integrated circuits or with matched transistor pairs (BCM847). With discrete transistors emitter resistors of a few $100\,\Omega$ improve the matching. The output current changes with the collector voltage because of the Early effect; the Wilson mirror and cascode mirrors reduce this. Mirrors with several output transistors distribute one reference current to many stages; by scaling the emitter areas, the output current becomes a multiple of the reference current.

Exercise 3.7: BJT current source for a resistive sensor

A resistive sensor ($R_S = 1.0\,\text{k}\Omega \ldots 1.4\,\text{k}\Omega$) shall be excited with $1\,\text{mA}$ by a BJT current source at $V_{CC} = 12\,\text{V}$ ($V_{BE} = 0.65\,\text{V}$, $\beta = 100$).

a) The base voltage comes from $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$. Calculate V_B with and without the base current, and choose R_E from E12.

b) Calculate the voltage across the sensor at $1.0\,\text{k}\Omega$ and $1.4\,\text{k}\Omega$, and the compliance range.

c) The base divider is replaced by two red LEDs in series ($V_F = 1.8\,\text{V}$ each) fed by $4.7\,\text{k}\Omega$. Recalculate R_E.

d) Read the collector currents for $1\,\text{k}\Omega$, $4.7\,\text{k}\Omega$ and $15\,\text{k}\Omega$ in Falstad.

Solution 3.7

a) Unloaded: $V_B = 12\,\text{V} \cdot 2.2/12.2 = 2.16\,\text{V}$. $I_B \approx 10\,\mu\text{A}$ through $R_1 \parallel R_2 = 1.8\,\text{k}\Omega$ lowers it by 18 mV: $V_B = 2.15\,\text{V}$. $R_E = (2.15\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 1.5\,\text{k}\Omega$ (E12).

b) $V_S = 1.0\,\text{V}$ and $1.4\,\text{V}$. With $V_E = 1.5\,\text{V}$ and $V_{CE} \geq 0.3\,\text{V}$: $R_{L,\text{max}} = (12 - 1.5 - 0.3)\,\text{V}/1\,\text{mA} = 10.2\,\text{k}\Omega$; the sensor is far inside.

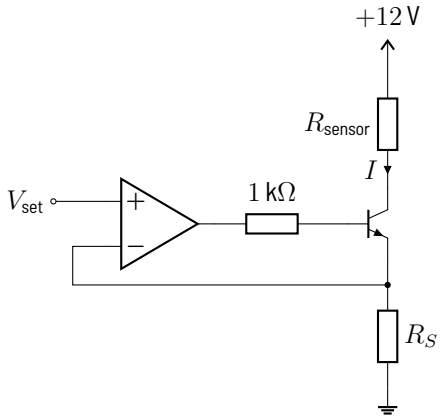
c) $V_B = 3.6\,\text{V}$, $R_E = (3.6\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 2.95\,\text{k}\Omega \Rightarrow$ E24 $3.0\,\text{k}\Omega$ ($0.98\,\text{mA}$). V_B no longer depends on V_{CC}.

d) ▶ Falstad: `examples/sources/bjt_current_source.txt`: $V_B = 2.15\,\text{V}$, $V_E = 1.55\,\text{V}$, $I_C = 1.023\,\text{mA}$ for $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$ ($V_C = 7.19\,\text{V}$), $0.71\,\text{mA}$ for $15\,\text{k}\Omega$ (saturated).

3.7 Op-Amp Current Sources and Sensor Excitation

The BJT current source depends on V_{BE} and β. An op-amp in the loop removes the V_{BE} error completely: it forces the voltage across the sense resistor to equal a set voltage. Such voltage-to-current converters are used for sensor excitation, for $4\,\text{mA} \ldots 20\,\text{mA}$ current loops and as programmable current sinks. As for the voltage source, we only need the golden rules (chapter 2): an op-amp with negative feedback makes the voltage between its inputs zero.

Op-amp and transistor

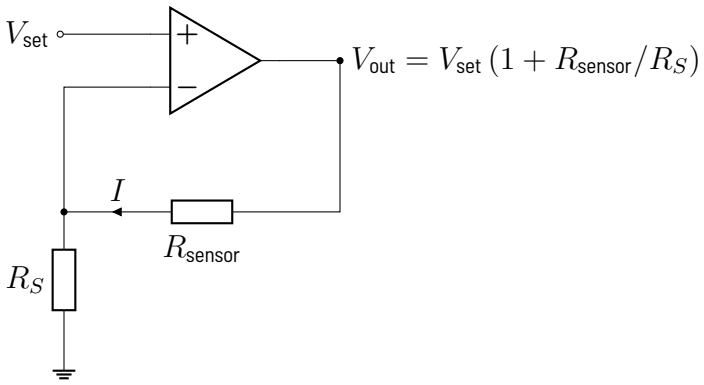


Op-amp current source

$$V_{R_S} = V_{\text{set}} \quad \Rightarrow \quad I_E = \frac{V_{\text{set}}}{R_S} \qquad I = I_C = I_E \, \frac{\beta}{\beta + 1}$$

- V_{BE} and its drift are inside the loop and drop out; the base current is lost ($1\,\%$ at $\beta = 100$, a MOSFET avoids it)

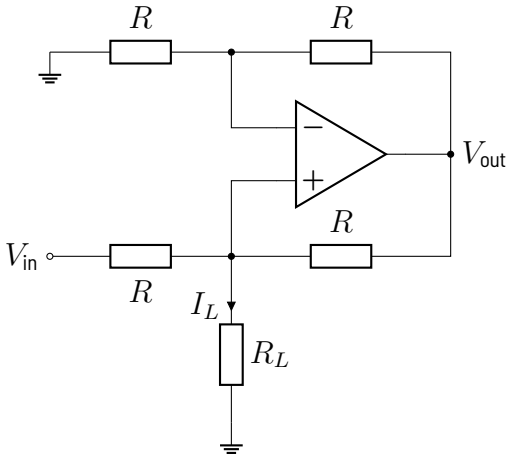
Floating load in the feedback path



- non-inverting amplifier: the current V_{set}/R_S through R_S must flow through the sensor
- exact current (no base current); but neither sensor terminal is at ground
- compliance: V_{out} must stay inside the op-amp output range

▶ Falstad: `examples/sources/opamp_current_source.txt`

Howland current pump (grounded load)



- four equal resistors: $I_L = V_{in}/R$, independent of R_L (within the compliance range)
- the resistors must be matched well: a mismatch of $0.1\,\%$ limits the output resistance

With four equal resistors the inverting input is at $V_{out}/2$. Because the op-amp makes both inputs equal, the load voltage is $V_L = V_{out}/2$. KCL at the non-inverting input gives $I_L = (V_{in} - V_L)/R + (V_{out} - V_L)/R = V_{in}/R$: the load voltage cancels. The Howland circuit is elegant but sensitive to resistor tolerances; integrated difference amplifiers with laser-trimmed resistors (e.g. INA105) are a good basis for it.

Sensor excitation: why 1 mA?

- resistive temperature sensor (Pt1000 like): $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 1\,\text{k}\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$
  - $0\,°\text{C}$: $1000\,\Omega$, $100\,°\text{C}$: $1385\,\Omega$
- with a constant current I: $V_{sensor} = I \cdot R(T)$ is linear in T: $3.85\,\text{mV/K}$ at $1\,\text{mA}$
- more current gives more signal, but **self-heating**: $\Delta T = P/G_{th}$
  - $P = I^2R = (1\,\text{mA})^2 \cdot 1.385\,\text{k}\Omega = 1.39\,\text{mW}$; with a typical $0.4\,\text{K/mW}$: $0.55\,\text{K}$ error
- the signal offset ($1\,\text{V}$ at $0\,°\text{C}$) is removed by a **bridge**

The power in the sensor heats it above the temperature it should measure. The self-heating coefficient depends strongly on the mounting: a thin-film sensor in still air heats up by some tenths of a kelvin per milliwatt, in flowing water ten times less. Since the power grows with the square of the current, halving the current reduces the error to one quarter, but also halves the signal. 1 mA for a $1\,\text{k}\Omega$ sensor (and 0.3 mA for a Pt1000 in precise applications, 1 mA for a Pt100) are common compromises.
A current source makes the sensor voltage proportional to its resistance, but the large constant part (IR_0) remains. The capstone task therefore uses a Wheatstone bridge: a second branch with fixed resistors produces the voltage that the sensor has at the lower end of the range, and the instrumentation amplifier only amplifies the difference. The bridge is fed from a stable voltage, and its resistors are chosen so that the sensor current stays below 1 mA. The chapter on bridge circuits treats this in detail.

Example: 1 mA for a sensor

- $V_{set} = 1.00\,\text{V}$ (from a reference), $R_S = 1.00\,\text{k}\Omega$ ($0.1\,\%$): $I = 1.000\,\text{mA}$
- op-amp and transistor: sensor current 0.990 mA (base current missing)
- floating sensor in the feedback path: 1.000 mA, $V_{out} = 1\,\text{V}\,(1 + 1.385) = 2.385\,\text{V}$ at $100\,°\text{C}$
- accuracy of the current = accuracy of V_{set} and R_S

▶ Falstad: `examples/sources/opamp_current_source.txt` Falstad shows exactly these numbers: VS $= 1.000$ V, the sensor in the collector carries 0.990 mA, the floating sensor 1.000 mA with VOUT2 $= 2.385$ V. The $1\,\mathrm{k}\Omega$ resistor between op-amp and base limits the output current of the op-amp and decouples the capacitive load of the base. For precise sensor measurements the ratiometric principle is often even better than a precise current: the same reference voltage that sets the current is also used as the reference of the ADC, so its errors cancel.

4 Voltage References

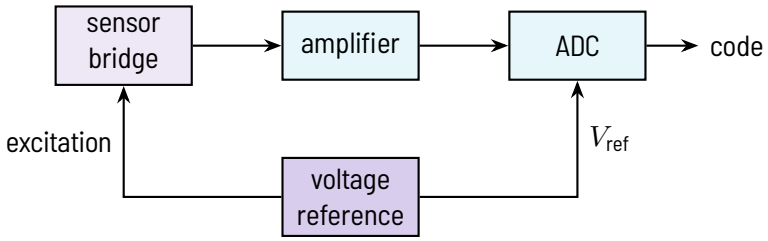
4.1 Why Voltage References?

Every measurement compares an unknown quantity with a known one. In an electronic measurement chain the known quantity is almost always a voltage: the *reference voltage*. An analog-to-digital converter does not measure volts, it measures the ratio of its input voltage to its reference voltage. A bridge circuit does not deliver millivolts per kelvin, it delivers a fraction of its excitation voltage. If the reference is $1\,\%$ too high, the result is $1\,\%$ wrong, no matter how good the rest of the circuit is.

A supply voltage from a voltage regulator is not good enough for this job. A typical regulator has an initial tolerance of $2\,\%$ … $5\,\%$, changes with load and temperature and carries ripple and noise from the digital part of the circuit. A voltage reference is a circuit (usually one small chip) that is optimised for a single purpose: a voltage that is accurate, stable over temperature and time, and quiet. It is usually not meant to deliver much current.

In this chapter we look at the specifications of references, at the two physical principles behind almost all of them (the Zener diode and the bandgap of silicon), at the shunt reference TL431 and at the practical questions: how to buffer and scale a reference, how to drive the reference ladder of a flash converter, and how a *ratiometric* design removes the reference error from the result altogether. The capstone task needs exactly this: a $4.00\,\mathrm{V}$ reference for the ADC ladder and for the sensor bridge [1, 6].

The reference sets the scale of the whole chain



- ADC: code $= 2^N \cdot V_{in}/V_{ref}$ (an ADC measures a *ratio*)
- bridge: output is a fraction of the excitation voltage
- an error of V_{ref} is a **gain error** of the complete chain

Regulator or reference?

	voltage regulator	voltage reference
purpose	supply power	define a voltage
initial accuracy	$1\,\%$ ... $5\,\%$	$0.02\,\%$ ... $1\,\%$
temperature coefficient	100 ppm/K ... 300 ppm/K	1 ppm/K ... 50 ppm/K
output current	mA ... A	µA ... 10 mA
noise	high	low, specified

- a reference is a "voltage standard" on the board, not a power supply
- if a load needs current: buffer the reference with an op-amp

4.2 Specifications

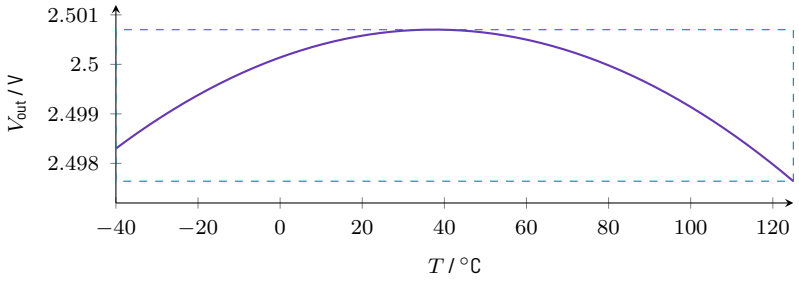
Data sheets of voltage references list a handful of parameters. They all describe how far the actual output voltage can be from its nominal value, but for different reasons: production spread, temperature, supply voltage, load current, noise and ageing. For an error budget we add them up (see the end of this chapter). The temperature coefficient is usually given in ppm/K (parts per million per kelvin): 1 ppm $= 10^{-6}$, so 100 ppm $= 0.01\,\%$.

Data sheet parameters

- **initial accuracy**: tolerance at $25\,°\mathrm{C}$, e.g. $\pm 0.1\,\%$
- **temperature coefficient** (TC): e.g. 20 ppm/K
- **line regulation**: $\Delta V_{\mathrm{out}}/\Delta V_{\mathrm{in}}$, e.g. 50 µV/V
- **load regulation**: $\Delta V_{\mathrm{out}}/\Delta I_{\mathrm{out}}$, e.g. 100 ppm/mA
- **noise**: 0.1 Hz … 10 Hz peak-to-peak, e.g. 10 µV pp
- **long-term drift**: e.g. 50 ppm per 1000 h
- **thermal hysteresis**: change after a temperature cycle

The low-frequency noise from 0.1 Hz … 10 Hz is given separately because it cannot be filtered: it looks like a slow drift of the reference. Broadband noise above that can be reduced with a capacitor at a noise-reduction pin or with an RC filter after the reference.

Temperature coefficient: the box method



Box method

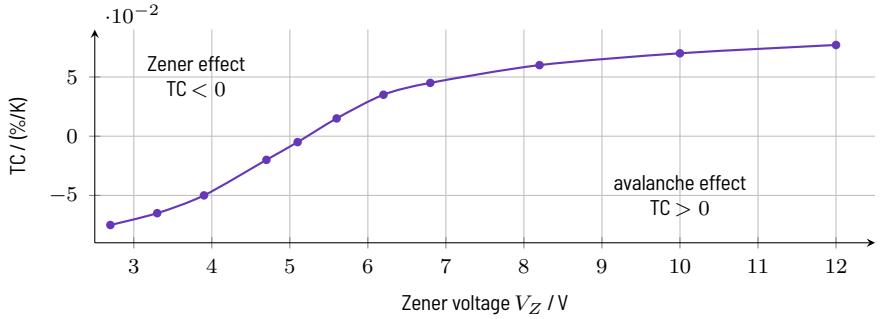
$$TC = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{V_{\mathrm{nom}}\,(T_{\mathrm{max}} - T_{\mathrm{min}})} \cdot 10^{6}\ \text{ppm}$$

The curve in the figure is typical for a bandgap reference: it is a parabola with its maximum near room temperature. The box method draws the smallest rectangle around the curve over the specified temperature range and divides its height by its width. In the example the box is 3.06 mV high and 165 K wide, so $TC = 3.06\,\mathrm{mV}/(2.5\,\mathrm{V} \cdot 165\,\mathrm{K}) \approx 7.4$ ppm/K. The TC does *not* mean that the voltage changes linearly; it only guarantees the box.

4.3 The Zener Reference

The oldest and simplest reference is a Zener diode operated in reverse breakdown. Above the breakdown voltage the current rises steeply, so the voltage stays nearly constant over a wide range of currents. Two physical effects cause the breakdown. Below about 5 V it is the *Zener effect* (quantum mechanical tunnelling through a thin, heavily doped junction); its breakdown voltage falls with temperature. Above about 6 V it is the *avalanche effect* (carriers gain enough energy to ionise further atoms); its breakdown voltage rises with temperature. Between 5 V and 6 V both effects are present and their temperature coefficients nearly cancel. This is why 5.6 V and 6.2 V Zener diodes are preferred as references [2].

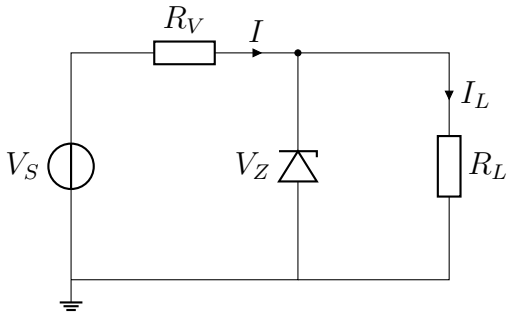
Temperature coefficient of Zener diodes



- minimum $|TC|$ near 5 V … 6 V: preferred reference Zeners 5.6 V, 6.2 V
- 6.2 V Zener ($+2$ mV/K) in series with a forward diode (-2 mV/K): compensated reference diode (1N821 family)

The curve shows typical values of small-signal Zener diodes; the spread between data sheets is large. The compensated reference diodes achieve 5 ppm/K … 100 ppm/K, but only at the specified current (typically 7.5 mA). The best references ever built (LM399, LTZ1000) use a *buried* Zener below the chip surface, which is far less noisy, together with an on-chip heater that keeps the die at a constant temperature.

Feeding a Zener through a resistor



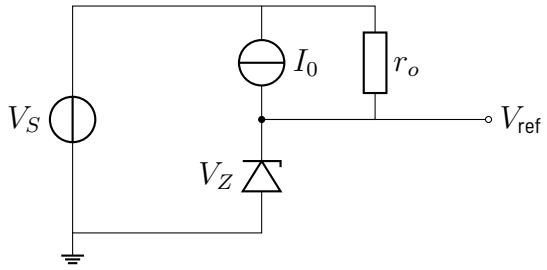
$$I_Z = I - I_L$$

- $R_V = \dfrac{V_S - V_Z}{I_Z + I_L}$, I_Z must stay above $I_{Z,\text{min}}$ for all V_S and I_L
- dynamic resistance $r_Z = \mathrm{d}V_Z/\mathrm{d}I_Z$: $5\,\Omega$ … $50\,\Omega$
- line regulation $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{R_V + r_Z}$, load regulation $\Delta V_Z \approx -r_Z\,\Delta I_L$

▶ Falstad: `examples/references/zener_resistor.txt`

The Zener and the series resistor form a voltage divider whose lower part has the small dynamic resistance r_Z. A supply change is divided down by $r_Z/(R_V + r_Z)$, typically 1:100. The load current is taken from the Zener current: if the load draws more, the Zener carries less, and its voltage drops by $r_Z\,\Delta I_L$. Falstad models the Zener with an ideal exponential characteristic $I_Z = 5\,\text{mA}\cdot e^{(V-V_Z)/V_T}$, so the nominal voltage is reached at 5 mA and the dynamic resistance is only $V_T/I_Z \approx 5\,\Omega$. Real 5.6 V Zeners have $10\,\Omega$ … $40\,\Omega$, so in reality the regulation is several times worse than in the simulation.

Feeding a Zener from a current source



- the Zener current no longer depends on V_S
- only the output resistance r_o of a real current source remains: $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{r_o}$
- e.g. transistor current source from chapter 3, $r_o = 100\,\text{k}\Omega$

▶ Falstad: `examples/references/zener_current_source.txt`

With $r_Z = 5\,\Omega$ and $r_o = 100\,\text{k}\Omega$ the line regulation improves from $1 : 240$ (with $R_V = 1.2\,\text{k}\Omega$) to $1 : 20000$. The same trick is used inside every integrated reference: the reference element is fed from a current that is derived from the reference voltage itself, so that supply changes do not reach it.

Exercise 4.1: Zener reference with series resistor

A Zener diode with $V_Z = 5.6\,\text{V}$ (at 5 mA) is fed from $V_S = 12\,\text{V}$ through $R_V = 1.2\,\text{k}\Omega$. Use the Falstad model of the Zener, $I_Z = 5\,\text{mA} \cdot e^{(V - V_Z)/V_T}$ with $V_T = 25.9\,\text{mV}$.

a) Calculate the Zener current and the dynamic resistance r_Z.

b) How much does V_{ref} change if V_S rises to $14\,\text{V}$?

c) A load $R_L = 2.2\,\text{k}\Omega$ is connected at $12\,\text{V}$. How much does V_{ref} drop?

Solution 4.1

a) $I_Z \approx (12\,\text{V} - 5.6\,\text{V})/1.2\,\text{k}\Omega = 5.33\,\text{mA}$, so $V_{\text{ref}} = V_Z + V_T \ln(5.33/5) = 5.602\,\text{V}$ and $r_Z = V_T/I_Z = 25.9\,\text{mV}/5.33\,\text{mA} = 4.9\,\Omega$.

b) At $14\,\text{V}$: $I_Z = 7.0\,\text{mA}$, $\Delta V_{\text{ref}} = V_T \ln(7.0/5.33) = 7.0\,\text{mV}$ (small-signal estimate with the mean $r_Z \approx 4.2\,\Omega$: $2\,\text{V} \cdot 4.2/1204 = 7.0\,\text{mV}$). Line regulation $3.5\,\text{mV/V}$, i.e. $0.06\,\%/\text{V}$.

c) The load takes $I_L = 5.6\,\text{V}/2.2\,\text{k}\Omega = 2.55\,\text{mA}$, the Zener keeps $I_Z = 5.33 - 2.55\,\text{mA} = 2.79\,\text{mA}$: $\Delta V_{\text{ref}} = V_T \ln(2.79/5.33) = -16.6\,\text{mV}$.

▶ Falstad: `examples/references/zener_resistor.txt`: the labels show `Vz12` $= 5.602\,\text{V}$, `Vz14` $= 5.609\,\text{V}$ ($+7.0\,\text{mV}$) and `Vz12L` $= 5.585\,\text{V}$ ($-16.6\,\text{mV}$).

Exercise 4.2: Zener with current-source feed

The Zener of the previous exercise is fed from a current source of $5\,\text{mA}$ with an output resistance $r_o = 100\,\text{k}\Omega$. How much does V_{ref} change when V_S rises from $12\,\text{V}$ to $14\,\text{V}$? Compare with the series resistor.

Solution 4.2

a) The current through r_o changes by $2\,\text{V}/100\,\text{k}\Omega = 20\,\mu\text{A}$, the Zener current from $5.064\,\text{mA}$ to $5.084\,\text{mA}$. $\Delta V_{\text{ref}} = V_T \ln(5.084/5.064) = 0.10\,\text{mV}$, 70 times less than with $R_V = 1.2\,\text{k}\Omega$ ($7.0\,\text{mV}$).

▶ Falstad: `examples/references/zener_current_source.txt`: at $12\,\text{V}$ `VzI` $= 5.600\,\text{V}$; moving the slider to $14\,\text{V}$ changes `VzR` by $7.0\,\text{mV}$ and `VzI` by only $0.1\,\text{mV}$.

4.4 The Bandgap Reference

Zener diodes need more than 6 V of supply and several milliamperes. Modern circuits run from 3.3 V or less and from microamperes. Almost all integrated references (and the references inside microcontroller ADCs) therefore use the *bandgap* principle, invented by Widlar [8] and brought to its classic form by Brokaw [9]. The idea: the base-emitter voltage of a transistor falls with temperature, and the difference of two base-emitter voltages at different current densities rises with temperature. A suitably weighted sum of the two does not depend on temperature. The sum turns out to be close to the bandgap voltage of silicon, 1.2 V, which gives the circuit its name.

Two voltages with opposite temperature coefficients

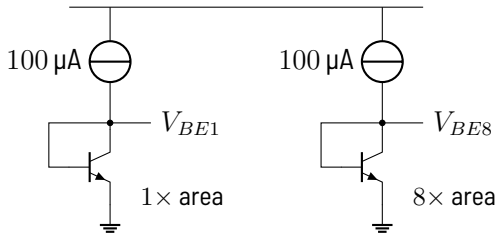
- base-emitter voltage: $V_{BE} = V_T \ln(I_C/I_S)$ with $V_T = kT/q = 25.7\,\text{mV}$ at $25\,°\text{C}$
  - I_S rises strongly with T: V_{BE} falls by about $2\,\text{mV/K}$ (CTAT: complementary to absolute temperature)
- two transistors with current densities in the ratio N:

Delta-VBE

$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln N = \dfrac{kT}{q} \ln N$ (PTAT: proportional to absolute temperature)

  - $k/q = 86.2\,\mu\text{V/K}$; $N = 8$: $\Delta V_{BE} = 53.5\,\text{mV}$ at $25\,°\text{C}$, $+0.179\,\text{mV/K}$
  - I_S cancels: ΔV_{BE} depends only on T and N

Delta-VBE in Falstad

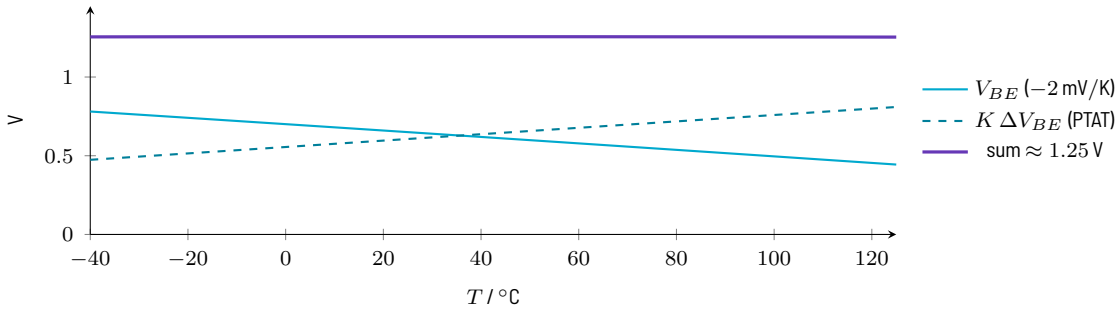


- same current, 8 transistors in parallel: current density $1/8$
- Falstad has no temperature, but ΔV_{BE} at room temperature can be measured: $V_{BE1} = 0.536\,\text{V}$, $V_{BE1} - V_{BE8} = 53.8\,\text{mV}$
- current ratio 10: $\Delta V_{BE} = V_T \ln 10 = 59.6\,\text{mV}$ (“60 mV per decade”)

▶ Falstad: `examples/references/delta_vbe.txt`

Falstad uses $V_T = 25.865\,\text{mV}$ ($27\,°\text{C}$) and a transistor model with $I_S = 10^{-13}\,\text{A}$. The absolute value of V_{BE} is therefore lower than in a real small-signal transistor (about $0.65\,\text{V}$ at $100\,\mu\text{A}$), but the difference $V_T \ln N$ is exact, because I_S cancels. In an integrated circuit the area ratio is made by placing N identical transistors in parallel, exactly as in the simulation; this is the most precise ratio a chip can offer.

The bandgap principle

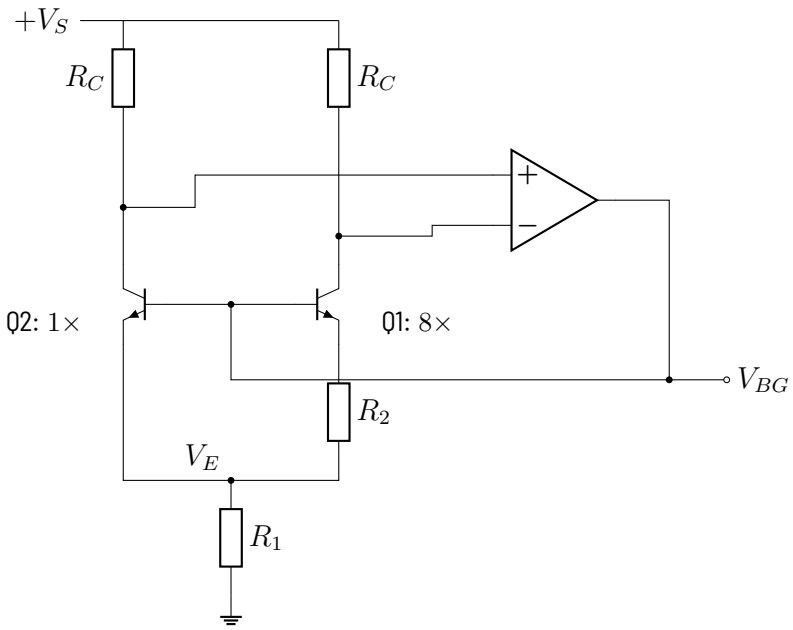


Bandgap reference

$V_{BG} = V_{BE} + K \cdot V_T \ln N \approx 1.25\,\text{V}$, K chosen so that $\mathrm{d}V_{BG}/\mathrm{d}T = 0$ at room temperature

The PTAT voltage must rise by $2\,\mathrm{mV/K}$. With $N = 8$ one ΔV_{BE} rises by $0.179\,\mathrm{mV/K}$, so it must be amplified by $K \approx 11$. At room temperature $K\,\Delta V_{BE} \approx 11 \cdot 54\,\mathrm{mV} \approx 0.6\,\mathrm{V}$, and the sum is $1.2\,\mathrm{V} \ldots 1.25\,\mathrm{V}$. Extrapolated to $0\,\mathrm{K}$, the sum is exactly the bandgap voltage of silicon, $V_{G0} = 1.205\,\mathrm{V}$. The remaining curvature (the $T \ln T$ term in V_{BE}) gives the parabola seen in the box-method figure; simple bandgaps reach $10\,\mathrm{ppm/K} \ldots 50\,\mathrm{ppm/K}$, curvature-corrected ones $1\,\mathrm{ppm/K} \ldots 5\,\mathrm{ppm/K}$.

The Brokaw cell



Q1 consists of eight transistors in parallel, Q2 of one. Both bases are driven by the output V_{BG}. The op-amp compares the voltages at the two equal collector resistors R_C and adjusts V_{BG} until both collector currents are equal. Then the current density in Q2 is eight times that in Q1, and the difference of the base-emitter voltages appears across R_2.

Brokaw cell: how it works

- the op-amp forces equal collector currents I (R_C equal, inputs at the collectors)
- current density ratio $N = 8$: $\Delta V_{BE} = V_T \ln 8$ appears across R_2
  - $I = V_T \ln 8 / R_2$ (PTAT current)
- R_1 carries both emitter currents: $V_E = 2 I R_1 = 2\,\dfrac{R_1}{R_2}\,V_T \ln 8$
- output: $V_{BG} = V_{BE2} + 2\dfrac{R_1}{R_2}\,V_T \ln 8$ ($K = 2R_1/R_2$)
- Falstad, $R_2 = 1\,\mathrm{k\Omega}$, $R_1 = 5.6\,\mathrm{k\Omega}$: $I = 53.8\,\mu\mathrm{A}$, $V_E = 0.602\,\mathrm{V}$, $V_{BE2} = 0.520\,\mathrm{V}$, $V_{BG} = 1.122\,\mathrm{V}$

▶ Falstad: `examples/references/brokaw_cell.txt`

With $K = 2 \cdot 5.6 = 11.2$ the PTAT part rises by $11.2 \cdot 0.179\,\mathrm{mV/K} = 2.0\,\mathrm{mV/K}$ and compensates V_{BE}. With a real transistor ($V_{BE} \approx 0.65\,\mathrm{V}$) the output would be $1.25\,\mathrm{V}$. Falstad's transistor model has a lower V_{BE} and no temperature dependence, so it shows $1.122\,\mathrm{V}$; move the slider of R_1 to see how the PTAT part scales ($R_1 = 6.8\,\mathrm{k\Omega}$ gives $1.25\,\mathrm{V}$). In a real chip R_1 is trimmed at the wafer test until the output has zero TC. A resistive divider between V_{BG} and the op-amp output (not shown) gives larger outputs such as $2.5\,\mathrm{V}$ or $4.096\,\mathrm{V}$.

Exercise 4.3: Bandgap design

A Brokaw cell has an area ratio $N = 8$, $R_2 = 1\,\text{k}\Omega$ and $R_1 = 5.6\,\text{k}\Omega$. Use $V_T = 25.865\,\text{mV}$ (as Falstad).

a) Calculate ΔV_{BE}, the collector current I and V_E.

b) The Falstad transistor model gives $V_{BE2} = 0.520\,\text{V}$ at this current. Calculate V_{BG}.

c) Calculate the temperature coefficient of the PTAT part and compare it with $\text{d}V_{BE}/\text{d}T \approx -2\,\text{mV/K}$.

Solution 4.3

a) $\Delta V_{BE} = 25.865\,\text{mV} \cdot \ln 8 = 53.8\,\text{mV}$, $I = 53.8\,\text{mV}/1\,\text{k}\Omega = 53.8\,\mu\text{A}$, $V_E = 2 \cdot 53.8\,\mu\text{A} \cdot 5.6\,\text{k}\Omega = 0.602\,\text{V}$.

b) $V_{BG} = 0.520\,\text{V} + 0.602\,\text{V} = 1.122\,\text{V}$.

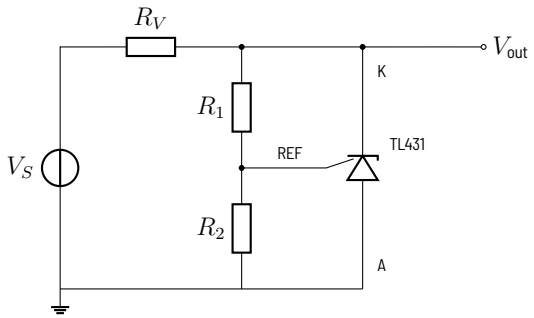
c) $\text{d}V_E/\text{d}T = 2\frac{R_1}{R_2}\frac{k}{q}\ln 8 = 11.2 \cdot 86.2\,\mu\text{V/K} \cdot 2.08 = +2.0\,\text{mV/K}$: it cancels the TC of V_{BE}.

▶ Falstad: `examples/references/brokaw_cell.txt`: VBG $= 1.122$ V, VE $= 0.602$ V; C1 and C2 are equal (4.47 V), the voltage across R_2 is 53.8 mV.

4.5 Shunt and Series References

Integrated references come in two kinds. A *shunt* reference behaves like a precise Zener diode: it is connected in parallel to the load and draws exactly the current needed to keep its voltage constant; it is fed through a series resistor. A *series* reference behaves like a precise voltage regulator: it has an input, a ground and an output pin, and it only draws a small quiescent current itself. The most widely used shunt reference is the TL431, an adjustable bandgap reference with an internal 2.495 V reference, an error amplifier and a shunt transistor.

The adjustable shunt reference TL431



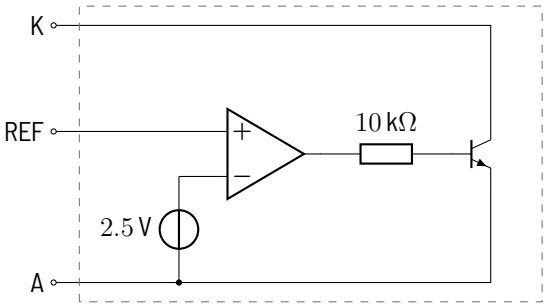
- the shunt element conducts until $V_{REF} = 2.5\,\text{V}$

TL431

$$V_{out} = V_{REF}\left(1 + \frac{R_1}{R_2}\right) + I_{REF}\,R_1, \quad V_{REF} = 2.495\,\text{V},\ I_{REF} \approx 2\,\mu\text{A}$$

- cathode current between $1\,\text{mA}$ and $100\,\text{mA}$

Inside the TL431 (behavioural model)



- error amplifier compares REF with the internal $2.5\,\text{V}$ bandgap
- V_{REF} too high $\Rightarrow$ transistor conducts more $\Rightarrow$ V_{out} falls
- $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$: $V_{out} = 2.5 \cdot 1.6 = 4.00\,\text{V}$

▶ Falstad: `examples/references/shunt_ref.txt`

The Falstad model sets the TL431 to $4.00\,\text{V}$, fed from $12\,\text{V}$ through $1\,\text{k}\Omega$. The cathode current is $(12\,\text{V} - 4\,\text{V})/1\,\text{k}\Omega - 4\,\text{V}/32\,\text{k}\Omega = 7.875\,\text{mA}$. Move the supply slider down to $6\,\text{V}$: the output stays at $4.00\,\text{V}$, only the cathode current falls. The internal reference is modelled as an ideal $2.5\,\text{V}$ source; the real chip uses a bandgap cell.

Shunt or series reference?

- **shunt** (TL431, LM4040, LM385)
  - two terminals, works at any supply above V_{out}, can be used as a negative reference
  - the series resistor wastes current: R_V must deliver $I_{min} + I_{L,max}$ at the lowest supply
- **series** (REF50xx, ADR4xx, MAX6xxx)
  - three terminals, quiescent current only $0.1\,\text{mA}$ … $1\,\text{mA}$
  - better for varying loads and battery operation, needs $V_{in} > V_{out} + V_{dropout}$
- both: decouple according to the data sheet (some are unstable with the wrong capacitor)

Exercise 4.4: TL431 set to 4.00 V

A TL431 ($V_{\text{REF}} = 2.495$ V, $I_{\text{REF}} = 2$ µA, $I_{K,\text{min}} = 1$ mA) shall provide 4.00 V from a supply of 6 V … 15 V.

a) Choose R_1 and R_2 from the E24 series.

b) How large is the error caused by I_{REF} and by $V_{\text{REF}} = 2.495$ V instead of 2.5 V?

c) Choose R_V such that a load of 1 mA can be supplied at 6 V.

Solution 4.4

a) $1 + R_1/R_2 = 1.6 \Rightarrow R_1/R_2 = 0.6$: $R_1 = 12$ kΩ, $R_2 = 20$ kΩ (both E24).

b) $I_{\text{REF}} R_1 = 2$ µA·12 kΩ $= 24$ mV ($+0.6$ %); 2.495 V·$1.6 = 3.992$ V (-0.2 %). Together $V_{\text{out}} = 4.016$ V: for an accurate 4.00 V choose smaller divider resistors or trim.

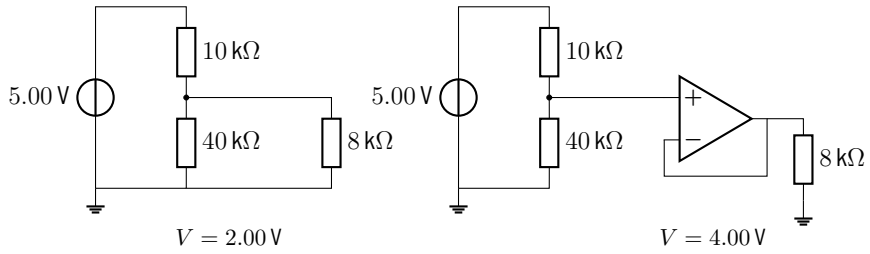
c) At 6 V R_V must carry $I_{K,\text{min}} + I_L + I_{\text{div}} = 1 + 1 + 0.125 = 2.125$ mA: $R_V \le (6\,\text{V} - 4\,\text{V})/2.125$ mA $= 941\ \Omega$, choose $910\ \Omega$. At 15 V the TL431 then carries $(15 - 4)/0.91 - 0.125 - 1 = 11$ mA (44 mW): fine.

▶ Falstad: `examples/references/shunt_ref.txt` (ideal model with 2.5 V, $R_V = 1$ kΩ, no load): `VK` $= 4.00$ V at 12 V and at 6 V, `REF` $= 2.50$ V.

4.6 Buffering and Scaling a Reference

References are available in a few standard voltages: 1.25, 2.048, 2.5, 3.0, 4.096 and 5.0 V. If the circuit needs a different value, or more current than the reference can deliver, an op-amp buffers and scales it. The capstone needs exactly 4.00 V for a reference ladder of eight 1 kΩ resistors, so that one LSB of the 3-bit flash converter is 0.5 V. The ladder draws $4\,\text{V}/8\,\text{k}\Omega = 0.5$ mA: far too much for a voltage divider with high-value resistors, and a noticeable load for many references. The op-amp circuits are treated in detail in chapter 6; here we only use the results.

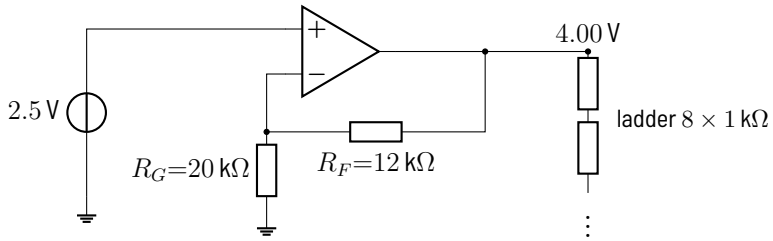
A divider is no reference



- divider $10\,\text{k}\Omega/40\,\text{k}\Omega$: 4.00 V open, but only 2.00 V with the $8\,\text{k}\Omega$ ladder
- source resistance $R_1 \parallel R_2 = 8\,\text{k}\Omega$ = load resistance
- follower: input current ≈ 0, output resistance ≈ 0: 4.00 V

▶ Falstad: `examples/references/ref_divider_loaded.txt`

Scaling with a non-inverting amplifier



- $G = 1 + R_F/R_G = 1 + 12/20 = 1.6$: $\quad 2.5\,\text{V} \cdot 1.6 = 4.00\,\text{V}$
- the op-amp delivers ladder current 0.5 mA plus 0.125 mA for $R_F + R_G$
- gain error from resistor tolerance: $\frac{\Delta G}{G} = \frac{G-1}{G}\left(\frac{\Delta R_F}{R_F} - \frac{\Delta R_G}{R_G}\right)$

▶ Falstad: `examples/references/ref_buffer_ladder.txt`

The resistor tolerances enter the result only with the factor $(G-1)/G = 0.375$: two $0.1\,\%$ resistors give at most $0.375 \cdot 0.2\,\% = 0.075\,\%$. The offset voltage of the op-amp is multiplied by G: 1 mV offset gives 1.6 mV at the output ($0.04\,\%$). Alternatively a 4.096 V reference with a follower can be used; the LSB is then 0.512 V instead of 0.5 V.

Design card

Design card: Reference for an ADC ladder

- full scale V_{FS} and LSB: $\text{LSB} = V_{FS}/2^N$ $\quad$ (4.00 V, 3 bit: 0.5 V)
- ladder current $I = V_{FS}/(2^N R)$ $\quad$ ($8 \times 1\,\text{k}\Omega$: 0.5 mA)
- reference voltage V_R from a standard value, gain $G = V_{FS}/V_R$ $\quad$ (2.5 V, $G = 1.6$)
- non-inverting amplifier: $R_F/R_G = G - 1$, $R_F + R_G \gg$ ladder ($12\,\text{k}\Omega + 20\,\text{k}\Omega$)
- check: op-amp output current, $(G-1)/G \cdot$ resistor tolerance, $G \cdot V_{\text{offset}}$
- decouple the ladder top with 100 nF … 1000 nF close to the comparators

Exercise 4.5: Reference for the 3-bit flash converter

The flash converter of the capstone needs 4.00 V at the top of a ladder of $8 \times 1\,\text{k}\Omega$. A 2.5 V reference is available.

a) Calculate the ladder current and the tap voltages.

b) Design a non-inverting amplifier with E24 resistors.

c) Why can the $4.00\,\text{V}$ not be taken from a divider $10\,\text{k}\Omega/40\,\text{k}\Omega$ at a $5\,\text{V}$ reference?

d) How large is the worst-case error of V_{ref} with $1\,\%$ resistors?

Solution 4.5

a) $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; taps every $0.5\,\text{V}$: $T_1 = 0.5\,\text{V}$, …, $T_4 = 2.0\,\text{V}$, …, $T_7 = 3.5\,\text{V}$.

b) $G = 4/2.5 = 1.6$, $R_F/R_G = 0.6$: $R_F = 12\,\text{k}\Omega$, $R_G = 20\,\text{k}\Omega$.

c) The divider has a source resistance of $10\,\text{k}\Omega \parallel 40\,\text{k}\Omega = 8\,\text{k}\Omega$, equal to the ladder. The loaded voltage is $5 \cdot \frac{40 \parallel 8}{10 + 40 \parallel 8} = 5 \cdot \frac{6.67}{16.67} = 2.00\,\text{V}$.

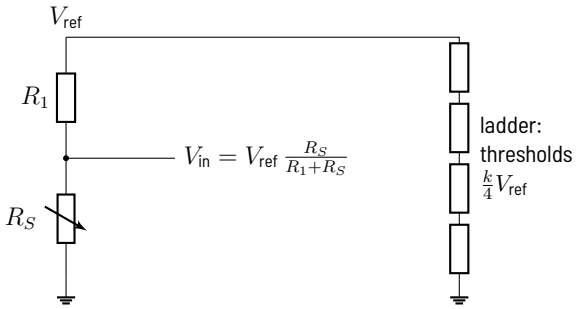
d) $\Delta G/G = 0.375 \cdot 2 \cdot 1\,\% = 0.75\,\%$, i.e. $\pm 30\,\text{mV}$; for the 3-bit converter (LSB $0.5\,\text{V}$) irrelevant, for a 12-bit converter 30 LSB.

▶ Falstad: `examples/references/ref_buffer_ladder.txt`: `VREF4` $= 4.00\,\text{V}$, `T4` $= 2.00\,\text{V}$, `T1` $= 0.50\,\text{V}$, ladder current $0.5\,\text{mA}$; ▶ Falstad: `examples/references/ref_divider_loaded.txt`: `Vopen` $= 4.00\,\text{V}$, `Vload` $= 2.00\,\text{V}$, `Vbuf` $= 4.00\,\text{V}$.

4.7 Ratiometric Measurement

Resistive sensors do not produce a voltage themselves; they need an excitation, and their output is a fraction of it. If the same reference excites the sensor and sets the full scale of the ADC, the reference appears in the numerator and in the denominator of the result and cancels. Such a measurement is called *ratiometric*. It is the most effective way to get an accurate result from an inexpensive reference: the reference only needs to be quiet and stable during one conversion, not accurate. The capstone uses this principle: the $4.00\,\text{V}$ reference feeds both the sensor bridge and the ladder of the flash converter [6].

The reference cancels



Ratiometric

code $= 2^N \dfrac{V_{\text{in}}}{V_{\text{ref}}} = 2^N \dfrac{R_S}{R_1 + R_S}$ independent of V_{ref}

Ratiometric versus absolute

- Falstad: two 2-bit flash converters (3 comparators), sensor divider $1\,\text{k}\Omega/2.7\,\text{k}\Omega$: $V_{\text{in}} = 0.730\,V_{\text{exc}}$
- left: sensor and ladder from V_{ref}: code 2 for $V_{\text{ref}} = 3.6, 4.0, 4.4$ V
- right: sensor from a fixed 4.00 V, ladder from V_{ref}
  - $V_{\text{ref}} = 4.0$ V: $V_{\text{in}} = 2.92$ V < 3.0 V: code 2
  - $V_{\text{ref}} = 3.6$ V: threshold $\frac{3}{4} V_{\text{ref}} = 2.7$ V: code 3 (wrong)
- ratiometric works only if both see the *same* voltage at the *same* time
  - a common connection with voltage drops, or noise on one path only, breaks it

▶ Falstad: `examples/references/ratiometric.txt`

4.8 Error Budget

An error budget lists all contributions to the error of a quantity and adds them up. Independent random errors are usually added as the root of the sum of squares (RSS), which is the realistic expectation; the simple sum is the worst case, which a guaranteed specification must meet. We look at the 4.00 V reference of the capstone, built from a 2.5 V series reference and the non-inverting amplifier, over an ambient temperature of $0\,^\circ$C … $50\,^\circ$C.

Error budget of the 4.00 V reference

contribution	data	error
initial accuracy	$\pm 0.2\,\%$	$0.20\,\%$
temperature	25 ppm/K, ± 25 K	$0.06\,\%$
gain resistors	$2 \times 0.1\,\%$, factor 0.375	$0.08\,\%$
op-amp offset	1 mV $\times$ 1.6	$0.04\,\%$
long-term drift	50 ppm / 1000 h	$0.01\,\%$
worst case (sum)		$0.39\,\%$
RSS		$0.23\,\%$

- compare: 3 bit, LSB $= 12.5\,\%$ of full scale: no problem at all
- 12 bit, LSB $= 0.024\,\%$: 16 LSB worst case $\Rightarrow$ calibrate or measure ratiometrically

The initial accuracy dominates. It can be removed by a one-time calibration (trimming, or a correction factor in the software). What remains after calibration is temperature drift and ageing, and these are the parameters that distinguish an expensive reference from a cheap one. In a ratiometric design all reference terms drop out; only the ratio accuracy of the resistors remains.

Exercise 4.6: Ratiometric measurement and error budget

A sensor divider $R_1 = 1\,\text{k}\Omega$, $R_S = 2.7\,\text{k}\Omega$ is read by a 2-bit flash converter with a ladder of $4 \times 1\,\text{k}\Omega$.

a) The divider and the ladder are fed from the same V_{ref}. Which code results for $V_{\text{ref}} = 4.0$ V and 3.6 V?

b) The divider is fed from a fixed 4.00 V, the ladder from V_{ref}. Repeat a).

c) A 12-bit converter uses a reference with $\pm 0.2\,\%$ initial accuracy and 25 ppm/K. How many LSB can the result be off at $50\,^{\circ}$C after calibration at $25\,^{\circ}$C?

Solution 4.6

a) $V_{\text{in}}/V_{\text{ref}} = 2.7/3.7 = 0.730$; the thresholds are at 0.25, 0.5 and $0.75\,V_{\text{ref}}$. $0.5 < 0.730 < 0.75$: code 2 for every V_{ref}.

b) $V_{\text{in}} = 2.92$ V. $V_{\text{ref}} = 4.0$ V: thresholds 1, 2, 3 V: code 2. $V_{\text{ref}} = 3.6$ V: thresholds 0.9, 1.8, 2.7 V: code 3. A $10\,\%$ reference error changes the result by one code.

c) After calibration only the drift remains: 25 ppm/K $\cdot\, 25$ K $= 625$ ppm $= 0.000625 \cdot 4096 = 2.6$ LSB. Without calibration add $0.002 \cdot 4096 = 8.2$ LSB.

▶ Falstad: `examples/references/ratiometric.txt`: with the VREF slider at 4.0 V both converters show D = E = code 2 (VIN1 $= 2.92$ V); at 3.6 V the right converter switches E3 on (code 3), the left stays at code 2.

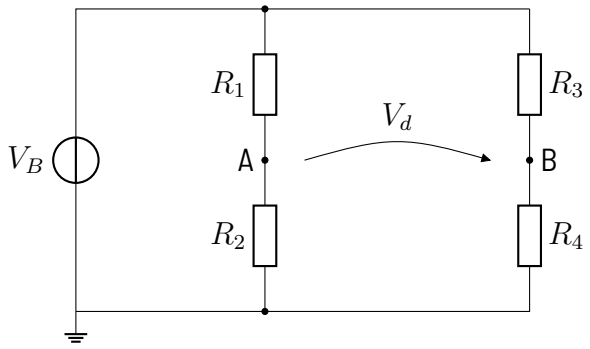
5 Bridge Circuits

5.1 The Wheatstone Bridge

Many sensors are resistors whose value changes a little with the measured quantity: a platinum resistor changes by $0.385\,\%$ per kelvin, a strain gauge by $0.2\,\%$ at the elastic limit of steel. If we put such a sensor into a simple voltage divider, the interesting change is a tiny ripple on a large constant voltage. Amplifying it would amplify the constant part as well, and any drift of the supply would look like a change of the measured quantity.

The bridge circuit solves this problem with a second voltage divider that produces the constant part on its own. The difference between the two divider outputs is zero when the sensor has its reference value and contains only the change. The circuit was described by Christie in 1833 and made popular by Wheatstone in 1843 as a method to measure an unknown resistance by adjusting a known one until the galvanometer between the two dividers shows zero (*null method*). Today the bridge is mostly used in the *deflection mode*: the resistors are fixed and the bridge output voltage is amplified and digitised. This is the first stage of the capstone task [2, 6].

Two voltage dividers



Bridge output

$$V_d = V_A - V_B = V_B \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right)$$

Here V_B is the excitation (supply) voltage of the bridge. Each leg is an unloaded voltage divider from chapter 1; the output is the difference of the two tap voltages. The output is taken between two nodes of which neither is at ground: a *differential* voltage.

Balance condition

- the bridge is **balanced** if $V_d = 0$:

Balance condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \Leftrightarrow \quad R_1 R_4 = R_2 R_3$$

- the balance does **not** depend on V_B: supply noise and drift do not appear at the output
- null method: adjust a known resistor until $V_d = 0$, then $R_2 = R_1 R_4 / R_3$
- deflection method: fixed resistors, balanced at the reference point (e.g. $0\,^\circ$C), measure V_d

At balance the output is zero, independent of the supply. Away from balance the output is proportional to V_B: the bridge converts a relative resistance change into a fraction of the excitation voltage. This is why the excitation must be a stable reference, or the measurement must be ratiometric (chapter 4).

5.2 Output Voltage and Linearity

In most applications only one resistor of the bridge is a sensor (*quarter bridge*). We write its value as $R_2 = R_0(1 + x)$, where $x = \Delta R / R_0$ is the relative change, and choose the other resistors such that the bridge is balanced at $x = 0$. The output voltage is then a function of x only. For small x it is proportional to x, for larger x it bends: the bridge is not linear. The reason is that the current in the sensor leg changes with the sensor resistance.

Quarter bridge with equal resistors

- $R_1 = R_3 = R_4 = R_0$, sensor $R_2 = R_0(1 + x)$:

$$V_d = V_B \left(\frac{1+x}{2+x} - \frac{1}{2} \right) = \frac{V_B}{4} \cdot \frac{x}{1 + x/2}$$

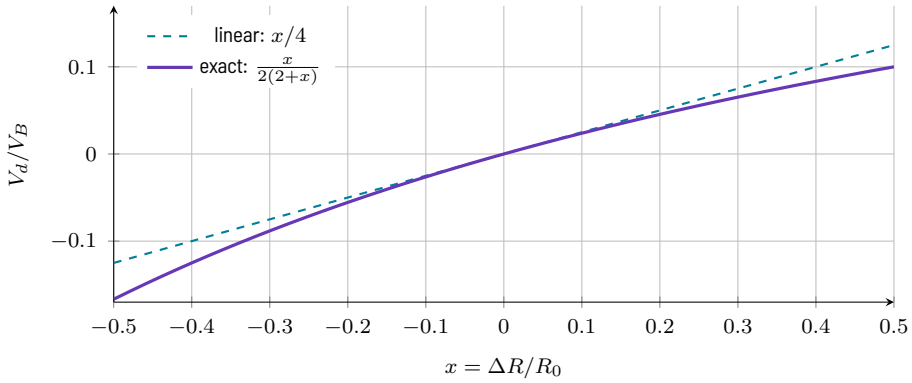
Quarter bridge, small signal

$V_d \approx \frac{V_B}{4} \frac{\Delta R}{R_0}$ relative error $\approx -\frac{x}{2}$

- example: $V_B = 5\,\text{V}$, $x = 1\,\%$: $V_d = 12.44\,\text{mV}$ (estimate $12.5\,\text{mV}$)
- $x = 50\,\%$: $V_d = 0.500\,\text{V}$ instead of $0.625\,\text{V}$

▶ Falstad: `examples/bridges/wheatstone.txt`

The bridge is not linear



- positive x: output too small, negative x: too large
- strain gauges ($x < 0.5\,\%$): negligible; Pt100 over $100\,\text{K}$ ($x = 38.5\,\%$): $16\,\%$

Unequal arms: trade sensitivity for linearity

- $R_1 = R_3 = m\,R_0$, $R_4 = R_0$, sensor $R_0(1 + x)$:

$$V_d = V_B \frac{m\,x}{(m+1)(m+1+x)} \approx V_B \frac{m}{(m+1)^2}\,x$$

- sensitivity $m/(m+1)^2$: maximum $1/4$ at $m = 1$; $m = 39$: 0.024
- relative nonlinearity $\approx -x/(m+1)$: $m = 39$ makes it 20 times smaller
- large m also limits the sensor current: $I = V_B/(R_1 + R_{\text{sensor}})$ (self-heating!)

A large ratio m turns the sensor leg into an almost ideal current source: the current hardly changes with the sensor resistance, so the voltage across the sensor is nearly proportional to its resistance. The price is a smaller output. Since the amplifier needs a gain anyway, a few tens of millivolts are easy to handle, while a nonlinearity would have to be corrected in software. Other ways to linearise a bridge are a constant current excitation or an op-amp that keeps the bridge at balance [6].

Exercise 5.1: Wheatstone bridge

A bridge with $R_1 = R_3 = R_4 = 1\,\text{k}\Omega$ is supplied with $V_B = 5\,\text{V}$. The resistor $R_2 = R_x$ is the sensor.

a) Calculate V_d exactly and with the small-signal formula for $R_x = 1010\,\Omega$.

b) Repeat for $R_x = 1500\,\Omega$. How large is the error of the formula?

c) Check both values in Falstad.

Solution 5.1

a) $V_d = 5\,(1010/2010 - 0.5) = 12.44\,\text{mV}$; formula: $\frac{5}{4}\cdot 0.01 = 12.5\,\text{mV}$ ($0.5\,\%$ too high, $\approx x/2$).

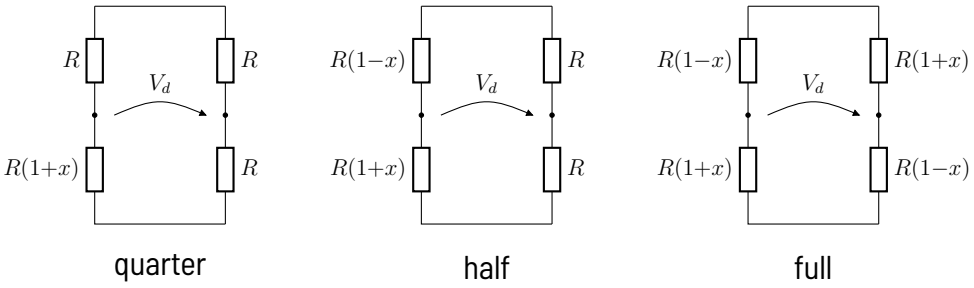
b) $V_d = 5\,(1500/2500 - 0.5) = 0.500\,\text{V}$; formula: $\frac{5}{4}\cdot 0.5 = 0.625\,\text{V}$: $25\,\%$ too high.

c) ▶ Falstad: `examples/bridges/wheatstone.txt`: the voltmeter shows $12.44\,\text{mV}$; with the slider at $1500\,\Omega$ it shows $0.500\,\text{V}$.

5.3 Half and Full Bridges: Strain Gauges

A strain gauge is a thin metal foil meander on a plastic carrier, glued onto a mechanical part. When the part is stretched, the conductor becomes longer and thinner and its resistance rises. The relative change is proportional to the strain $\varepsilon = \Delta l/l$; the factor is the *gauge factor* k, about 2 for metal foil gauges. Typical values are $120\,\Omega$, $350\,\Omega$ and $1\,\text{k}\Omega$; a strain of $1000\,\mu\text{m/m}$ ($0.1\,\%$, close to the elastic limit of steel) changes the resistance by only $0.2\,\%$. On a bending beam, gauges on the top are stretched and gauges on the bottom are compressed by the same amount. Placing them in adjacent arms of the bridge doubles or quadruples the signal and makes the bridge exactly linear. Load cells of kitchen and industrial scales are full bridges.

Quarter, half and full bridge



- quarter: $V_d \approx \frac{V_B}{4} x$ half: $V_d = \frac{V_B}{2} x$ full: $V_d = V_B\, x$ $(x = k\,\varepsilon)$
- half and full bridge are exactly linear: the leg current stays constant
- temperature changes all gauges equally: compensated

▶ Falstad: `examples/bridges/bridge_types.txt`

In the half bridge the left leg contains $R(1-x)$ and $R(1+x)$; the sum is $2R$, so the leg current does not change, and the tap voltage is $V_B(1+x)/2$, exactly linear. In the full bridge the right leg moves in the opposite direction and doubles the output once more. A temperature change alters all four gauges by the same factor and leaves the ratios, and thus the output, unchanged. In a quarter bridge the same effect is obtained with an unstrained *dummy gauge* on the same material as R_4.

Exercise 5.2: Strain gauge bridge

Strain gauges with $R = 350\,\Omega$ and $k = 2$ are glued to a beam. The strain is $\varepsilon = 1000\,\mu\text{m/m}$, the bridge supply $V_B = 5\,\text{V}$.

a) Calculate ΔR.

b) Calculate the output of a quarter, a half and a full bridge.

c) Which gain is needed to get 2 V at this strain from the full bridge?

Solution 5.2

a) $x = k\varepsilon = 2 \cdot 10^{-3}$, $\Delta R = 0.002 \cdot 350\,\Omega = 0.7\,\Omega$.

b) Quarter: $V_d = 5\,(350.7/700.7 - 0.5) = 2.4975\,\text{mV}$ ($\approx \frac{5}{4} \cdot 0.002 = 2.5\,\text{mV}$); half: $\frac{5}{2} \cdot 0.002 = 5.00\,\text{mV}$; full: $5 \cdot 0.002 = 10.0\,\text{mV}$.

c) $G = 2\,\text{V}/10\,\text{mV} = 200$, with an instrumentation amplifier (chapter 6).

▶ Falstad: `examples/bridges/bridge_types.txt`:
the three voltmeters show $2.50\,\text{mV}$, $5.00\,\text{mV}$ and $10.0\,\text{mV}$.

5.4 Resistive Temperature Sensors

The resistance of metals rises with temperature. Platinum is used for precision sensors because it is chemically stable and its characteristic is reproducible and standardised (IEC 60751). A Pt100 has $100\,\Omega$ at $0\,°\text{C}$, a Pt1000 $1000\,\Omega$. Between $0\,°\text{C}$ and $100\,°\text{C}$ the mean temperature coefficient is $\alpha = 3.85 \times 10^{-3}/\text{K}$, so a Pt100 changes by $0.385\,\Omega/\text{K}$. Over wider ranges the Callendar–Van Dusen equation describes the slight curvature. Thermistors (NTC: negative temperature coefficient) are semiconductor sensors: cheap, ten times more sensitive, but strongly nonlinear and less accurate.

Platinum sensors Pt100 and Pt1000

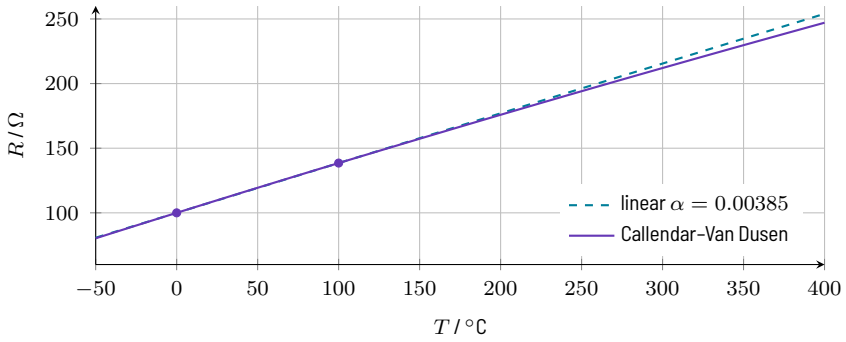
- linear model ($0\,°\text{C} \ldots 100\,°\text{C}$):

Pt100 / Pt1000

$$R(T) = R_0\,(1 + \alpha T), \quad R_0 = 100\,\Omega \,/\, 1000\,\Omega, \quad \alpha = 3.85 \times 10^{-3}/\text{K}$$

- Pt100: $100\,\Omega$ at $0\,°\text{C}$, $138.5\,\Omega$ at $100\,°\text{C}$, $0.385\,\Omega/\text{K}$
- Callendar–Van Dusen ($T \geq 0\,°\text{C}$): $R(T) = R_0(1 + AT + BT^2)$
  - $A = 3.9083 \times 10^{-3}/\text{K}$, $B = -5.775 \times 10^{-7}/\text{K}^2$
- tolerance class B: $\pm(0.3\,\text{K} + 0.005\,|T|)$, class A: $\pm(0.15\,\text{K} + 0.002\,|T|)$

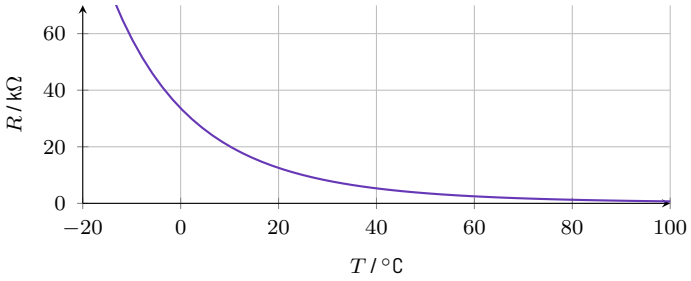
Pt100 characteristic



- $0\,°\text{C}$ … $100\,°\text{C}$: linear model and standard agree at both ends, $0.4\,\text{K}$ apart at $50\,°\text{C}$
- $400\,°\text{C}$: $254\,\Omega$ (linear) vs. $247\,\Omega$: $18\,\text{K}$ error

For $T < 0\,°\text{C}$ the Callendar–Van Dusen equation has an additional term $C\,(T - 100\,°\text{C})\,T^3$ with $C = -4.183 \times 10^{-12}/\text{K}^4$. At $50\,°\text{C}$ the standard gives $119.40\,\Omega$, the linear model $119.25\,\Omega$; the difference of $0.15\,\Omega$ corresponds to $0.4\,\text{K}$. For the capstone with its resolution of $12.5\,\text{K}$ per LSB the linear model is more than good enough.

NTC thermistors



- $R(T) = R_{25}\,\exp\left(B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right)$, e.g. $R_{25} = 10\,\text{k}\Omega$, $B = 3950\,\text{K}$ (T in kelvin)
- sensitivity at $25\,°\text{C}$: $-B/T^2 = -4.4\,\%/\text{K}$ (Pt: $+0.39\,\%/\text{K}$)
- strongly nonlinear: linearise with a series resistor, or a table in software

Self-heating: why the sensor current is limited

- the measuring current heats the sensor: $P = I^2 R$

- self-heating coefficient E: e.g. 0.1 K/mW (thin film, in water) … 1 K/mW (in still air)

  Self-heating error
  $\Delta T = E \cdot I^2 R$

- Pt100, 1 mA: $P = 0.1$ mW, $\Delta T = 0.01$ K … 0.1 K
- Pt100, 10 mA: $P = 10$ mW, $\Delta T = 1$ K … 10 K
- rule: Pt100 ≤ 1 mA, Pt1000 ≤ 0.3 mA (capstone: $I \leq 1$ mA)

The sensor measures its own temperature, not that of the medium. As long as the self-heating is constant it could be calibrated out, but it depends on how well the sensor is coupled to the medium: a sensor in flowing water is cooled much better than the same sensor in still air. A smaller current reduces the error with the square, but also the signal linearly. For a Pt1000 the same power is reached at a $\sqrt{10}$ times smaller current.

Exercise 5.3: Pt100 values

a) Calculate the resistance of a Pt100 at $25\,^\circ$C and $37\,^\circ$C with the linear model.

b) Calculate $R(100\,^\circ\text{C})$ and $R(200\,^\circ\text{C})$ with the Callendar–Van Dusen equation. Which temperature does the linear model assign to these resistances?

c) A Pt100 is read with 5 mA in still air ($E = 0.5$ K/mW). How large is the self-heating error at $0\,^\circ$C?

Solution 5.3

a) $R(25) = 100\,(1 + 0.00385 \cdot 25) = 109.63\,\Omega$, $R(37) = 114.25\,\Omega$.

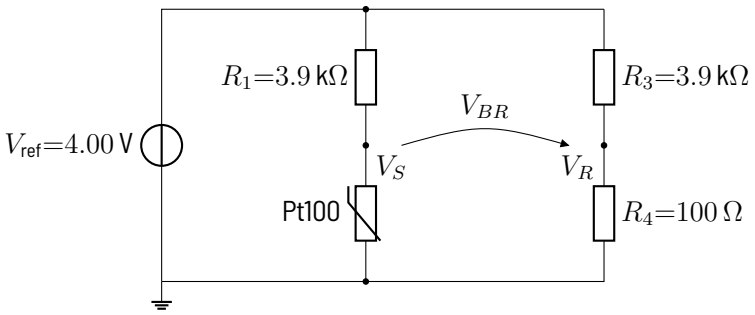
b) $R(100) = 100\,(1 + 0.39083 - 0.005775) = 138.51\,\Omega$, linear model: $T = 0.3851/0.00385 = 100.0\,^\circ$C. $R(200) = 100\,(1 + 0.78166 - 0.0231) = 175.86\,\Omega \Rightarrow$ linear: $T = 197.0\,^\circ$C (3 K too low).

c) $P = (5\text{ mA})^2 \cdot 100\,\Omega = 2.5$ mW, $\Delta T = 1.25$ K: four times the class B tolerance at $0\,^\circ$C (0.3 K).

5.5 Designing the Sensor Bridge

We now design the bridge of the capstone task. The requirements: a Pt100 for $0\,^\circ$C … $100\,^\circ$C, a sensor current of at most 1 mA, zero output at $0\,^\circ$C, and an amplifier that produces exactly 4.00 V at $100\,^\circ$C. The bridge is excited from the 4.00 V reference that also feeds the ladder of the flash converter, so that the measurement is ratiometric (chapter 4). The same design is used in the capstone chapter.

The capstone bridge



- $R_4 = R_0 = 100\,\Omega$: balanced at $0\,°\text{C}$
- $I = V_{\text{ref}}/(R_1 + R) \le 1\,\text{mA} \Rightarrow R_1 \ge 3.9\,\text{k}\Omega$; $R_3 = R_1$ ($m = 39$)
- $V_{BR}(T) = V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right)$

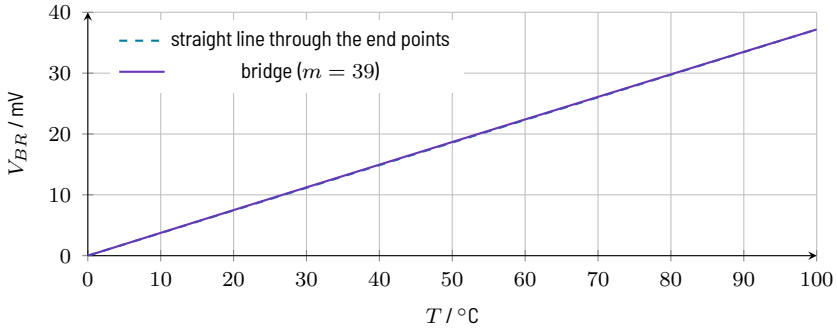
Numbers of the capstone bridge

	$0\,°\text{C}$	$50\,°\text{C}$	$100\,°\text{C}$
sensor R	$100\,\Omega$	$119.25\,\Omega$	$138.5\,\Omega$
sensor current	$1.000\,\text{mA}$	$0.995\,\text{mA}$	$0.990\,\text{mA}$
V_S	$100.0\,\text{mV}$	$118.7\,\text{mV}$	$137.2\,\text{mV}$
V_R	$100.0\,\text{mV}$	$100.0\,\text{mV}$	$100.0\,\text{mV}$
V_{BR}	$0\,\text{mV}$	$18.68\,\text{mV}$	$37.18\,\text{mV}$

- sensitivity $\approx 0.37\,\text{mV/K}$; nonlinearity at $50\,°\text{C}$: $+0.09\,\text{mV}$ ($0.25\,\%$ of full scale)
- required gain: $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- self-heating: $P = 0.1\,\text{mW}$

▶ Falstad: `examples/bridges/pt100_bridge.txt`

Bridge output over temperature



- with $m = 39$ the curve is almost straight: maximum deviation $0.09\,\text{mV} \approx 0.24\,\text{K}$
- with $m = 1$ (all arms $100\,\Omega$) the deviation would be $4.4\,\text{K}$

Design card

Design card: Sensor bridge

- excitation V_B: the ADC reference (ratiometric), here $4.00\,\text{V}$
- reference arm: $R_4 = R_{\text{sensor}}(T_{\text{min}}) \Rightarrow V_d = 0$ at T_{min}
- current limit: $R_1 \geq V_B / I_{\text{max}} - R_{\text{sensor}}$, round up to E-series; $R_3 = R_1$
- full-scale output $V_d(T_{\text{max}})$ with the exact formula, gain $G = V_{\text{FS}} / V_d(T_{\text{max}})$
- check nonlinearity ($\approx x/(m+1)$), self-heating ($I^2 R \cdot E$), common-mode voltage ($\approx V_B/(m+1)$)

Exercise 5.4: Design of the capstone bridge

A Pt100 ($\alpha = 3.85 \times 10^{-3}/\text{K}$) shall measure $0\,^\circ\text{C} \ldots 100\,^\circ\text{C}$. The bridge is fed from $V_{\text{ref}} = 4.00\,\text{V}$, the sensor current must not exceed $1\,\text{mA}$.

a) Choose R_1, R_3 and R_4.

b) Calculate the bridge output at $0\,^\circ\text{C}$, $50\,^\circ\text{C}$ and $100\,^\circ\text{C}$ and the sensor currents.

c) How large is the deviation from a straight line at $50\,^\circ\text{C}$, in millivolt and in kelvin?

d) Which total amplifier gain gives $4.00\,\text{V}$ at $100\,^\circ\text{C}$?

Solution 5.4

a) $R_4 = 100\,\Omega$ (balance at $0\,^\circ\text{C}$); $R_1 \geq 4\,\text{V}/1\,\text{mA} - 100\,\Omega = 3.9\,\text{k}\Omega$ (E12), $R_3 = R_1 = 3.9\,\text{k}\Omega$.

b) $V_R = 4 \cdot 100/4000 = 100.0\,\text{mV}$. $0\,^\circ\text{C}$: $V_{BR} = 0$, $I = 1.000\,\text{mA}$. $50\,^\circ\text{C}$: $V_S = 4 \cdot 119.25/4019.25 = 118.68\,\text{mV}$, $V_{BR} = 18.68\,\text{mV}$. $100\,^\circ\text{C}$: $V_S = 4 \cdot 138.5/4038.5 = 137.18\,\text{mV}$, $V_{BR} = 37.18\,\text{mV}$, $I = 0.990\,\text{mA}$.

c) Straight line: $37.18/2 = 18.59\,\text{mV}$; deviation $0.09\,\text{mV}$, with $0.372\,\text{mV/K}$: $0.24\,\text{K}$.

d) $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$ (capstone: 10×10.76).

▶ Falstad: `examples/bridges/pt100_bridge.txt`: with the slider at $100\,\Omega$ the voltmeter shows $0\,\text{V}$ and $V_R = 100\,\text{mV}$; at $119.25\,\Omega$ $18.68\,\text{mV}$; at $138.5\,\Omega$ $37.18\,\text{mV}$.

Exercise 5.5: Alternative: Pt1000 at 5 V

A Pt1000 is used in a bridge with $V_B = 5\,\text{V}$, $R_1 = R_3 = 4.7\,\text{k}\Omega$, $R_4 = 1\,\text{k}\Omega$.

a) Calculate the sensor current and the self-heating power at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

b) Calculate the bridge output at $100\,^\circ\text{C}$ and the required gain for $4.00\,\text{V}$.

c) How large is the deviation from the straight line at $50\,^\circ\text{C}$? Compare with the Pt100 design.

Solution 5.5

a) $I = 5/5700 = 0.877\,\text{mA}$, $P = I^2 \cdot 1\,\text{k}\Omega = 0.77\,\text{mW}$; at $100\,^\circ\text{C}$: $I = 5/6085 = 0.822\,\text{mA}$, $P = 0.94\,\text{mW}$, almost ten times the Pt100.

b) $V_d = 5\,(1385/6085 - 1000/5700) = 260.9\,\text{mV}$, $G = 4.00/0.2609 = 15.33$.

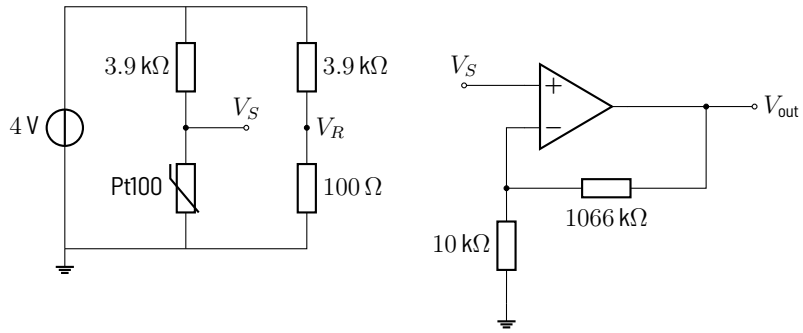
c) $V_d(50\,^\circ\mathrm{C}) = 5\,(1192.5/5892.5 - 0.17544) = 134.7\,\mathrm{mV}$, straight line $130.4\,\mathrm{mV}$: $4.3\,\mathrm{mV} = 1.6\,\%$ of full scale ($m = 4.7$), almost seven times worse than the Pt100 design with $m = 39$; in exchange the signal is seven times larger.

▶ Falstad: `examples/bridges/pt1000_bridge.txt`: voltmeter $0\,\mathrm{V}$, $134.7\,\mathrm{mV}$ and $260.9\,\mathrm{mV}$ at the slider positions 1000, 1192.5 and $1385\,\Omega$.

5.6 Reading the Bridge: Common-Mode Voltage

The bridge output is the difference of two node voltages, each of which is far from zero. In the capstone bridge both nodes sit at about $100\,\mathrm{mV}$; with $m = 1$ they would be at half the supply. The mean of the two node voltages is the *common-mode voltage* $V_{\mathrm{CM}} = (V_S + V_R)/2$, the difference is the *differential voltage* $V_d = V_S - V_R$. The amplifier must amplify V_d by about 100 and ignore V_{CM}. An ordinary amplifier referenced to ground cannot do this.

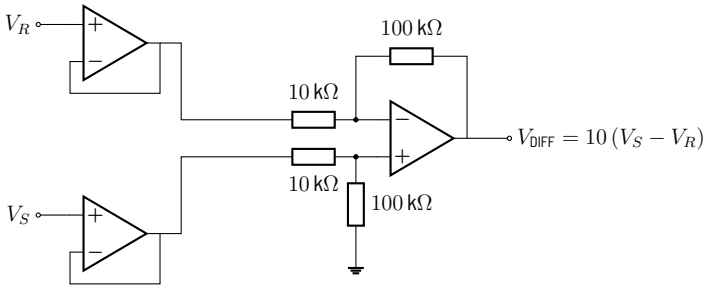
A single-ended amplifier fails



- gain 107.6 applied to $V_S = 100$ mV: $V_{out} = 10.75$ V at 0 °C (wanted: 0 V)
- at 100 °C: 14.76 V; the useful signal is a small change on a large offset
- any drift of V_{ref} appears amplified at the output

▶ Falstad: `examples/bridges/bridge_single_ended.txt`

Difference and instrumentation amplifier



- difference amplifier: amplifies $V_S - V_R$, rejects V_{CM} (if the resistors match)
- buffers: no current from the bridge, the bridge is not loaded
- second stage $G_2 = 10.76$: 0 V … 4.00 V (details: chapter 6)

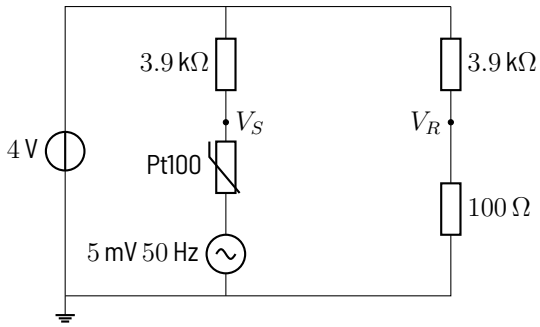
▶ Falstad: `examples/bridges/bridge_ina.txt`

Without the buffers the difference amplifier would load the bridge with its input resistors. Here the bridge has a source resistance of only $3.9\,\text{k}\Omega \parallel 100\,\Omega = 97.5\,\Omega$, so $10\,\text{k}\Omega$ inputs would cause about $1\,\%$ gain error, and worse, a mismatch between the two inputs that converts common-mode voltage into an output error. The two buffers and the difference amplifier together form the classic three-op-amp instrumentation amplifier (INA), treated in chapter 6. In Falstad the complete front end delivers 0 V, 2.01 V and 4.00 V at 0 °C, 50 °C and 100 °C.

5.7 Interference and Lead Resistance

The sensor is often far away from the electronics, connected by a cable of several metres. The cable brings two problems. First, the mains wiring and the fluorescent lamps in the lab create 50 Hz electric and magnetic fields. The loop formed by the two sensor wires acts as the secondary winding of a transformer, and the induced voltage appears in series with the sensor: a differential signal that the instrumentation amplifier amplifies just like the measurement signal. Interference that couples equally into both bridge nodes is common-mode and rejected. In the capstone the overhead lamps are modelled by a 5 mV, 50 Hz source in series with the sensor. Second, the resistance of the cable adds to the sensor resistance. For a Pt100 one ohm corresponds to 2.6 K.

50 Hz in series with the sensor

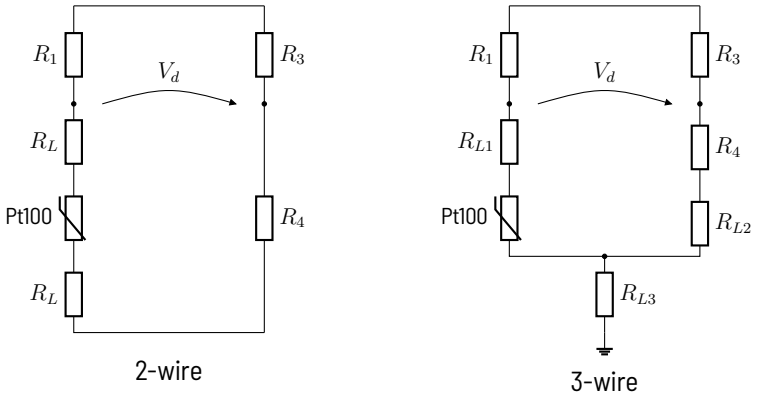


- at V_S: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} = 5\,\text{mV} \cdot \frac{3900}{4000} = 4.88\,\text{mV}$ amplitude
- after $G = 107.6$: $0.52\,\text{V}$ amplitude: about one LSB ($0.5\,\text{V}$)
- remedies: twisted pair, shield, short leads, low-pass filter before the ADC (chapters 6, 7)

▶ Falstad: `examples/bridges/bridge_50hz.txt`

Twisting the two sensor wires reduces the loop area and makes the induced voltages of neighbouring twists cancel. A shield connected to ground at the amplifier side diverts capacitively coupled currents. What remains must be removed by filtering: the temperature changes slowly, so everything above a few hertz can be cut off. Chapter 7 designs the anti-aliasing filter and chapter 8 shows what happens if the interference reaches the sampler unfiltered.

Lead resistance: 2-wire and 3-wire connection



- 2-wire: $2R_L$ in series with the sensor; $R_L = 0.5\,\Omega$: $1\,\Omega \mathrel{\hat{=}} 2.6\,\text{K}$
- 3-wire: R_{L1} in the sensor arm, R_{L2} in the neighbouring arm R_4: they cancel at balance; R_{L3} only carries the supply current
- 4-wire: current source through two wires, voltage sensed by the other two (precision instruments)

▶ Falstad: `examples/bridges/bridge_3wire.txt`

Exercise 5.6: Interference and lead resistance

The capstone bridge ($V_\text{ref} = 4\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100) is followed by an amplifier with gain 107.6.

a) A $5\,\text{mV}$, $50\,\text{Hz}$ interference voltage is induced in series with the sensor. Calculate its amplitude at V_S and at the amplifier output. Why does the instrumentation amplifier not reject it?

b) The sensor is connected with two leads of $0.5\,\Omega$ each. Which bridge output and temperature error result at $0\,°\text{C}$?

c) Repeat b) for the 3-wire connection, at $0\,°\text{C}$ and $100\,°\text{C}$.

Solution 5.6

a) The source sees R_1 above and the sensor below: $5\,\text{mV} \cdot 3900/4000 = 4.875\,\text{mV}$ at V_S, $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$ at the output. It changes only V_S, not V_R: it is a differential signal.

b) $R = 101\,\Omega$: $V_d = 4\,(101/4001 - 0.025) = 0.975\,\text{mV}$; with $0.372\,\text{mV/K}$: $2.6\,\text{K}$.

c) $0\,°\text{C}$: both lower arms have $100.5\,\Omega$, $V_d = 0$. $100\,°\text{C}$: $V_d = 37.16\,\text{mV}$ instead of $37.18\,\text{mV}$ (2-wire: $38.13\,\text{mV}$); the remaining error is $0.05\,\text{K}$.

▶ Falstad: `examples/bridges/bridge_50hz.txt`: the scope shows $4.88\,\text{mV}$ amplitude at V_S and $0.52\,\text{V}$ at `Vout`;

▶ Falstad: `examples/bridges/bridge_3wire.txt`: voltmeters $0.975\,\text{mV}$ (2-wire) and $0\,\text{V}$ (3-wire) at $0\,°\text{C}$, $38.13\,\text{mV}$ and $37.16\,\text{mV}$ with both sliders at $138.5\,\Omega$.

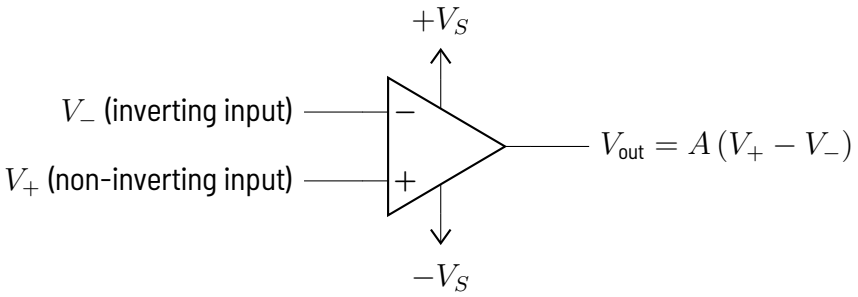
6 Operational Amplifiers

6.1 The Ideal Operational Amplifier

A bridge circuit (chapter 5) turns a small change of a sensor resistance into a voltage of a few millivolts. An analog-to-digital converter needs volts. In between sits the operational amplifier (op-amp): a differential amplifier with a very high gain that is almost never used on its own. Together with a few resistors it becomes a buffer, an amplifier with an exactly defined gain, a summing or a difference amplifier, an instrumentation amplifier or an active filter (chapter 7). The trick behind all of these circuits is negative feedback: the resistors, not the op-amp, define the behaviour.

Chapter 2 introduced the ideal op-amp, the golden rules, the voltage follower and the non-inverting amplifier. This chapter repeats them briefly and derives the other basic circuits with the same two rules. Then we look at the limits of real op-amps: finite open-loop gain, finite bandwidth, output swing, offset voltage and bias currents. At the end we design the measurement amplifier of the lab capstone task "From Sensor to Digital Value". Good introductions are [1] (chapter 4), [5] and [2]; [4] collects many practical details.

The op-amp: symbol and model



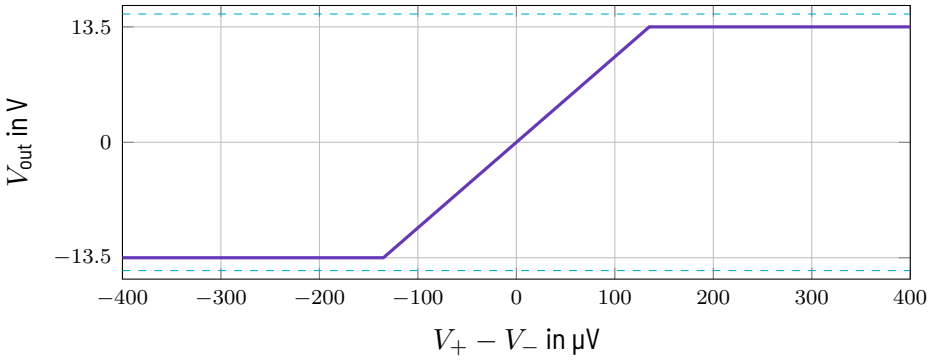
- amplifies the **difference** of its two input voltages
- open-loop gain A: typically $10^5 \ldots 10^7$ (100 dB ... 140 dB)
- supply $\pm V_S$ (or single supply V_S and ground); the output cannot leave the supply range
- supply pins are often omitted in drawings – but they are always there

Properties of the ideal op-amp

property	ideal	real (typical)
open-loop gain A	∞	$10^5 \ldots 10^7$
input resistance	∞	1 MΩ ...1 TΩ
input currents	0	pA (CMOS) ...100 nA (bipolar)
output resistance	0	10 Ω ... 100 Ω (open loop)
offset voltage	0	10 µV ...5 mV
bandwidth	∞	GBW 1 MHz ...100 MHz

The ideal values are good approximations in most circuits of this lecture. Section 6.4 shows when they are not: high gain, high frequency, high source resistance and output voltages close to the supply rails.

Open loop: the op-amp as a comparator



- with $A = 10^5$ and ± 13.5 V output swing the linear range is only ± 135 µV
- any larger difference drives the output to a rail: V_{out} tells only *which input is higher*
- this is a **comparator** – the building block of the flash ADC (chapter 9)

Without feedback an op-amp is useless as a linear amplifier: a difference of a few hundred microvolts is enough to saturate it, and the gain A varies from part to part by a factor of ten and with temperature. As a comparator it works, although dedicated comparator ICs switch much faster and have logic-compatible outputs. The ADC chapter treats comparators in detail.

Negative feedback and the golden rules

- **negative feedback**: a part of V_{out} is fed back to the **inverting** input
- if V_{out} rises, V_- rises, the difference $V_+ - V_-$ shrinks and pulls V_{out} back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$

Rule 6.1 Golden rules for an op-amp with negative feedback:

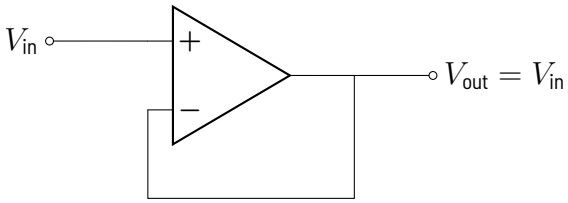
1. The output does whatever is necessary to make the voltage difference between the inputs zero: $V_+ = V_-$.
2. The inputs draw no current: $I_+ = I_- = 0$.

The two rules come from [1]. Rule 1 holds only as long as the feedback loop works, i.e. the output is not saturated at a rail and the feedback goes to the inverting input. With feedback to the non-inverting input (positive feedback) the circuit becomes a Schmitt trigger or an oscillator. Rule 2 is a property of the op-amp itself and holds always (within the bias current).

6.2 Basic Amplifier Circuits

With the golden rules the analysis of the basic circuits takes only a few lines. All circuits in this section exist as Falstad examples. Falstad uses an op-amp with $A = 10^5$ and an output clamped to the supply, which is close enough to ideal for these circuits.

Voltage follower (buffer)



- rule 1: $V_- = V_+$, and $V_- = V_{out} \Rightarrow V_{out} = V_{in}$, gain 1
- input current ≈ 0: the source is **not loaded**
- output: low resistance, can drive mA into a load

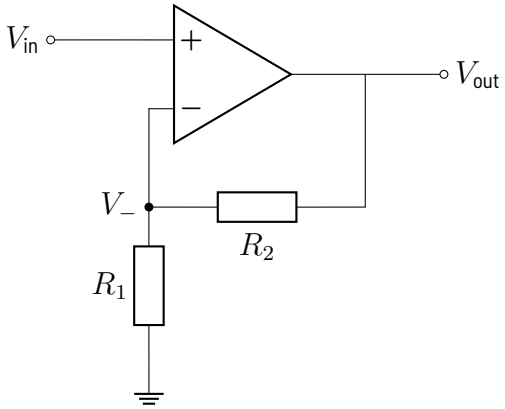
The follower does not amplify the voltage, it amplifies the current: it copies a voltage from a high-impedance node to a low-impedance output. This is exactly what a bridge needs. A bridge tap is a voltage divider with a source resistance of $R_1 \parallel R_2$ (chapter 1); a following stage with a finite input resistance would load it and shift the measured voltage. The same holds for the sample-and-hold circuit (chapter 8): the voltage on the hold capacitor must not be discharged by the following stage, so it is read by a follower with a very small input current.

Why buffering matters

- divider $100\,\text{k}\Omega/100\,\text{k}\Omega$ at $10\,\text{V}$, load $R_L = 10\,\text{k}\Omega$
- directly: $V_{out} = 10\,\text{V} \cdot \frac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L} = 0.833\,\text{V}$ instead of $5\,\text{V}$
- behind a follower: $5.00\,\text{V}$, the divider does not notice the load
- uses in this course
  - bridge taps → instrumentation amplifier inputs
  - reference ladder of the flash ADC, hold capacitor of the S&H

▶ Falstad: `examples/opamps/opamp_follower.txt`

Non-inverting amplifier

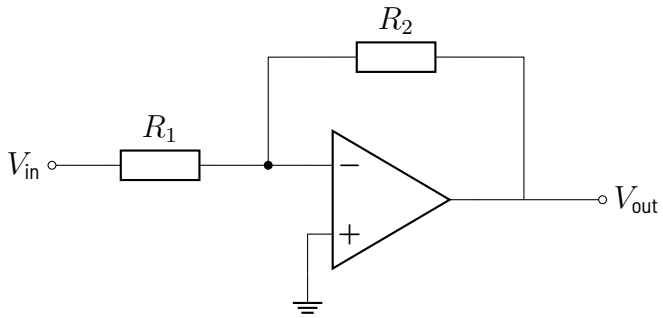


- R_1, R_2 form a divider: $V_- = V_{out} \frac{R_1}{R_1 + R_2}$; rule 1: $V_- = V_{in}$

Non-inverting amplifier

$$G = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1} \qquad \text{input resistance} \approx \infty$$

Inverting amplifier

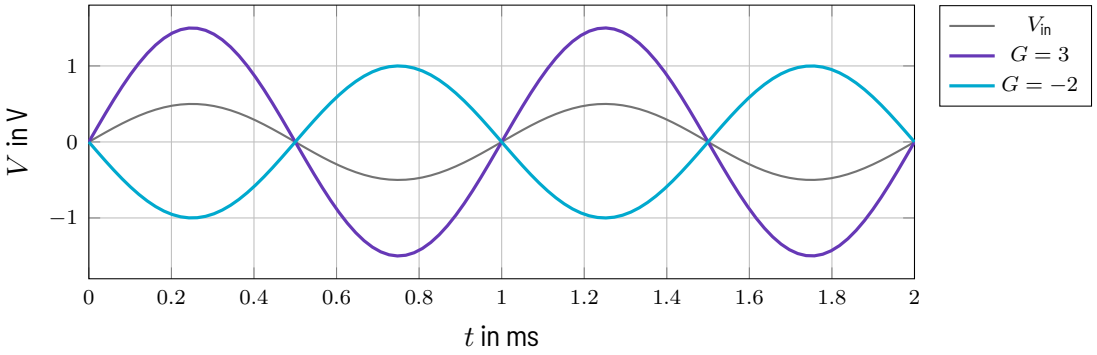


- rule 1: $V_- = V_+ = 0$, the inverting input is a **virtual ground**
- rule 2: the current V_{in}/R_1 continues through R_2: $V_{\text{out}} = -R_2 \cdot V_{\text{in}}/R_1$
- input resistance $= R_1$ (it loads the source!)

Inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1}$$

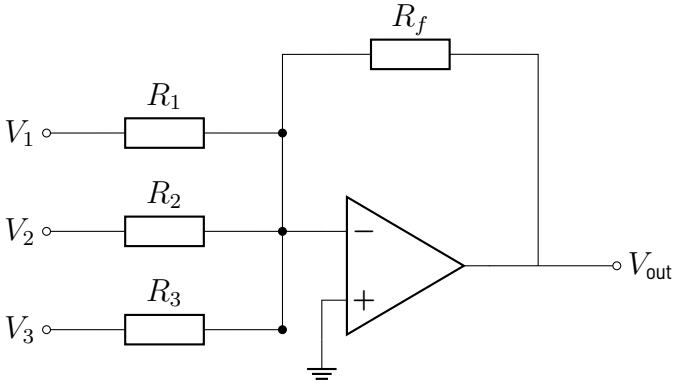
Non-inverting and inverting in Falstad



- V_{in}: 0.5 V amplitude, 1 kHz; $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ in both circuits
- non-inverting: $G = 1 + 2 = 3$, 1.5 V in phase; inverting: $G = -2$, 1.0 V, phase $180°$

▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`

Summing amplifier



- the virtual ground decouples the inputs; the currents add up in R_f

Inverting summing amplifier

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Because the summing node is held at 0 V, every input sees only its own resistor and the inputs do not influence each other. Weighted sums are easy: the weight of input k is $-R_f/R_k$. Applications are audio mixers, binary weighted DACs and adding an offset voltage to a signal (level shifting). The Falstad example adds $V_1 = 1$ V through $10\,\text{k}\Omega$, $V_2 = 2$ V through $20\,\text{k}\Omega$ and $V_3 = -0.5$ V through $10\,\text{k}\Omega$ with $R_f = 10\,\text{k}\Omega$ (exercise 6.2).

6.3 Difference and Instrumentation Amplifiers

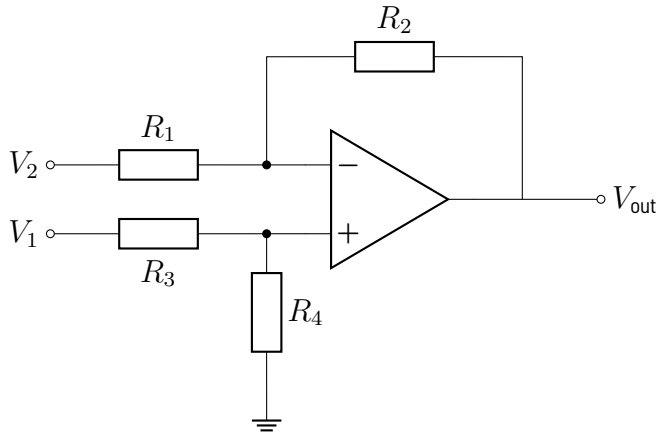
A bridge delivers its signal as the difference of two voltages that both sit at about half the bridge supply. We need an amplifier that amplifies the small difference (the differential-mode signal) and ignores the large common part (the common-mode signal). The difference amplifier does this with one op-amp and four resistors; the instrumentation amplifier adds buffers so that the bridge is not loaded.

Differential and common-mode signal

- two input voltages V_1, V_2 can be written as
  - differential-mode voltage $V_D = V_1 - V_2$ (the measurement signal)
  - common-mode voltage $V_{CM} = (V_1 + V_2)/2$ (not wanted)
- symmetric bridge at $5\,\text{V}$: V_D some mV, $V_{CM} \approx 2.5\,\text{V}$; capstone bridge: $V_D = 0\,\text{mV} \ldots 37\,\text{mV}$, $V_{CM} \approx 0.1\,\text{V}$
- ideal: $V_\text{out} = A_D\, V_D$; real: $V_\text{out} = A_D\, V_D + A_{CM}\, V_{CM}$

Definition 6.1 The **common-mode rejection ratio** is the ratio of differential gain to common-mode gain: $\text{CMRR} = A_D / A_{CM}$, in decibels $20 \log_{10}(A_D/A_{CM})$.

Difference amplifier



- superposition: $V_\text{out} = V_1 \dfrac{R_4}{R_3 + R_4}\left(1 + \dfrac{R_2}{R_1}\right) - V_2 \dfrac{R_2}{R_1}$
- with $R_4/R_3 = R_2/R_1$: $V_\text{out} = \dfrac{R_2}{R_1}(V_1 - V_2)$

The derivation uses superposition. With $V_1 = 0$ the circuit is an inverting amplifier for V_2 (the $+$ input sits at 0 V through R_3 and R_4): $V_\text{out} = -V_2 R_2/R_1$. With $V_2 = 0$ the voltage V_1 is divided by R_3, R_4 and then amplified by a non-inverting amplifier with gain $1 + R_2/R_1$. If the two ratios are equal, the factor in front of V_1 becomes $\frac{R_2/R_1}{1+R_2/R_1}(1 + R_2/R_1) = R_2/R_1$ and only the difference remains. The input resistances are not equal (R_1 for V_2, $R_3 + R_4$ for V_1) and not high: the circuit loads a bridge.

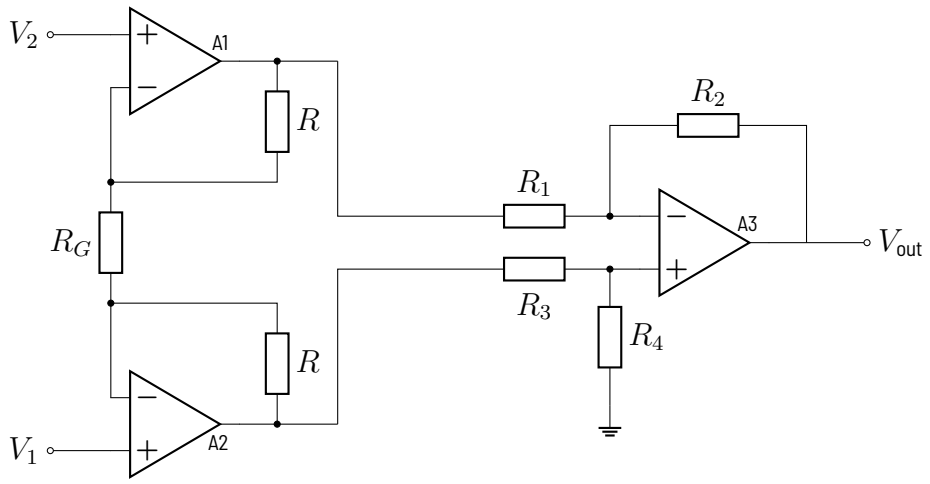
Resistor matching limits the CMRR

- the common-mode rejection depends only on how well R_4/R_3 equals R_2/R_1
- example $G = 10$, $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$: ideal $V_\text{out} = 50.0\,\text{mV}$
- R_4 only $1\,\%$ too high: $V_\text{out} = 72.6\,\text{mV}$ – an error of $45\,\%$!
- worst case with resistor tolerance ε: $\text{CMRR} \approx \dfrac{1+G}{4\varepsilon}$
  - $1\,\%$ resistors, $G = 10$: $\text{CMRR} \approx 275$ (49 dB)
  - remedy: $0.1\,\%$ resistors, matched resistor networks, a trimmer, or an integrated difference amplifier (e.g. INA105, AD8276)

▶ Falstad: `examples/opamps/opamp_diffamp.txt`

The common-mode voltage of a symmetric bridge (2.5 V) is more than a hundred times larger than its signal (some 10 mV), so a small common-mode gain already gives a large error. The worst case formula follows if all four resistors deviate by $\pm\varepsilon$ in the unfavourable direction [4]. An op-amp has its own CMRR (typically 80 to 120 dB), but in a discrete difference amplifier the resistors dominate. A higher gain G improves the CMRR, because the differential gain grows while the common-mode error stays the same.

Instrumentation amplifier (three op-amps)



Instrumentation amplifier

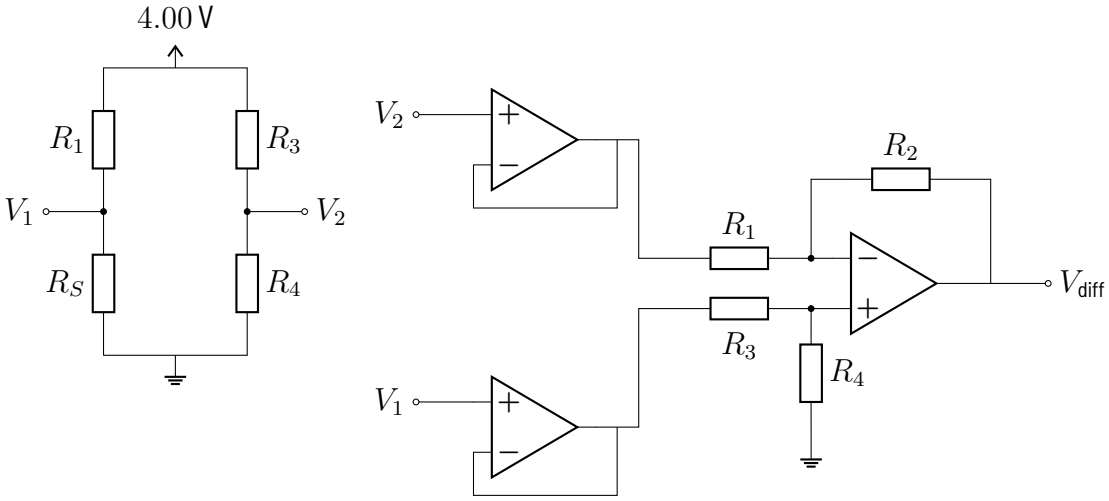
$$V_{\text{out}} = \left(1 + \frac{2R}{R_G}\right) \frac{R_2}{R_1} (V_1 - V_2)$$

How the three op-amp INA works

- A1, A2 are non-inverting: the inputs draw no current, the bridge is not loaded
- rule 1 puts V_1 and V_2 across R_G: current $(V_1 - V_2)/R_G$ flows through R, R_G, R
- difference at the outputs of A1/A2: $(V_1 - V_2)(1 + 2R/R_G)$; the common mode passes with gain 1
- A3 is a difference amplifier with gain R_2/R_1 and removes the common mode
- one resistor R_G sets the gain; integrated INAs (INA128, AD620, AD8221) contain everything except R_G

The input stage amplifies the differential signal by $1 + 2R/R_G$ but the common-mode signal only by one. This improves the CMRR of the complete amplifier by the factor $1 + 2R/R_G$ compared to the difference stage alone. Integrated instrumentation amplifiers have laser-trimmed resistors and reach a CMRR of 100 dB and more. Their reference pin (the lower end of R_4) shifts the output voltage, which is used for level shifting and offset trimming (section 6.5).

The lab version: two buffers and a difference amplifier



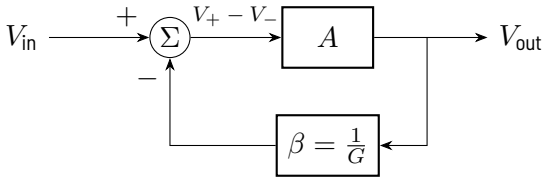
- followers isolate the bridge; the difference stage sets the gain $G_1 = R_2/R_1$
- simpler than the three op-amp version, but all gain must come from R_2/R_1

This is the circuit of the lab: the two buffers make the input resistance practically infinite, so the bridge voltages are not changed by the difference amplifier. Compared to the three op-amp INA the input stage has gain one, so the CMRR is determined by the matching of $R_1 \ldots R_4$ alone. In the capstone circuit the gain is distributed over the difference stage and a following non-inverting stage (section 6.5).

6.4 Limits of Real Op-Amps

The golden rules assume an infinite open-loop gain, zero input currents and an output that can reach any voltage. A measurement amplifier for a bridge works with a total gain of about 100 and an input signal of a few tens of millivolts. Here the deviations of a real op-amp become visible: the gain is a little lower than calculated, the output shows a voltage although the input is zero, and the output cannot reach the supply rails. This section quantifies these effects and shows how to design around them [5, 1].

Finite open-loop gain



- $V_{\text{out}} = A\,(V_{\text{in}} - \beta\,V_{\text{out}})$, with the feedback factor $\beta = 1/G$ (non-inverting: $\beta = R_1/(R_1+R_2)$)
- loop gain $A\beta = A/G$: the larger, the closer the circuit comes to the ideal gain G

Closed-loop gain with finite open-loop gain A

$$G_{\text{real}} = \frac{A}{1+A\beta} = \frac{G}{1+G/A} = \frac{1}{1/G+1/A} \qquad \text{relative error} \approx -\frac{G}{A}$$

How large is the error?

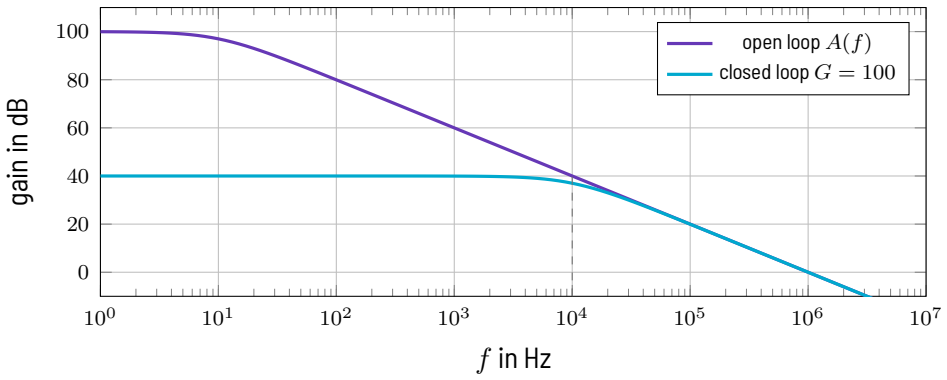
ideal gain G	G_{real} ($A = 10^5$)	error	V_{out} for 1 mV
10	9.999	$-0.01\,\%$	$10.0\,\text{mV}$
100	99.90	$-0.1\,\%$	$99.9\,\text{mV}$
1000	990.1	$-1\,\%$	$0.990\,\text{V}$
10000	9091	$-9.1\,\%$	$9.09\,\text{V}$

- the error grows proportionally to G: one stage with $G > 1000$ shows a visible error
- **split the gain**: $G = 10 \cdot 100$ gives 998.9 ($-0.11\,\%$), $G = 100 \cdot 100$ gives 9980 ($-0.2\,\%$)
- rule of the lab: aim for $G < 100$ per stage

▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

The relative errors of cascaded stages add up, but each stage has a small error because its own gain is small: $10/A + 100/A = 110/A$ instead of $1000/A$. Note that A itself is not a precise number: the data sheet gives a minimum, and A decreases with temperature, with the load and, most importantly, with frequency. A circuit that relies on a loop gain of only a few is therefore not precise, even if the formula above corrects for it.

Gain-bandwidth product



- above a low corner frequency (here $10\,\text{Hz}$) A falls by $20\,\text{dB}$ per decade: $A(f) \approx \text{GBW}/f$

- closed-loop bandwidth $f_c \approx \text{GBW}/G$ (here 1 MHz/100 = 10 kHz)

Consequences of the GBW

- on the falling slope gain and frequency are inversely proportional:

Gain-bandwidth product
GBW $= A(f) \cdot f = G \cdot f_c$ (constant above the corner frequency)

- at 1 kHz an op-amp with GBW = 1 MHz has only $A = 1000$: a $G = 100$ stage then has $10\,\%$ error
- splitting the gain also helps here: two stages $G = 10$ each have 100 kHz bandwidth
- our sensor signal is slow (a few Hz): the GBW is no problem in the capstone
- **Falstad does not simulate the GBW**: its op-amp has a constant $A = 10^5$ at all frequencies

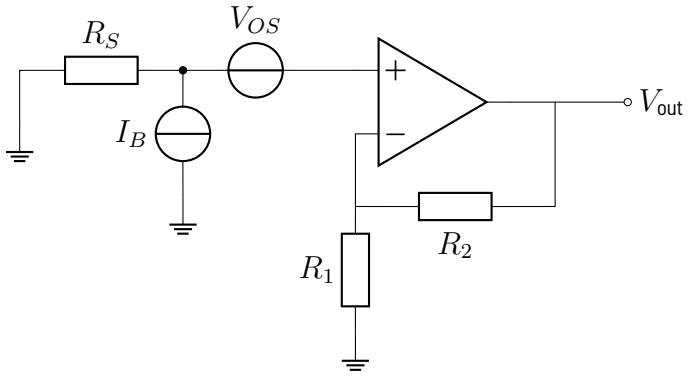
The open-loop gain of almost every op-amp is deliberately rolled off by an internal capacitor (dominant-pole compensation), so that the amplifier stays stable with any amount of negative feedback. The result is a first-order low-pass with a very low corner frequency. The product of gain and frequency is constant on the falling slope and is listed in the data sheet as gain-bandwidth product (GBW) or unity-gain frequency f_T. For the closed loop the gain that counts is the noise gain $1/\beta$, which is $1 + R_2/R_1$ for both the non-inverting and the inverting amplifier. A related limit for large signals is the slew rate, the maximum rate of change of the output voltage (e.g. $0.5\,\text{V}/\mu\text{s}$). In Falstad and in our simulator the op-amp has no bandwidth limit, so results of fast circuits must be checked against the data sheet.

Output swing and rail-to-rail

- the output cannot exceed the supply; a classic op-amp stays 1 V … 2 V away from the rails
  - LM358 at 5 V: output from about 0 V to 3.5 V
- **rail-to-rail output**: reaches the rails within some 10 mV (light load) to a few 100 mV (mA load)
- the **input common-mode range** is limited as well: not every op-amp works with inputs near the rails (**rail-to-rail input**); the capstone bridge sits at only 0.1 V
- consequence for the capstone: an output of exactly 0 V is not possible on a single 5 V supply – use a negative supply, or a small offset and a range of e.g. 0.1 V … 4.1 V

When the output hits a rail the feedback loop is broken, rule 1 no longer holds, and the signal is clipped. Clipping cannot be repaired by later stages: a clipped 50 Hz interference becomes a DC error after the low-pass filter. Each stage therefore needs headroom for the signal plus all interference it has to carry. Falstad's op-amp model clamps exactly at the values `vmax` and `vmin`, which behaves like an ideal rail-to-rail output.

Input offset voltage and bias currents



- offset V_{OS}: a small voltage source in series with an input, amplified by $1 + R_2/R_1$
- bias current I_B flows through the source resistance R_S and adds $R_S I_B$
- $V_{\text{out}} = (1 + R_2/R_1)\,(V_{OS} + R_S I_B)$ with $V_{\text{in}} = 0$

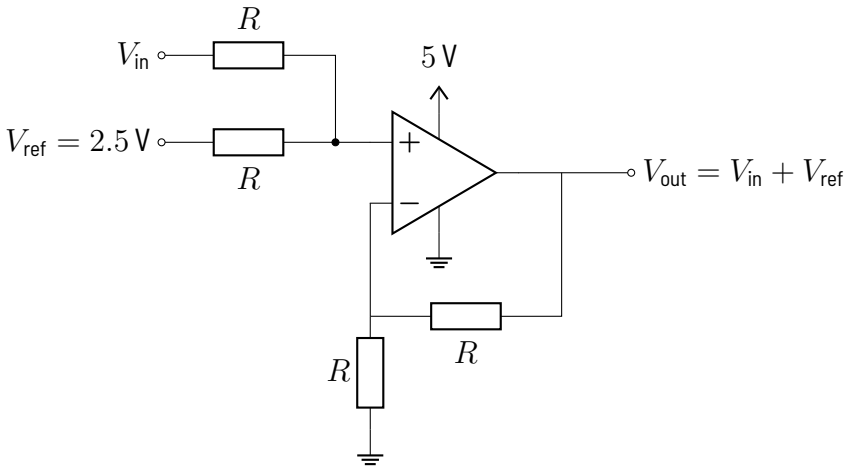
Offset after a high gain

- typical values
  - general purpose bipolar (LM358): $V_{OS} \approx 2$ mV, $I_B \approx 20$ nA
  - CMOS (MCP6002, TLV9002): $V_{OS} \approx 1$ mV, $I_B \approx 1$ pA
  - zero-drift / chopper (OPA2333, MCP6V02): $V_{OS} < 10$ µV
- Falstad example with $G = 101$: $V_{OS} = 1$ mV gives 0.101 V; $I_B = 100$ nA through 100 kΩ gives -1.01 V
- capstone: 2 mV offset $\times$ 107.6 = 0.22 V – 5 % of the full scale, almost half an LSB!
- remedies: zero-drift op-amps, CMOS inputs, low source resistances, **offset trimming**

▶ Falstad: `examples/opamps/opamp_offset.txt`

The offset voltage is a property of the input transistor pair: the two transistors are never perfectly equal. It is specified as a typical and a maximum value and it drifts with temperature (typically 1 to 10 µV/K). Because it appears at the input, it is amplified by the full noise gain. A bridge signal of 37 mV and an offset of 2 mV make an error of more than 5 %. Bias currents are the base currents (bipolar) or gate leakage currents (CMOS) of the input transistors. In a bridge amplifier the source resistance is low (about 100 Ω for the capstone bridge), so I_B matters little; but behind a sample-and-hold capacitor or a high-ohmic filter it does. The Falstad model of the op-amp has no offset and no bias current; the example models them with an extra voltage source and current source.

Single-supply operation and level shifting



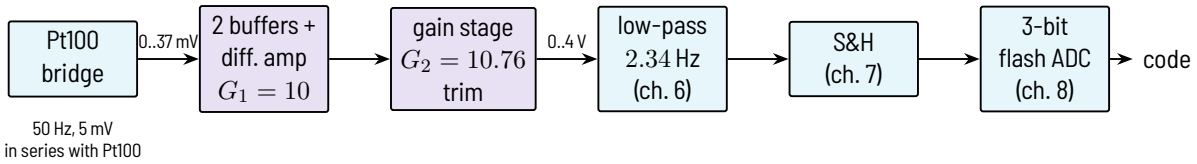
- single supply: the output can only be positive; bipolar signals need a **bias point** (often $V_S/2$)
- here $V_+ = (V_{\text{in}} + V_{\text{ref}})/2$, gain 2: $V_{\text{in}} = \pm 2$ V becomes 0.5 V … 4.5 V
- a follower on the same supply clips the negative half wave at 0 V

▶ Falstad: `examples/opamps/opamp_level_shift.txt`

6.5 Designing the Measurement Amplifier

Now we design the front end of the capstone task (chapter 10). The sensor is a Pt100 ($100\,\Omega$ at $0\,^\circ$C, $138.5\,\Omega$ at $100\,^\circ$C) in a Wheatstone bridge (chapter 5). The bridge is fed from the same 4.00 V reference that later supplies the ADC ladder, with $R_1 = R_3 = 3.9\,\text{k}\Omega$ and $R_4 = 100\,\Omega$. It is balanced at $0\,^\circ$C and delivers 0 to 37.18 mV over the measurement range. The instrumentation amplifier amplifies this to 0 V at the bottom and exactly 4.00 V at the top of the range. The following low-pass filter (chapter 7), the sample-and-hold (chapter 8) and the 3-bit flash ADC (chapter 9) expect exactly this range: with 8 codes over 4 V one LSB is 0.5 V.

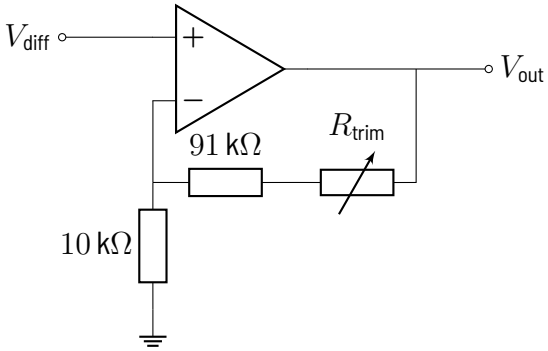
The signal chain of the capstone



- bridge from $4.00\,\text{V}$: $V_{BR} = V_{\text{ref}} \left(\frac{R}{R_1+R} - \frac{R_4}{R_3+R_4} \right) = 0\,\text{mV} \ldots 37.18\,\text{mV}$ for $R = 100\,\Omega \ldots 138.5\,\Omega$
- total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: difference amplifier $G_1 = 100\,\text{k}\Omega/10\,\text{k}\Omega = 10$, non-inverting stage $G_2 = 10.76$ – both < 100
- zero: set by the bridge ($R_4 = R(0\,°\text{C})$); span: one resistor R_F in the second stage

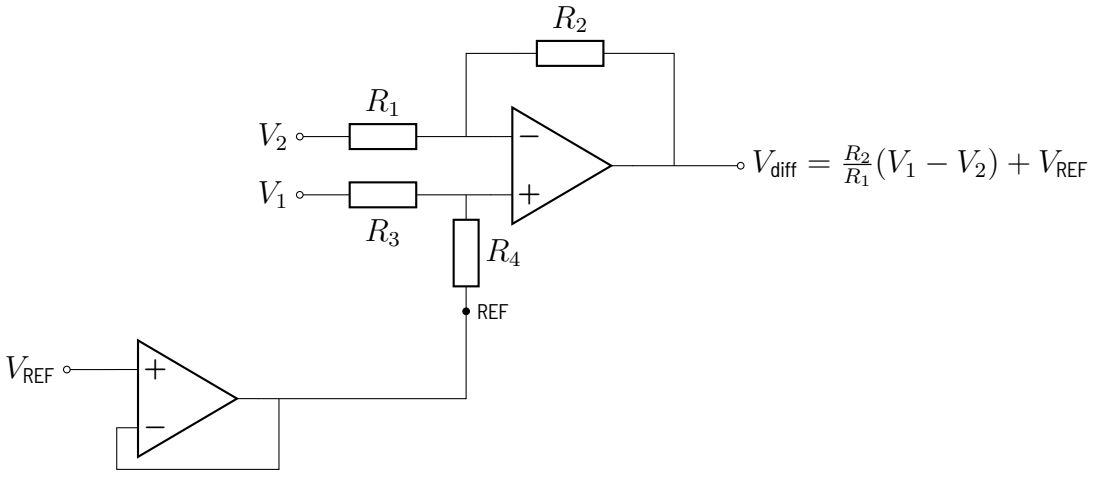
The common-mode voltage of this bridge is only about $0.1\,\text{V}$ (R_4 is small compared with R_3), so the op-amps must accept input voltages close to their negative supply. With a symmetric supply or an op-amp whose input range includes ground this is no problem.

Gain stage with trimmer



- required: $G_2 = 4.00\,\text{V}/(10 \cdot 37.18\,\text{mV}) = 10.758 \Rightarrow R_F = 97.58\,\text{k}\Omega$
- fixed E96 value: $R_F = 97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$ (used in Falstad)
- in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer: $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V} \ldots 4.13\,\text{V}$; setting $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- the fixed resistor carries most of the value: fine trimmer resolution, enough range for the tolerances

Zero adjustment at the reference input



- the lower end of R_4 is the **reference input**: V_{REF} adds directly to the output
- it must be driven from a low impedance (follower), otherwise it changes R_4 and the CMRR
- in the capstone the zero is set by the bridge ($R_4 = R(0\,°\text{C})$); a REF adjustment of a few mV is needed only if op-amp offsets or resistor tolerances shift it

Trimming procedure

- two adjustments: zero (offset) first, then span (gain)

Design card: Measurement amplifier (zero and span)

1. Calculate the bridge span $V_{BR,\text{max}}$ and the total gain $G = V_{\text{FS}}/V_{BR,\text{max}}$ (here $4.00\,\text{V}/37.18\,\text{mV} = 107.6$).
2. Split the gain: input stage (buffers + difference amplifier) $G_1 = 10$, second stage $G_2 = G/G_1 = 10.76$, each stage $G < 100$.
3. Use a fixed resistor for most of R_F and a trimmer for the rest ($91\,\text{k}\Omega$ + $10\,\text{k}\Omega$).
4. **Zero**: set the sensor to the bottom of the range ($100\,\Omega$, or a precise resistor) and check $V_{\text{out}} = 0.00\,\text{V}$; correct with R_4 or V_{REF} if necessary.
5. **Span**: set the sensor to the top of the range ($138.5\,\Omega$), adjust R_{trim} until $V_{\text{out}} = 4.00\,\text{V}$.
6. Check the zero again.

The order matters: the span trimmer changes the gain of everything that arrives at the second stage, including a remaining offset. If the zero is correct first, a later change of the gain no longer moves the zero point. In the capstone the zero is fixed by the bridge and the span by R_F, so the two adjustments do not influence each other.

The complete front end in Falstad

- bridge $4.00\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100 with a slider from $100\,\Omega$ to $138.5\,\Omega$
- two followers, difference amplifier $10\,\text{k}\Omega/100\,\text{k}\Omega$, gain stage $1+(91\,\text{k}\Omega+R_{\text{trim}})/10\,\text{k}\Omega$, $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- readings: $V_R = 0.100\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$, $V_{\text{diff}} = 372\,\text{mV}$, $V_{\text{out}} = 4.00\,\text{V}$ at $138.5\,\Omega$; $V_{\text{out}} = 0.00\,\text{V}$ at $100\,\Omega$
- the 50 Hz interference of $5\,\text{mV}$ in series with the Pt100 is a *differential* signal: $4.9\,\text{mV}$ at the tap, $\times 107.6 = 0.52\,\text{V}$ amplitude
  - no CMRR helps here – the low-pass filter of chapter 7 has to remove it
  - the output must have headroom: $4\,\text{V} + 0.52\,\text{V}$ peak

▶ Falstad: `examples/opamps/opamp_ina.txt`

6.6 Exercises

Exercise 6.1: Non-inverting and inverting amplifier

A sine voltage with $0.5\,\text{V}$ amplitude and $1\,\text{kHz}$ is to be amplified. Resistors: $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$.

a) Calculate gain and output amplitude of the non-inverting amplifier.

b) Calculate gain and output amplitude of the inverting amplifier. What is the phase shift?

c) What input resistance does the source see in both circuits?

d) Which resistor would you change to get an inverting gain of -4.7?

e) Check a) and b) in Falstad.

Solution 6.1

a) $G = 1 + R_2/R_1 = 1 + 20/10 = 3$; output amplitude $3 \cdot 0.5\,\text{V} = 1.5\,\text{V}$, in phase with the input.

b) $G = -R_2/R_1 = -2$; output amplitude $1.0\,\text{V}$, phase shift $180°$.

c) Non-inverting: practically infinite (the op-amp input, MΩ to TΩ). Inverting: $R_1 = 10\,\text{k}\Omega$, because the inverting input is a virtual ground.

d) $R_2 = 4.7 \cdot R_1 = 47\,\text{k}\Omega$ (E12 value).

e) ▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`
the scope shows `Vout_ni` with $\pm 1.50\,\text{V}$ and `Vout_inv` with $\pm 1.00\,\text{V}$, inverted.

Exercise 6.2: Summing amplifier

An inverting summing amplifier has $R_f = 10\,\text{k}\Omega$ and three inputs: $V_1 = 1\,\text{V}$ via $R_1 = 10\,\text{k}\Omega$, $V_2 = 2\,\text{V}$ via $R_2 = 20\,\text{k}\Omega$, $V_3 = -0.5\,\text{V}$ via $R_3 = 10\,\text{k}\Omega$.

a) Calculate V_{out} and the voltage at the inverting input.

b) Calculate the current through R_f.

c) How must R_3 be changed so that V_3 is weighted with -0.25?

d) Check a) in Falstad.

Solution 6.2

a) $V_{\text{out}} = -10\,\text{k}\Omega\left(\frac{1\,\text{V}}{10\,\text{k}\Omega} + \frac{2\,\text{V}}{20\,\text{k}\Omega} + \frac{-0.5\,\text{V}}{10\,\text{k}\Omega}\right) = -(1 + 1 - 0.5)\,\text{V} = -1.50\,\text{V}$. The inverting input is a virtual ground: $V_- = 0\,\text{V}$.

b) $I = 0.1\,\text{mA} + 0.1\,\text{mA} - 0.05\,\text{mA} = 0.15\,\text{mA}$, which gives $V_{\text{out}} = -10\,\text{k}\Omega \cdot 0.15\,\text{mA} = -1.5\,\text{V}$.

c) Weight $-R_f/R_3 = -0.25 \Rightarrow R_3 = 40\,\text{k}\Omega$ (e.g. $39\,\text{k}\Omega$ + $1\,\text{k}\Omega$).

d) ▶ Falstad: `examples/opamps/opamp_summing.txt`
`Vout` $= -1.50\,\text{V}$, `V_sum_node` $= 0.00\,\text{V}$.

Exercise 6.3: Difference amplifier and resistor matching

A difference amplifier with $R_1 = R_3 = 10\,\text{k}\Omega$ and $R_2 = R_4 = 100\,\text{k}\Omega$ amplifies a bridge signal: $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$.

a) Calculate differential and common-mode voltage and the output voltage.

b) R_4 is $1\,\%$ too high ($101\,\text{k}\Omega$). Calculate the output voltage.

c) Calculate the common-mode gain and the CMRR for b).

d) Estimate the worst-case CMRR with $1\,\%$ resistors.

e) Check a) and b) in Falstad.

Solution 6.3

a) $V_D = 5\,\text{mV}$, $V_{CM} = 2.5025\,\text{V}$; $V_{\text{out}} = 10 \cdot 5\,\text{mV} = 50.0\,\text{mV}$.

b) $V_+ = 2.505\,\text{V} \cdot \frac{101}{111} = 2.279\,32\,\text{V}$, $V_{\text{out}} = V_+\,(1+10) - 2.5\,\text{V} \cdot 10 = 25.0726\,\text{V} - 25\,\text{V} = 72.6\,\text{mV}$: $45\,\%$ too high.

c) $A_{CM} = 11 \cdot \frac{101}{111} - 10 = 0.0090$ (input both at $2.5\,\text{V}$ gives $22.5\,\text{mV}$); $A_D \approx 10$; CMRR $= 10/0.009 \approx 1100$ ($61\,\text{dB}$).

d) CMRR $\approx (1+G)/(4\varepsilon) = 11/0.04 = 275$ ($49\,\text{dB}$): with $2.5\,\text{V}$ common mode up to $9\,\text{mV}$ error at the input, i.e. almost half of the full bridge span. Matched resistors or an integrated difference amplifier are needed.

e) ▶ Falstad: `examples/opamps/opamp_diffamp.txt`
`Vout_matched` $= 50.0\,\text{mV}$, `Vout_mismatch` $= 72.6\,\text{mV}$.

Exercise 6.4: Finite open-loop gain

A signal of $1\,\text{mV}$ DC is amplified with non-inverting stages ($R_1 = 1\,\text{k}\Omega$). The op-amps have $A = 10^5$ (as in Falstad).

a) Calculate V_{out} for one stage with $G = 1000$ and one stage with $G = 10000$. Give R_2 and the errors.

b) Calculate V_{out} for $G = 10 \cdot 100$ and $G = 100 \cdot 100$ in two stages.

c) How large is the error of the $G = 100$ stage at $1\,\text{kHz}$ if the op-amp has GBW $= 1\,\text{MHz}$?

d) Check a) and b) in Falstad.

Solution 6.4

a) $G = 1000$: $R_2 = 999\,\text{k}\Omega$, $G_{\text{real}} = 1000/(1+0.01) = 990.1$, $V_{\text{out}} = 0.990\,\text{V}$ ($-1\,\%$). $G = 10000$: $R_2 = 9.999\,\text{M}\Omega$, $G_{\text{real}} = 10000/1.1 = 9091$, $V_{\text{out}} = 9.09\,\text{V}$ ($-9.1\,\%$).

b) $10/(1+10^{-4}) \cdot 100/(1+10^{-3}) = 9.999 \cdot 99.90 = 998.9$: $V_{\text{out}} = 0.999\,\text{V}$ ($-0.11\,\%$). $(100/1.001)^2 = 9980$: $V_{\text{out}} = 9.98\,\text{V}$ ($-0.2\,\%$).

c) $A(1\,\text{kHz}) = 1\,\text{MHz}/1\,\text{kHz} = 1000$, error $\approx G/A = 10\,\%$ (and a phase shift). Falstad does not show this, because it does not simulate the GBW.

d) ▶ Falstad: `examples/opamps/opamp_finite_gain.txt`
`Vout_1000` $= 0.990\,\text{V}$, `Vout_10x100` $= 0.999\,\text{V}$,
`Vout_10000` $= 9.09\,\text{V}$, `Vout_100x100` $= 9.98\,\text{V}$.

Exercise 6.5: Offset voltage and bias current

A non-inverting amplifier has $R_1 = 1\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$; its input is at $0\,\text{V}$.

a) The op-amp has an offset voltage of $1\,\text{mV}$. Calculate V_{out}.

b) The input is connected to ground through $R_S = 100\,\text{k}\Omega$ and a bias current of $100\,\text{nA}$ flows into the non-inverting input (no offset). Calculate V_{out}.

c) The capstone amplifier has $G = 107.6$ and the op-amps have 2 mV offset (referred to the input). Compare the output error with the full scale of 4 V and with one LSB of the 3-bit ADC.

d) Check a) and b) in Falstad.

Solution 6.5

a) $V_{\text{out}} = (1 + 100) \cdot 1\,\text{mV} = 0.101\,\text{V}$.

b) $V_+ = -R_S I_B = -100\,\text{k}\Omega \cdot 100\,\text{nA} = -10\,\text{mV}$, $V_{\text{out}} = 101 \cdot (-10\,\text{mV}) = -1.01\,\text{V}$. A CMOS op-amp with $I_B = 1\,\text{pA}$ would give only 10 µV.

c) $107.6 \cdot 2\,\text{mV} = 0.215\,\text{V}$: 5.4 % of the full scale, 0.43 LSB (LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$). The zero must be checked and trimmed, or a zero-drift op-amp used.

d) ▶ Falstad: `examples/opamps/opamp_offset.txt`
`Vout_offset` $= 0.101\,\text{V}$, `V_plus_b` $= -10.0\,\text{mV}$, `Vout_bias` $= -1.01\,\text{V}$.

Exercise 6.6: Capstone: instrumentation amplifier with 4.00 V full scale

The Pt100 of the capstone ($100\,\Omega$ at 0 °C, $138.5\,\Omega$ at 100 °C) sits in a bridge fed from the 4.00 V reference: R_1 (3.9 kΩ) above the Pt100 gives V_S, $R_3 = 3.9\,\text{k}\Omega$ above $R_4 = 100\,\Omega$ gives V_R. The amplifier consists of two followers, a difference amplifier and a non-inverting stage. Its output shall be 0 V at 0 °C and exactly 4.00 V at 100 °C.

a) Calculate V_R, V_S and the bridge output $V_{BR} = V_S - V_R$ at 0 °C and 100 °C. What is the common-mode voltage?

b) Calculate the total gain. Why is it split into two stages?

c) Choose $G_1 = 10$ for the difference amplifier and give its resistors.

d) Dimension the second stage with $R_g = 10\,\text{k}\Omega$. Give the exact R_F, the nearest E96 value and a version with 91 kΩ plus a 10 kΩ trimmer (setting and full-scale range).

e) The three op-amp INA could replace the followers. Which R_G gives an input stage gain of 10 with $R = 10\,\text{k}\Omega$?

f) A 50 Hz interference of 5 mV amplitude is in series with the Pt100. How large is it at the output? What follows for the supply voltage and the next stage?

g) Check the design in Falstad.

Solution 6.6

a) $V_R = 4\,\text{V} \cdot 100/4000 = 0.100\,\text{V}$. 0 °C: $V_S = 0.100\,\text{V}$, $V_{BR} = 0$ (sensor current 1.000 mA). 100 °C: $V_S = 4\,\text{V} \cdot 138.5/4038.5 = 0.137\,18\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$. The common mode is only about 0.1 V: the op-amp inputs must work close to the negative rail.

b) $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$. One stage would be possible (error $G/A = 0.11\,\%$), but two stages of about 10 keep each error at 0.01 % and the bandwidth high, follow the lab rule $G < 100$, and separate zero (bridge) and span (R_F).

c) $R_1 = R_3 = 10\,\text{k}\Omega$, $R_2 = R_4 = 100\,\text{k}\Omega$ (0.1 % or matched). $V_{\text{diff}} = 10 \cdot 37.18\,\text{mV} = 371.8\,\text{mV}$ at 100 °C.

d) $G_2 = 4.00\,\text{V}/371.8\,\text{mV} = 10.758$, $R_F = 9.758 \cdot 10\,\text{k}\Omega = 97.58\,\text{k}\Omega$. E96: 97.6 kΩ, $G_2 = 10.76$, full scale 4.0005 V. Trimmer version: 91 kΩ + $R_{\text{trim}} = 6.6\,\text{k}\Omega$; range $G_2 = 10.1 \ldots 11.1$, full scale 3.76 V … 4.13 V.

e) $1 + 2R/R_G = 10 \Rightarrow R_G = 2R/9 = 2.22\,\text{k}\Omega$; then the difference stage can have gain 1 and the second stage stays as it is.

f) At the tap: $5\,\text{mV} \cdot \frac{3.9\,\text{k}\Omega}{3.9\,\text{k}\Omega + 100\,\Omega} = 4.875\,\text{mV}$, at the output $\times 107.6 = 0.52\,\text{V}$ amplitude: the output swings up to $4.52\,\text{V}$. The op-amps need headroom (e.g. $\pm 15\,\text{V}$ as in Falstad); the low-pass of chapter 7 must attenuate the interference below $\text{LSB}/2 = 0.25\,\text{V}$.

g) ▶ Falstad: `examples/opamps/opamp_ina.txt`
with the Pt100 at $138.5\,\Omega$ the labels show `VB_minus` $= 0.100\,\text{V}$, `V_diff` $= 372\,\text{mV}$ and `Vout` $= 4.00\,\text{V}$; with the slider at $100\,\Omega$ `Vout` $= 0.00\,\text{V}$.

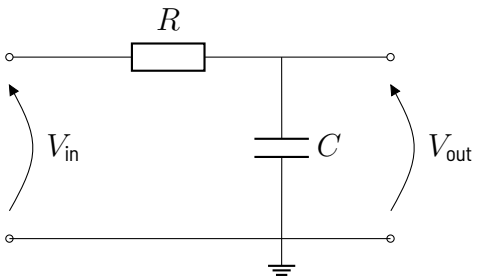
7 Active Filters

7.1 Passive RC Filters

A filter lets some frequencies pass and attenuates others. In a measurement system the wanted signal is usually slow, while interference (mains hum at 50 Hz, switching noise, radio frequencies) is faster. A low-pass filter in front of the analog-to-digital converter removes it. It has a second, more fundamental task: a sampling system with sample rate f_s cannot distinguish a signal at frequency f from one at $f \pm k f_s$. Everything above $f_s/2$ must be removed *before* sampling; this is the anti-aliasing filter (chapter 9 explains aliasing in detail).

We start with the passive RC filter and the tools to describe it: complex transfer function, magnitude and phase, decibels and the Bode plot. Then we add op-amps (chapter 6) to get first-order and second-order active filters, and finally design the 2.5 Hz anti-aliasing low-pass of the capstone task. Further reading: [2] (chapters on filters), [1] (chapter 6), [5] (active filter design) and [3].

RC low-pass



- low frequency: the capacitor is open, $V_{out} = V_{in}$
- high frequency: the capacitor is a short, $V_{out} \to 0$
- a frequency-dependent voltage divider with the impedance $Z_C = \dfrac{1}{j\omega C}$

RC low-pass

$$\underline{H}(j\omega) = \frac{\underline{V}_{out}}{\underline{V}_{in}} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$$

Magnitude, phase and cutoff frequency

- with $f_c = \dfrac{1}{2\pi RC}$: $\quad \underline{H} = \dfrac{1}{1 + jf/f_c}$
- magnitude (amplitude ratio) and phase:

First-order low-pass

$$|\underline{H}| = \frac{1}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = -\arctan(f/f_c) \qquad f_c = \frac{1}{2\pi RC}$$

- at $f = f_c$: $|\underline{H}| = 1/\sqrt{2} = 0.707$, $\varphi = -45°$
- far above f_c: $|\underline{H}| \approx f_c/f$ – ten times the frequency, one tenth of the amplitude

For a sine input the output of a linear circuit is a sine of the same frequency. Its amplitude is multiplied by $|\underline{H}|$ and its phase is shifted by φ. The cutoff frequency f_c (also corner frequency or -3 dB frequency) is the frequency at which the real part and the imaginary part of the denominator are equal; there the output power is half the power at low frequencies. The time constant $\tau = RC$ and the cutoff frequency are two views of the same thing: $f_c = 1/(2\pi\tau)$.

Decibels

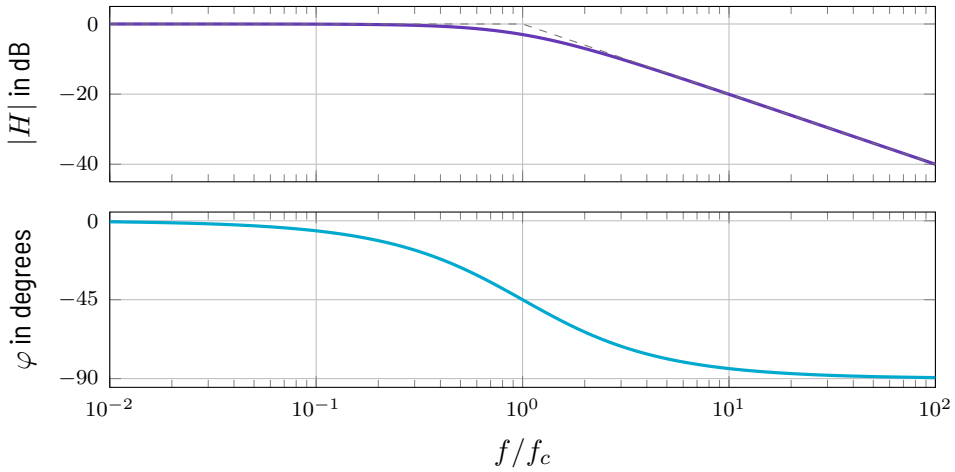
- gains and attenuations span many decades: a logarithmic measure is convenient

Definition 7.1 The gain in decibels is $|H|_{dB} = 20 \log_{10} |H|$ (voltage or current ratios).

$\lvert H\rvert$	dB	$\lvert H\rvert$	dB
1	0 dB	2	6 dB
$1/\sqrt{2} = 0.707$	-3 dB	10	20 dB
0.5	-6 dB	100	40 dB
0.1	-20 dB	1000	60 dB
0.01	-40 dB	10^5	100 dB

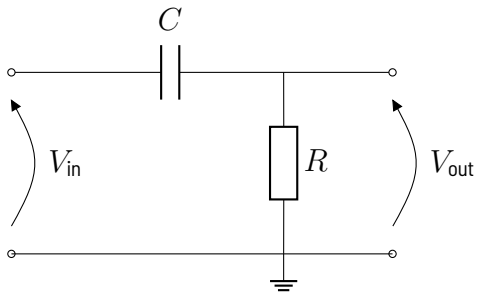
- gains of cascaded stages **multiply**; in dB they **add**
- a decade is a factor 10 in frequency, an octave a factor 2

Bode plot of the RC low-pass



- logarithmic frequency axis, magnitude in dB: the curve becomes two straight asymptotes
- 0 dB below f_c, -20 dB per decade above; the corner is at f_c (-3 dB)

RC high-pass



First-order high-pass

$$\underline{H} = \frac{jf/f_c}{1 + jf/f_c} \qquad |\underline{H}| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = 90° - \arctan(f/f_c)$$

- blocks DC (coupling capacitor), 20 dB per decade below f_c
- same cutoff frequency $f_c = 1/(2\pi RC)$, phase $45°$ at f_c

RC filters in Falstad

- $R = 1.6\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 995\,\text{Hz}$; three copies driven with $1\,\text{V}$ at $100\,\text{Hz}$, $1\,\text{kHz}$, $10\,\text{kHz}$

	$100\,\text{Hz}$	$1\,\text{kHz}$	$10\,\text{kHz}$
low-pass amplitude	$0.995\,\text{V}$	$0.705\,\text{V}\ (-45°)$	$0.099\,\text{V}$
high-pass amplitude	$0.100\,\text{V}$	$0.709\,\text{V}\ (45°)$	$0.995\,\text{V}$

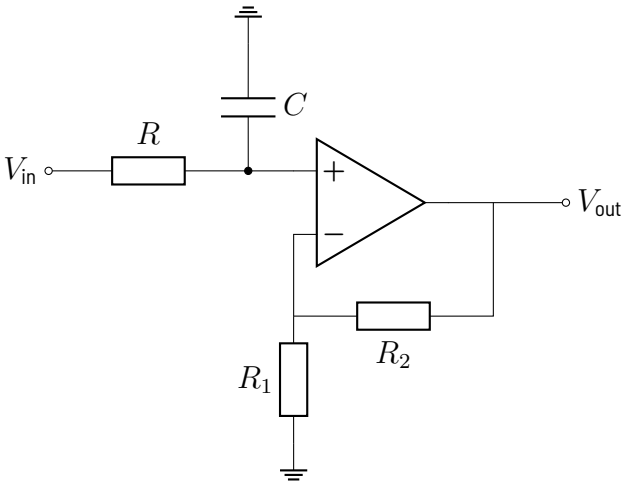
▶ Falstad: `examples/filters/filter_rc_lowpass.txt` ▶ Falstad: `examples/filters/filter_rc_highpass.txt`

- limits of passive filters: the load changes f_c (it is in parallel to C or R), no gain, steep filters need inductors
- remedy: an op-amp buffers the filter and adds gain $\rightarrow$ **active filter**

7.2 First-Order Active Filters

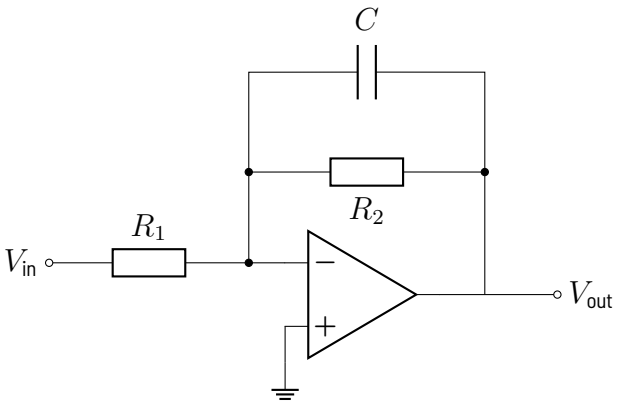
An op-amp turns the RC filter into an active filter: the filter is no longer loaded by the following stage, and it can have a gain. In the simplest case the RC network sits in front of a non-inverting amplifier; alternatively the capacitor is placed into the feedback path of an inverting amplifier. The cutoff frequency is still $1/(2\pi RC)$ for the resistor and capacitor that form the filter.

Non-inverting first-order low-pass



- RC low-pass, then a non-inverting amplifier (or a follower for $G = 1$)
- $\underline{H} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + j\omega RC}$, $f_c = \frac{1}{2\pi RC}$
- the op-amp input does not load C; the output can drive the next stage

Inverting first-order low-pass

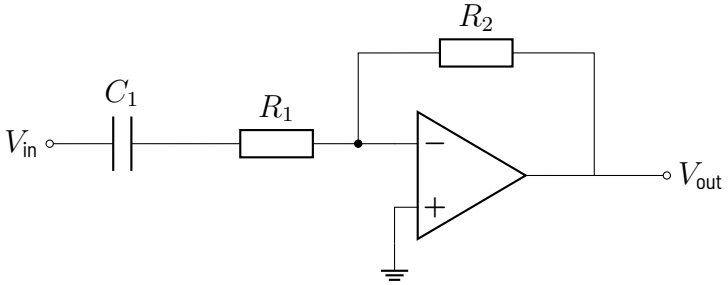


- feedback impedance $Z_2 = R_2 \parallel \frac{1}{j\omega C}$, gain $-Z_2/R_1$

Inverting first-order low-pass

$$\underline{H} = -\frac{R_2}{R_1} \frac{1}{1 + j\omega R_2 C} \qquad \text{DC gain } -\frac{R_2}{R_1}, \quad f_c = \frac{1}{2\pi R_2 C}$$

First-order high-pass (inverting)



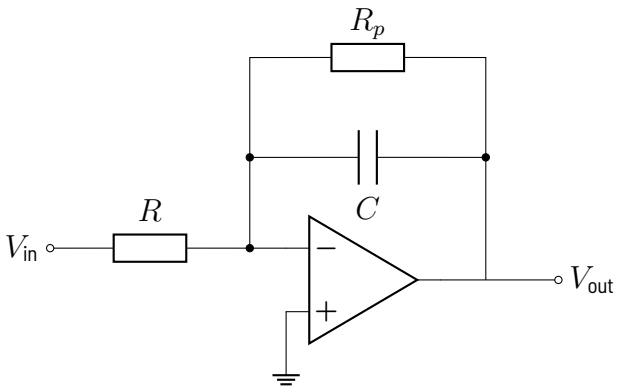
Inverting first-order high-pass

$$\underline{H} = -\frac{R_2}{R_1}\,\frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1} \qquad \text{HF gain } -\frac{R_2}{R_1}, \quad f_c = \frac{1}{2\pi R_1 C_1}$$

- Falstad: all three circuits with $f_c = 995$ Hz, input 0.1 V at 1 kHz
  - non-inverting LP ($G = 2$): 0.141 V; inverting LP (-10): 0.705 V; inverting HP (-10): 0.709 V

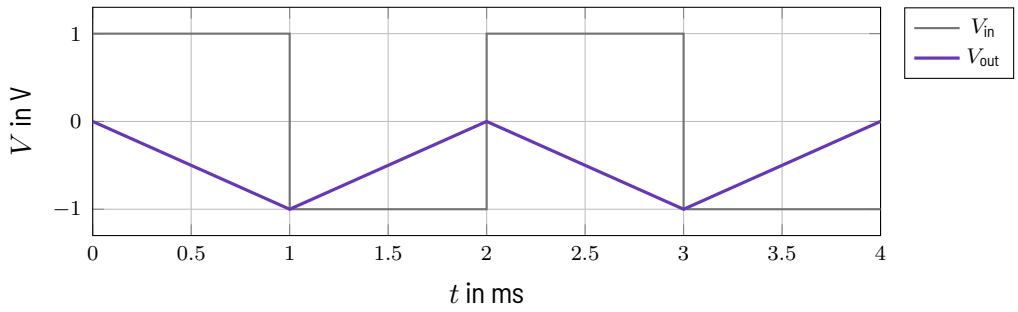
▶ Falstad: `examples/filters/filter_active_first_order.txt`

Integrator



- the current V_{in}/R charges C: $\quad V_{\text{out}}(t) = -\dfrac{1}{RC}\int V_{\text{in}}\,dt, \quad \underline{H} = -\dfrac{1}{j\omega RC}$
- -20 dB per decade at all frequencies; DC gain ∞: any offset drives it into saturation
- practical integrator: R_p limits the DC gain to $-R_p/R$ (it is then a low-pass with $f_c = 1/(2\pi R_p C)$)

Integrator: square wave in, triangle out



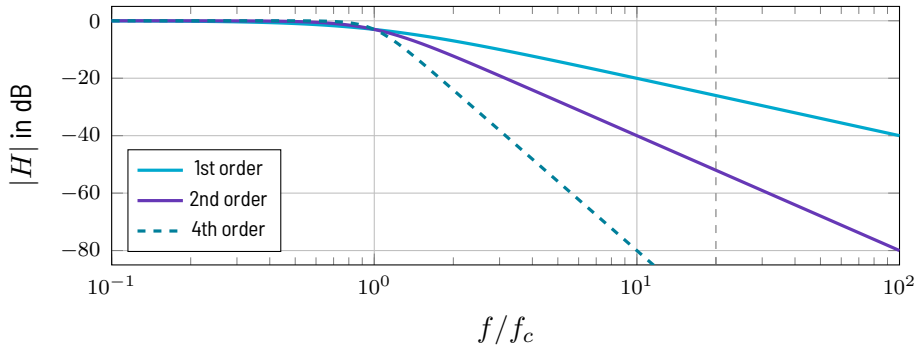
- $R = 10\,\text{k}\Omega$, $C = 100$ nF, $R_p = 1\,\text{M}\Omega$; $V_{\text{in}} = \pm 1$ V, 500 Hz
- slope $V_{\text{in}}/(RC) = 1000$ V/s: 1 V in 1 ms, triangle 1.0 V peak-to-peak
- the triangle starts between 0 V and -1 V; R_p centres it slowly ($\tau = R_p C = 0.1$ s)

▶ Falstad: `examples/filters/filter_integrator.txt`

7.3 Second-Order Filters

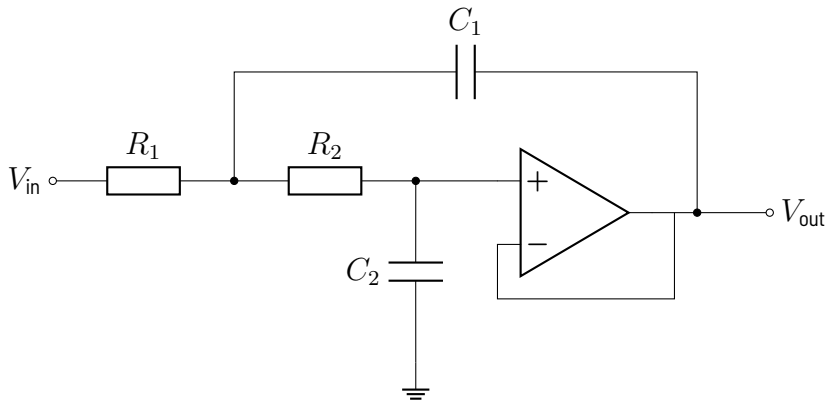
A first-order filter falls by only 20 dB per decade. If the interference is only one decade above the signal, a first-order filter attenuates it by a factor of ten – often not enough. Cascading RC stages helps little: passive RC stages load each other and the corner becomes soft. Active second-order filters give 40 dB per decade with one op-amp, and a sharp corner whose shape is set by the quality factor Q. The most popular circuit is the Sallen-Key filter [5, 2].

Filter order and attenuation



- Butterworth filter of order n: $|\underline{H}| = 1/\sqrt{1+(f/f_c)^{2n}}$; slope $-20n$ dB per decade
- dashed line: $f = 20\ f_c$ (50 Hz for $f_c = 2.5$ Hz): -26 dB, -52 dB, -104 dB

Sallen-Key low-pass (unity gain)



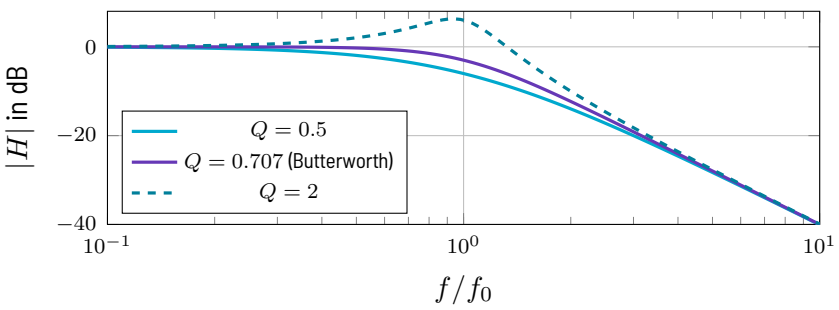
- the follower makes $V_{\text{out}} = V_+$; C_1 feeds the output back to the middle node (positive feedback near f_0)
- $\underline{H} = \dfrac{1}{1 + j\omega C_2(R_1+R_2) - \omega^2 R_1 R_2 C_1 C_2}$

Natural frequency and quality factor

- comparing with $1/(1 + j\frac{f}{Qf_0} - (f/f_0)^2)$ gives f_0 and Q:

Unity-gain Sallen-Key low-pass

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2\,(R_1+R_2)} \overset{R_1=R_2}{=} \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$



- $|\underline{H}(f_0)| = Q$; Butterworth $Q = 1/\sqrt{2}$: maximally flat, -3 dB at f_0; higher Q: peaking and ringing

Design of a Butterworth Sallen-Key low-pass

- capacitors come in coarse steps: choose them first, then calculate the resistors

Design card: Unity-gain Sallen-Key low-pass, $Q = 0.707$

1. Choose $R_1 = R_2 = R$. Then $Q = 0.707$ requires $C_1 = 2\,C_2$ (C_1 goes to the output).
2. Choose C_2 from the E6 series, $C_1 \approx 2C_2$ (e.g. 10 nF and 22 nF: $Q = 0.74$).
3. Calculate $R = \dfrac{1}{2\pi f_0 \sqrt{C_1 C_2}}$ and round to the E24 or E96 series.
4. Check f_0 and Q with the rounded values. Use film or C0G capacitors (stable, low leakage).

- example $f_0 = 1$ kHz: $C_2 = 10$ nF, $C_1 = 22$ nF, $R = 10.7$ kΩ (E96): $f_0 = 1003$ Hz, $Q = 0.742$

Capacitors are available in coarser steps than resistors, so they are chosen first and the resistors are calculated. Exact Butterworth behaviour is rarely needed; Q between 0.65 and 0.75 changes the response near f_0 by less than 1 dB. Higher-order filters are built as a cascade of second-order stages (and one first-order stage for odd orders) with tabulated values of f_0 and Q for each stage [2].

First order vs. second order in Falstad

	1 kHz	10 kHz
first order RC ($f_c = 995$ Hz)	0.705 V	0.099 V (-20 dB)
Sallen-Key ($f_0 = 1003$ Hz, $Q = 0.742$)	0.744 V	0.0101 V (-40 dB)

- input 1 V amplitude; one decade above the corner the second-order filter attenuates ten times more
- near f_0 both behave similarly

▶ Falstad: `examples/filters/filter_sallen_key.txt`

7.4 The Anti-Aliasing Filter of the Capstone

In the capstone the amplified bridge signal (0 to 4.00 V, chapter 6) is sampled by a sample-and-hold circuit with $f_s = 5$ Hz and converted by a 3-bit flash ADC. The sensor signal itself changes slowly. The problem is the 50 Hz interference of 5 mV in series with the Pt100: it is a differential signal (4.9 mV at the bridge tap), so the instrumentation amplifier amplifies it by the full gain of 107.6 to an amplitude of 0.52 V. This section designs the low-pass filter that removes it, with the values used in the capstone chapter: $R = 6.8$ kΩ, $C = 10$ µF.

Why filter before sampling? (preview of chapter 9)

- sampling theorem (Nyquist, Shannon [10]): only frequencies below $f_s/2$ are reproduced correctly
- with $f_s = 5$ Hz: $f_s/2 = 2.5$ Hz $\Rightarrow$ cutoff of the low-pass $f_c \approx 2.5$ Hz
- a component above $f_s/2$ is not removed by sampling – it appears at a wrong, low frequency (**aliasing**)
  - $50\,\text{Hz} = 10 \cdot f_s$: the samples see the interference always at the same phase – a constant, random-looking error
  - if the mains frequency drifts to 50.1 Hz, the error becomes a slow 0.1 Hz wave in the data
- after sampling nothing can separate the alias from the signal: the filter must act **before** the S&H

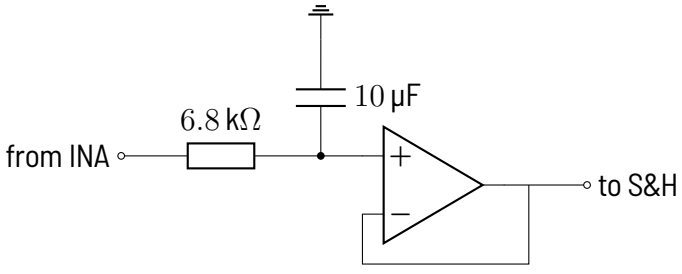
How much attenuation do we get at 50 Hz?

- first order, $f_c = 2.5$ Hz: $|\underline{H}(50\,\text{Hz})| = \dfrac{1}{\sqrt{1+(50/2.5)^2}} = \dfrac{1}{\sqrt{401}} = 0.050 \;\hat{=}\; -26$ dB
- second order (Butterworth): $\dfrac{1}{\sqrt{1+(50/2.5)^4}} = 0.0025 \;\hat{=}\; -52$ dB
- capstone filter $6.8\,\text{k}\Omega$, $10\,\mu$F: $f_c = 2.34$ Hz, $|\underline{H}| = 1/\sqrt{1+(50/2.34)^2} = 0.047 \;\hat{=}\; -26.6$ dB
- interference before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52$ V

	amplitude	in LSB (0.5 V)	
no filter	0.52 V	1 LSB	useless
first order, 2.34 Hz	25 mV	0.05 LSB	sufficient
Sallen-Key, 2.55 Hz	1.4 mV	0.003 LSB	large margin

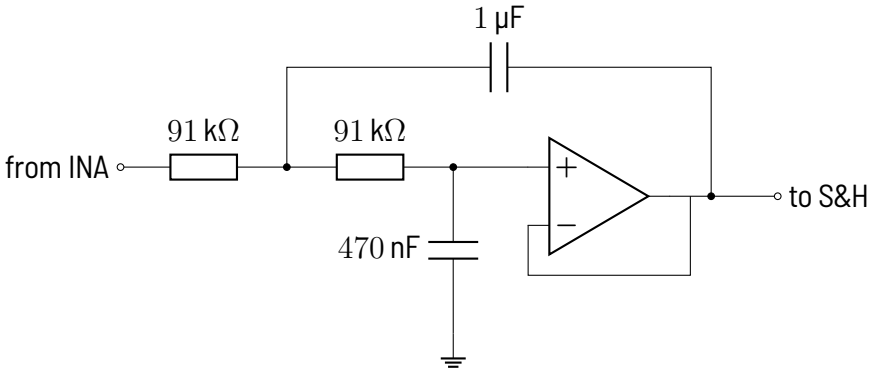
For a 3-bit converter with 4 V full scale one LSB is 0.5 V, and the quantization error is ± 0.25 V. A residual of 25 mV is far below LSB$/2$, so the first-order filter of the capstone is sufficient. A second-order filter costs only one more capacitor and gives a large margin, for example for a higher-resolution ADC: with 10 bits (LSB = 3.9 mV) the first-order filter leaves about 6 LSB of interference, the second-order filter less than half an LSB.

First-order anti-aliasing filter with E-series parts



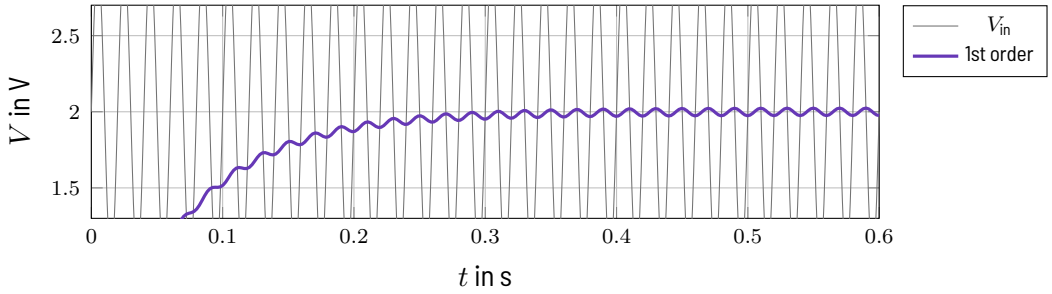
- $RC = 1/(2\pi \cdot 2.5\,\text{Hz}) = 63.7$ ms; choose $C = 10\,\mu$F $\Rightarrow R = 6.37\,\text{k}\Omega$
- round so that $f_c \le f_s/2$: $6.8\,\text{k}\Omega$ (E12) $\Rightarrow f_c = 2.34$ Hz, -26.6 dB at 50 Hz
- also usable: $62\,\text{k}\Omega$ with $1\,\mu$F ($f_c = 2.57$ Hz, slightly above $f_s/2$, still -26 dB at 50 Hz)
- the follower decouples the filter from the S&H input
- step response: $\tau = 68$ ms; settling to $1\,\%$ takes $4.6\,\tau = 0.31$ s

Second-order version (Sallen-Key)



- $C_1 = 1\,\mu\text{F}, C_2 = 470\,\text{nF}$: $Q = \frac{1}{2}\sqrt{1/0.47} = 0.729$
- $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot \sqrt{1\,\mu\text{F} \cdot 470\,\text{nF}}) = 92.8\,\text{k}\Omega \rightarrow$ E24 $91\,\text{k}\Omega$: $f_0 = 2.55\,\text{Hz}$

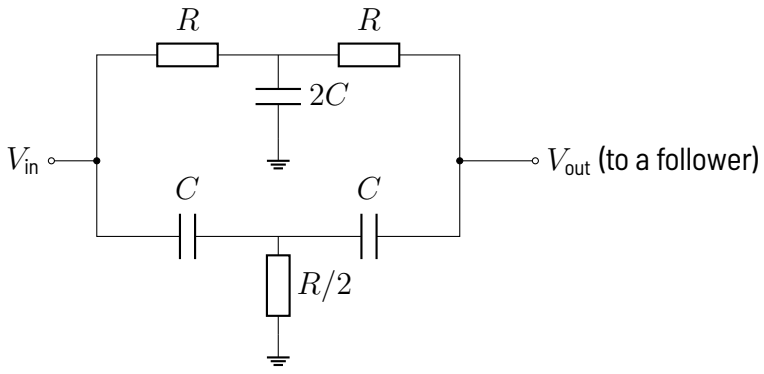
Both filters in Falstad



- input: 2 V signal + 0.52 V, 50 Hz interference (= 4.875 mV $\times$ 107.6)
- after settling: both outputs 2.00 V DC; remaining 50 Hz: 25 mV (1st order), 1.4 mV (Sallen-Key)

▶ Falstad: `examples/filters/filter_antialias.txt`

Alternative: twin-T notch at 50 Hz



- zero transmission at $f_0 = 1/(2\pi RC)$: $R = 31.8\,\text{k}\Omega, C = 100\,\text{nF} \Rightarrow 50\,\text{Hz}$
- Falstad: 50 Hz reduced from 1 V to 0.45 mV; but 5 Hz is also attenuated (0.927)
- needs precise parts, removes only one frequency – it does **not** replace the anti-aliasing low-pass

▶ Falstad: `examples/filters/filter_twin_t.txt`

Design card

Design card: Anti-aliasing low-pass

1. Cutoff frequency from the sample rate: $f_c \le f_s/2$ (capstone: $f_s = 5$ Hz, $f_c = 2.34$ Hz).
2. Required attenuation: interference amplitude at the ADC input compared with LSB$/2$; with $f/f_c = 20$ a first-order filter gives -26 dB, a second-order filter -52 dB (capstone: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < 250\,\text{mV}$).
3. First order: $RC = 1/(2\pi f_c)$; choose C, calculate R, round to the E-series so that $f_c \le f_s/2$ (capstone: 6.8 k, $10\,\mu$F). Buffer the output.
4. Second order: unity-gain Sallen-Key with $C_1 \approx 2C_2$, $R = 1/(2\pi f_0\sqrt{C_1 C_2})$.
5. Check: settling time of a step ($\approx 5\tau$) against the sample period, and headroom of the amplifier before the filter (signal + interference must not clip).

7.5 Exercises

Exercise 7.1: RC low-pass

An RC low-pass has $R = 1.6\,\text{k}\Omega$ and $C = 100$ nF.

a) Calculate the cutoff frequency.

b) Calculate magnitude (also in dB) and phase at 100 Hz, 1 kHz and 10 kHz.

c) What changes if a load of $1.6\,\text{k}\Omega$ is connected to the output?

d) Check b) in Falstad.

Solution 7.1

a) $f_c = 1/(2\pi \cdot 1.6\,\text{k}\Omega \cdot 100\,\text{nF}) = 995$ Hz.

b) 100 Hz: $f/f_c = 0.1005$, $|H| = 1/\sqrt{1.0101} = 0.995$ (-0.04 dB), $\varphi = -5.7°$.
1 kHz: $f/f_c = 1.005$, $|H| = 0.705$ (-3.0 dB), $\varphi = -45.2°$.
10 kHz: $f/f_c = 10.05$, $|H| = 0.099$ (-20.1 dB), $\varphi = -84.3°$.

c) The load is in parallel to C. Thevenin: $V_{\text{th}} = V_{\text{in}}/2$, $R_{\text{th}} = 0.8\,\text{k}\Omega$: the DC gain drops to 0.5 and f_c doubles to 1.99 kHz. A follower avoids this.

d) ▶ Falstad: `examples/filters/filter_rc_lowpass.txt`
amplitudes 0.995 V, 0.705 V (lagging by $45°$, i.e. 125 µs) and 0.099 V.

Exercise 7.2: First-order active filters

a) Design an inverting low-pass with DC gain -10, input resistance $10\,\text{k}\Omega$ and $f_c = 1$ kHz.

b) Calculate the output amplitude for 0.1 V at 1 kHz with C rounded to the E-series.

c) Design a non-inverting low-pass with $G = 2$ and the same f_c, using $C = 10$ nF.

d) Design an inverting high-pass with HF gain -10, $R_1 = 10\,\text{k}\Omega$, $f_c \approx 1$ kHz.

e) Check b) to d) in Falstad (input 0.1 V, 1 kHz).

Solution 7.2

a) $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $C = 1/(2\pi \cdot 100\,\text{k}\Omega \cdot 1\,\text{kHz}) = 1.59\,\text{nF}$.

b) $C = 1.6\,\text{nF}$ (E24): $f_c = 995\,\text{Hz}$; $0.1\,\text{V} \cdot 10 \cdot 0.705 = 0.705\,\text{V}$.

c) $R = 1/(2\pi \cdot 1\,\text{kHz} \cdot 10\,\text{nF}) = 15.9\,\text{k}\Omega \to 16\,\text{k}\Omega$; gain stage $R_1 = R_2 = 10\,\text{k}\Omega$. Output: $0.1\,\text{V} \cdot 2 \cdot 0.705 = 0.141\,\text{V}$.

d) $R_2 = 100\,\text{k}\Omega$, $C_1 = 1/(2\pi \cdot 10\,\text{k}\Omega \cdot 1\,\text{kHz}) = 15.9\,\text{nF} \to 16\,\text{nF}$: $f_c = 995\,\text{Hz}$, output $0.1\,\text{V} \cdot 10 \cdot 0.709 = 0.709\,\text{V}$.

e) ▶ Falstad: `examples/filters/filter_active_first_order.txt`
`Vout_a` $= 0.141$ V, `Vout_b` $= 0.705$ V, `Vout_c` $= 0.709$ V.

Exercise 7.3: Integrator

An integrator has $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$ and $R_p = 1\,\text{M}\Omega$ parallel to C. The input is a square wave of ± 1 V and 500 Hz, starting with $+1$ V at $t = 0$, $V_{\text{out}}(0) = 0$.

a) Calculate the slope of the output voltage and sketch $V_{\text{out}}(t)$ for the first two periods.

b) Calculate the peak-to-peak value of the triangle.

c) What is the purpose of R_p? What happens without it if the op-amp has an offset of 1 mV?

d) Above which frequency does the circuit behave as an integrator?

e) Check a) and b) in Falstad.

Solution 7.3

a) $dV_{\text{out}}/dt = -V_{\text{in}}/(RC) = -1\,\text{V}/1\,\text{ms} = -1000\,\text{V/s}$ while $V_{\text{in}} = +1$ V. Triangle: $0 \to -1$ V in the first 1 ms, back to 0 V at 2 ms, and so on.

b) 1 V per half period: 1.0 V peak-to-peak.

c) R_p limits the DC gain to $-R_p/R = -100$. Without it, 1 mV offset causes a current of 0.1 µA that charges C at 1 V/s: the output saturates after some seconds. With R_p: DC error 0.1 V.

d) Above $f_c = 1/(2\pi R_p C) = 1.6$ Hz; at 500 Hz the error is negligible.

e) ▶ Falstad: `examples/filters/filter_integrator.txt`
the output reaches about -1 V after 1 ms (-0.98 V in the simulation, R_p and the time step cost a little), the triangle has 1.0 V peak-to-peak and drifts slowly towards a symmetric position.

Exercise 7.4: Sallen-Key low-pass

Design a unity-gain Sallen-Key low-pass with $f_0 = 1$ kHz and Butterworth characteristic.

a) Start with $C_2 = 10\,\text{nF}$ and $R_1 = R_2$. Choose C_1 from the E6 series and calculate R (E96).

b) Calculate Q and f_0 with the chosen values.

c) Calculate the amplitude for 1 V input at 1 kHz and 10 kHz. Compare with a first-order filter.

d) Check c) in Falstad.

Solution 7.4

a) $Q = 0.707$ needs $C_1 = 2C_2 = 20$ nF; E6: 22 nF.
$R = 1/(2\pi \cdot 1\,\text{kHz} \cdot \sqrt{22\,\text{nF} \cdot 10\,\text{nF}}) = 10.73\,\text{k}\Omega \to 10.7\,\text{k}\Omega$.

b) $Q = \frac{1}{2}\sqrt{22/10} = 0.742$; $f_0 = 1/(2\pi \cdot 10.7\,\text{k}\Omega \cdot 14.83\,\text{nF}) = 1003$ Hz.

c) $|H| = 1/\sqrt{(1-x^2)^2 + (x/Q)^2}$ with $x = f/f_0$. 1 kHz: $x = 0.997$, $|H| = 0.744$ (0.744 V; first order: 0.705 V). 10 kHz: $x = 9.97$, $|H| = 1/\sqrt{98.4^2 + 13.4^2} = 0.0101$ (−40 dB; first order: 0.099, −20 dB).

d) ▶ Falstad: `examples/filters/filter_sallen_key.txt`
`VSK_1kHz` = 0.744 V, `VSK_10kHz` = 10.1 mV,
`V1st_1kHz` = 0.705 V, `V1st_10kHz` = 0.099 V.

Exercise 7.5: Capstone: anti-aliasing low-pass

The capstone samples with $f_s = 5$ Hz. The amplifier has a gain of 107.6 and its output carries a 50 Hz interference (4.875 mV at the bridge tap). The ADC has 3 bits and 4 V full scale.

a) Choose the cutoff frequency of the anti-aliasing filter and justify it.

b) Design a first-order filter (follower behind an RC) with $C = 10$ µF and an E12 resistor.

c) Calculate the attenuation at 50 Hz in dB for an ideal first-order and second-order filter with $f_c = 2.5$ Hz, and for your filter.

d) Calculate the remaining interference at the ADC input with your filter and compare it with LSB/2.

e) How long does the filter output need to settle within 1 % after a step of the sensor signal?

f) Check the result in Falstad (signal 2 V DC with 0.52 V at 50 Hz).

Solution 7.5

a) Nyquist: only frequencies below $f_s/2 = 2.5$ Hz can be sampled without aliasing, so $f_c \le 2.5$ Hz. The temperature changes much more slowly than that.

b) $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot 10\,\mu\text{F}) = 6.37\,\text{k}\Omega \to 6.8\,\text{k}\Omega$ (rounded up, so that f_c stays below $f_s/2$): $f_c = 1/(2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}) = 2.34$ Hz. (62 kΩ with 1 µF, $f_c = 2.57$ Hz, slightly above $f_s/2$, would also be acceptable.)

c) Ideal first order: $1/\sqrt{1+20^2} = 0.0499$, −26 dB; second order: $1/\sqrt{1+20^4} = 0.0025$, −52 dB. With $f_c = 2.34$ Hz: $1/\sqrt{1+(50/2.34)^2} = 0.047$, −26.6 dB.

d) Before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52$ V; after it: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250$ mV (0.05 LSB). Sufficient.

e) $\tau = RC = 68$ ms; 1 %: $t = \tau \ln 100 = 4.6\,\tau = 0.31$ s, i.e. about 1.5 sample periods.

f) ▶ Falstad: `examples/filters/filter_antialias.txt`
after about 0.6 s `Vout_1st` shows 2.00 V with 25 mV of 50 Hz ripple; the Sallen-Key output (91 kΩ, 1 µF, 470 nF) leaves only 1.4 mV.

Exercise 7.6: Twin-T notch

a) Dimension a twin-T notch for 50 Hz with $C = 100$ nF.

b) Its transfer function is $\underline{H} = \dfrac{1 - x^2}{1 - x^2 + j4x}$ with $x = f/f_0$. Calculate $|H|$ at 5 Hz and at 50.5 Hz.

c) Compare with the first-order low-pass with $f_c = 2.34$ Hz. Which filter would you use in the capstone?

d) Check a) and b) in Falstad.

Solution 7.6

a) $R = 1/(2\pi \cdot 50\,\text{Hz} \cdot 100\,\text{nF}) = 31.8\,\text{k}\Omega$, $R/2 = 15.9\,\text{k}\Omega$, $2C = 200\,\text{nF}$.

b) 5 Hz: $x = 0.1$, $|H| = 0.99/\sqrt{0.99^2 + 0.4^2} = 0.927$.
50.5 Hz: $x = 1.01$, $|H| = 0.0201/\sqrt{0.0201^2 + 4.04^2} = 0.0050$.

c) At 50 Hz the notch is much better (-46 dB at 50.5 Hz and more at exactly 50 Hz), but it attenuates the signal at 5 Hz as well, it needs precise components, and it does nothing against other frequencies above $f_s/2$. The anti-aliasing low-pass is required anyway; the notch is at most an addition.

d) ▶ Falstad: `examples/filters/filter_twin_t.txt`
`Vout_5Hz` $= 0.927$ V, `Vout_50Hz` $= 0.45$ mV (input 1 V).

8 Sample and Hold

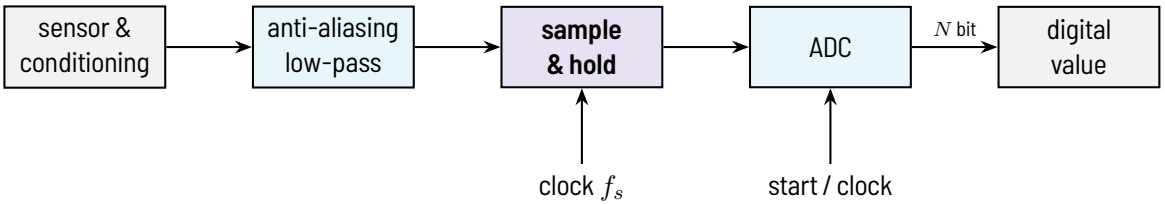
8.1 Why Sample and Hold?

The previous chapters produced a clean, filtered analog voltage. The next step is to turn it into numbers. A digital system cannot store a continuous signal; it takes one value per sampling period and works with this sequence of numbers. The circuit that takes these values is the *sample and hold* (S&H) stage. It sits between the anti-aliasing filter and the analog-to-digital converter (ADC).

The S&H stage has two jobs. First, it defines the *sampling instant*: the moment at which the signal is measured. Second, it keeps the measured voltage constant while the converter works on it. Many converters need some time for a conversion and give wrong results if their input changes during this time. A successive-approximation converter, for example, compares the input with a sequence of trial voltages; if the input moves in between, the result is a mixture of different input values.

In the lab and in the capstone task the sampling frequency is very low, $f_s = 5$ Hz, so that we can follow every sample by eye on the Falstad scope and on the LEDs. The principles are the same as in a converter with millions of samples per second, only the time scales differ [6, 7].

Position in the signal chain



- the S&H stage **defines the sampling instant** $t_k = k\,T_s$, $T_s = 1/f_s$
- it keeps the voltage **constant while the ADC converts**
- capstone: $f_s = 5$ Hz, 3-bit flash converter, full scale 4 V

What goes wrong without S&H

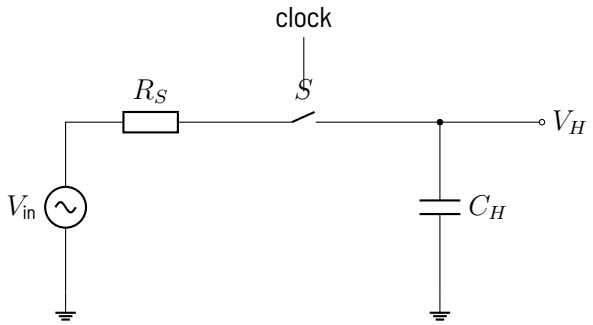
- the input changes during the conversion time t_{conv}
- the error must stay below $\frac{1}{2}$ LSB: $\left|\frac{dV}{dt}\right|_{\max} t_{\text{conv}} < \frac{1}{2}\,\text{LSB}$
- sine $V(t) = \hat{V}\sin(2\pi f t)$: maximum slope $2\pi f \hat{V}$
- example: 12-bit ADC, 4 V full scale (LSB $= 0.98$ mV), $t_{\text{conv}} = 10$ µs, $\hat{V} = 2$ V
  - $f < \frac{\text{LSB}/2}{2\pi \hat{V} t_{\text{conv}}} = 3.9$ Hz without S&H
  - with S&H only the much shorter *aperture time* counts (nanoseconds)
- a **flash** converter decides in one step, but the digital outputs still have to be read and displayed at a defined instant

The example shows how strict the requirement is. A 12-bit converter without S&H can only follow signals of a few hertz, even though its conversion time is short. With an S&H stage in front, the relevant time is the uncertainty of the sampling instant, which is many orders of magnitude shorter. In our capstone the flash converter is fast, but the S&H stage still gives us one well-defined value per 200 ms, so that the display does not flicker and the digital code belongs to a known instant.

8.2 The Track-and-Hold Principle

The simplest S&H stage is a switch and a capacitor. While the switch is closed, the capacitor voltage follows the input (track mode). When the switch opens, the capacitor keeps its charge and therefore its voltage (hold mode). Strictly speaking this circuit is a *track-and-hold* circuit, because it follows the input during the whole time the switch is closed. A true sample-and-hold would take a very short sample. In practice the two names are used for the same circuit.

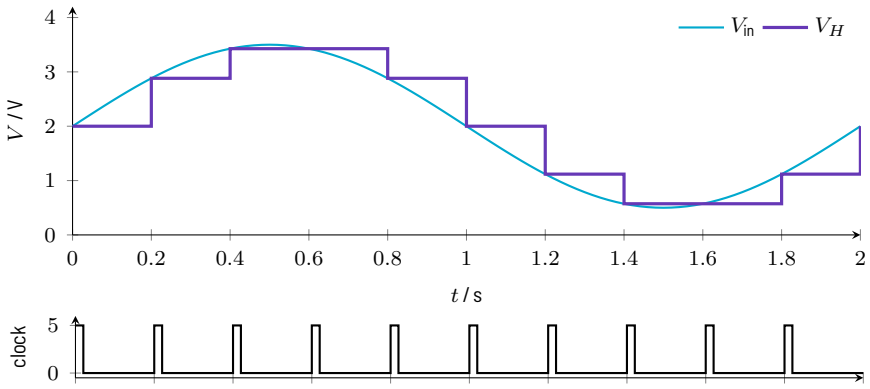
Switch and hold capacitor



- clock high: switch closed, C_H follows V_{in} (**track**)
- clock low: switch open, C_H keeps its charge (**hold**)
- capstone clock: 5 Hz, duty cycle $10\,\%$: track 20 ms, hold 180 ms

R_S is the total resistance in the charging path. It contains the output resistance of the source, the on-resistance of the switch and, in our circuits, a resistor of $1\,\text{k}\Omega$ that limits the charging current of the op-amp that drives the capacitor.

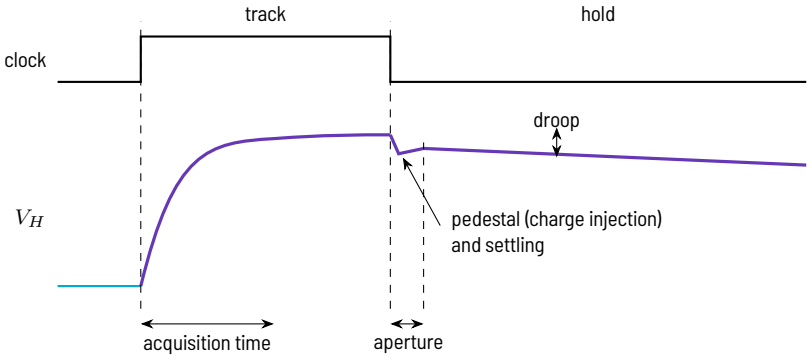
Timing diagram



- the output is a **staircase**: one step per sampling period
- the value of each step is V_{in} at the **end** of the track phase

The plot shows $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\,\sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$ sampled at $5\,\text{Hz}$. The clock in the timing row is high for one tenth of each period. The steps of V_H start at the falling clock edge, when the switch opens. (In the plot the short track phases are not resolved.)

Terms



- **acquisition time**: track time needed until V_H is within the error band (e.g. $\frac{1}{2}$ LSB)
- **aperture time**: delay between the hold command and the real opening of the switch; its variation is the **aperture jitter**
- **pedestal**: step at the transition to hold; **droop**: slow change during hold

Falstad: basic S&H

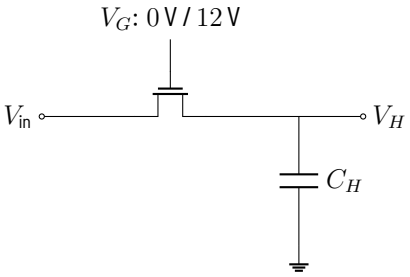
- $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\,\sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$, $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$
- clock rail: square wave 5 Hz, 0 V … 5 V, duty cycle 0.1
- held values: 2.094 V, 2.956 V, 3.452 V, 3.394 V, 2.804 V
- the scope shows V_{in}, V_H (label `Vhold`) and the clock

▶ Falstad: `examples/samplehold/sh_basic.txt` The held values are the input voltages at the end of each track window, $t = 0.02\,\text{s}, 0.22\,\text{s}, \ldots$ The simulation shows them about 5 mV lower, because the capacitor lags behind a rising input by roughly the slope times the time constant ($4.7\,\text{V/s} \cdot 1$ ms).

8.3 Analog Switches

The switch of an S&H stage is electronic. It must close and open on a logic signal, have a low resistance when closed and an extremely high resistance when open, and it must not disturb the stored charge. The standard solution is a MOSFET, or two complementary MOSFETs in parallel (transmission gate) [3, 1].

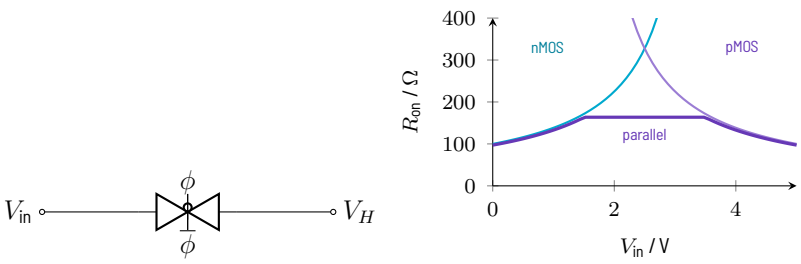
The MOSFET as a switch



- n-channel MOSFET: conducts if $V_{GS} = V_G - V_{\text{in}} > V_{th}$
- on-resistance $R_{\text{on}} \approx \frac{1}{k\,(V_{GS} - V_{th})}$: depends on V_{in}!
- near $V_{\text{in}} \approx V_G - V_{th}$ the switch does not close any more
- off: leakage currents in the pA ...nA range

Source and drain of a MOSFET switch are interchangeable; the current can flow in both directions. The gate voltage must be higher than the largest input voltage plus the threshold voltage. If the input rises, the gate-source voltage shrinks and the on-resistance grows. A single n-channel switch is therefore good for signals near ground, a single p-channel switch for signals near the positive supply.

CMOS transmission gate



- nMOS and pMOS in parallel, driven with complementary signals
- nMOS conducts well near 0 V, pMOS near V_{DD}: together almost constant R_{on}
- integrated as CD4066, 74HC4066 (4 switches, $R_{\text{on}} \approx 50\,\Omega$ ... $300\,\Omega$), DG4xx

The analog switch in Falstad

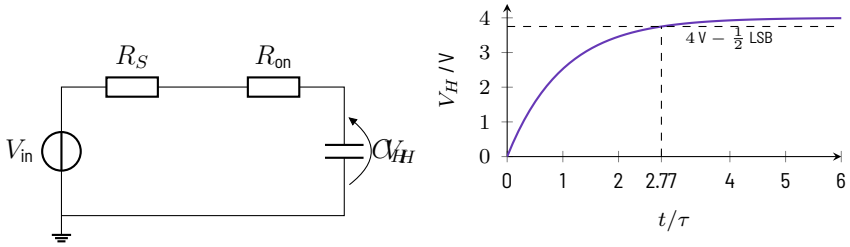
- element *Analog Switch* (models a FET switch / transmission gate)
  - closed if the control input is above 2.5 V
  - $R_{\text{on}} = 20\,\Omega$, $R_{\text{off}} = 10\,\text{G}\Omega$ (defaults, edit with a double click)
- no charge injection, no leakage: add them as extra elements if you want to see their effect
- in our circuits: $R_S = 1\,\text{k}\Omega$ in series
  - limits the current from the op-amp output into the empty capacitor
  - isolates the op-amp from the large capacitive load (stability)
  - real lab switches have R_{on} of some $100\,\Omega$ anyway

The Falstad switch is ideal apart from its two resistances. A real switch adds charge injection, leakage and a small coupling capacitance across the open switch. We add these effects one at a time with discrete elements (a capacitor from the clock to the hold node, a current source in parallel to C_H) to see their influence separately.

8.4 Designing the Hold Capacitor

The hold capacitor is the central design decision. Two requirements pull in opposite directions. A small capacitor charges quickly in the track phase, but it loses its voltage quickly through leakage currents in the hold phase. A large capacitor holds its voltage well, but it may not be charged completely in the short track window. Both requirements give a limit for C_H; any value in between works, and we choose one with margin on both sides.

Acquisition: charging in the track window



- $V_H(t) = V_{in}\,(1 - e^{-t/\tau})$ with $\tau = (R_S + R_{on})\,C_H$
- error after time t: $V_{in}\,e^{-t/\tau}$; worst case: step over the full scale V_{FS}

Settling to $\frac{1}{2}$ LSB

- requirement: $V_{FS}\,e^{-t/\tau} \le \frac{1}{2}\,\text{LSB} = \frac{V_{FS}}{2^{N+1}}$

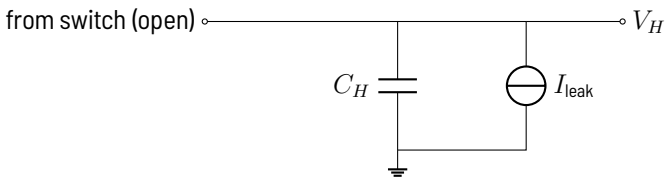
Acquisition time for N bit

$$t_{acq} = \tau\,\ln\left(2^{N+1}\right) = (N+1)\,\ln 2 \cdot \tau \approx 0.69\,(N+1)\,\tau$$

- $N = 3$: $t_{acq} = 2.77\,\tau$; $N = 8$: $6.2\,\tau$; $N = 12$: $9.0\,\tau$; $N = 16$: $11.8\,\tau$
- design rule: $t_{acq} \le t_{track} \quad\Rightarrow\quad C_H \le \dfrac{t_{track}}{(R_S + R_{on})\,\ln 2^{N+1}}$
- example: $R_S + R_{on} = 1.02\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$: $\tau = 1.02\,\text{ms}$, $t_{acq} = 2.83\,\text{ms} \ll 20\,\text{ms}$

▶ Falstad: `examples/samplehold/sh_acquisition.txt` The Falstad example charges $C_H = 1\,\mu\text{F}$ from a $4\,\text{V}$ source through $1\,\text{k}\Omega$ plus the $20\,\Omega$ of the switch. After one time constant ($t = 1.02\,\text{ms}$) the capacitor shows $2.53\,\text{V}$ ($63\,\%$), after $t_{acq} = 2.83\,\text{ms}$ it reaches $3.75\,\text{V}$, i.e. $\frac{1}{2}$ LSB below the final value, and at the end of the $20\,\text{ms}$ window it holds $4.00\,\text{V}$. Use a time constant of a few percent of the track window; the remaining time is reserve for the settling of the op-amp and for tolerances.

Droop: losing charge in the hold phase



- leakage currents: switch (off), capacitor dielectric, input bias current of the buffer, board surface

Droop

$$\frac{dV_H}{dt} = \frac{I_{leak}}{C_H} \quad\Rightarrow\quad \Delta V_{droop} = \frac{I_{leak}\,t_{hold}}{C_H} \quad\Rightarrow\quad C_H \ge \frac{I_{leak}\,t_{hold}}{\Delta V_{max}}$$

- t_{hold}: until the conversion is finished; worst case the whole period $T_s - t_{track}$

Too small, right, too large

- Falstad: $V_{in} = 3\,\text{V}$, $R_S = 1\,\text{k}\Omega$, leakage $I_{leak} = 50\,\text{nA}$ (exaggerated)

	$C_H = 10\,\text{nF}$	$C_H = 1\,\mu\text{F}$	$C_H = 100\,\mu\text{F}$
τ	$10\,\mu\text{s}$	$1.02\,\text{ms}$	$102\,\text{ms}$
charged in $20\,\text{ms}$?	yes	yes	only $18\,\%$ per sample
droop in $180\,\text{ms}$	$0.9\,\text{V}$	$9\,\text{mV}$	$0.09\,\text{mV}$
Falstad	$3.00\,\text{V} \to 2.10\,\text{V}$	$3.00\,\text{V} \to 2.99\,\text{V}$	$0.534\,\text{V}$, ..., $1.876\,\text{V}$ (5th)

▶ Falstad: `examples/samplehold/sh_capsize.txt` The too large capacitor is a low-pass filter rather than a hold stage: after every sample it has covered only $18\,\%$ of the remaining distance to the input voltage, $1 - e^{-20\,\text{ms}/102\,\text{ms}} = 0.178$. After five samples it shows $3\,\text{V}\,(1 - 0.822^5) = 1.88\,\text{V}$ instead of $3\,\text{V}$. The too small capacitor follows the input perfectly but loses $5\,\text{V/s}$ in the hold phase. A real leakage current is much smaller (pA to nA), so the lower limit for C_H is usually far away; we exaggerated it to make the effect visible.

Choosing C_H: the window

- upper limit from the acquisition, lower limit from the droop:

Hold capacitor

$$\frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}} \;\le\; C_H \;\le\; \frac{t_{\text{track}}}{(R_S + R_{\text{on}})\,\ln 2^{N+1}}$$

- capstone values: $I_{\text{leak}} = 10\,\text{nA}$, $\Delta V_{\text{max}} = \frac{1}{10}\,\text{LSB} = 50\,\text{mV}$, $t_{\text{hold}} = 180\,\text{ms}$, $t_{\text{track}} = 20\,\text{ms}$, $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$
  - $36\,\text{nF} \le C_H \le 7.1\,\mu\text{F} \quad \Rightarrow \quad C_H = 1\,\mu\text{F}$ (geometric middle, film capacitor)
- choose the capacitor type: polypropylene or polystyrene film (low leakage, low dielectric absorption), C0G ceramic for small values; no electrolytic capacitors

Choosing a value in the geometric middle of the allowed window gives the same safety factor (here about 7) against both limits. Electrolytic capacitors are unsuitable: their leakage current is in the microampere range, and their capacitance is poorly defined. Dielectric absorption is a memory effect of the insulating material: after a fast change of voltage a part of the charge comes back slowly from the dielectric, and the held voltage creeps towards the previous value. Film capacitors with polypropylene or polystyrene dielectric show very little of it.

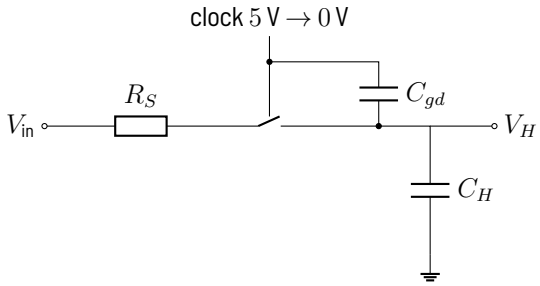
Design card: Sample and Hold

- structure: input follower $\to R_S$ + switch $\to C_H$ to ground $\to$ output follower
- clock: f_s, short duty cycle; track $t_{\text{track}} = D\,T_s$ (capstone: $5\,\text{Hz}$, $10\,\%$: $20\,\text{ms}$)
- charging: $\tau = (R_S + R_{\text{on}})\,C_H$, settled to $\frac{1}{2}\,\text{LSB}$ after $\tau\,\ln 2^{N+1}$ ($N = 3$: $2.77\,\tau$) $\le t_{\text{track}}$
- droop: $\Delta V = I_{\text{leak}}\, t_{\text{hold}}/C_H \le$ fraction of an LSB
- pedestal: $\Delta V = C_{gd}\,\Delta V_{\text{clk}}/(C_H + C_{gd})$; larger C_H helps
- capstone choice: $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$ film: $t_{\text{acq}} = 2.8\,\text{ms}$, droop $1.8\,\text{mV}$ at $10\,\text{nA}$

8.5 Errors of a Real Sample and Hold

Besides incomplete charging and droop, a real S&H stage shows a number of smaller errors. They matter in fast and high-resolution converters; in our slow 3-bit capstone they are negligible, but it is good to know where they come from and which design parameter influences them [6, 4].

Charge injection (pedestal)



- the gate of the switch is coupled to the hold node through C_{gd} (and the channel charge)
- when the switch opens, the clock step ΔV_{clk} is divided by C_{gd} and C_H:
  $\Delta V_{\text{ped}} = -\Delta V_{\text{clk}} \dfrac{C_{gd}}{C_{gd} + C_H} \approx -\dfrac{Q_{\text{inj}}}{C_H}$
- $C_{gd} = 10\,\text{pF}$, $5\,\text{V}$: $C_H = 1\,\text{nF}$: $-49.5\,\text{mV}$; $C_H = 100\,\text{nF}$: $-0.5\,\text{mV}$

▶ Falstad: `examples/samplehold/sh_pedestal.txt` The pedestal is a constant offset as long as the clock amplitude and C_{gd} are constant, so it can be calibrated. Its signal-dependent part (the channel charge depends on V_{GS}) cannot. Remedies are a larger C_H, small switches, a transmission gate (the charges of the n- and p-channel transistor partly cancel), and dummy switches driven with the inverted clock. In Falstad we model only the coupling capacitance; the labels `Vhold_1n` and `Vhold_100n` show $1.9505\,\text{V}$ and $1.9995\,\text{V}$ after the switch has opened, for an input of $2\,\text{V}$.

Feedthrough, aperture and jitter

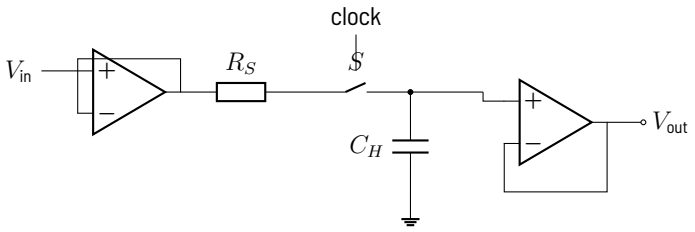
- **feedthrough**: the open switch has a capacitance C_{ds}; the input still reaches C_H: $\Delta V_H = \Delta V_{\text{in}} \frac{C_{\text{ds}}}{C_{\text{ds}} + C_H}$
- **aperture delay** t_a: the switch opens a little after the clock edge; constant, only shifts the sampling instant
- **aperture jitter** t_j: random variation of the sampling instant
  - error $\Delta V = \frac{dV}{dt}\, t_j$; sine: $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j$
  - requirement $\Delta V_{\text{max}} < \frac{1}{2}\,\text{LSB}$: $t_j < \dfrac{1}{2\pi f\, 2^{N+1}} \cdot \dfrac{V_{\text{FS}}}{\hat{V}}$
  - example: 12 bit, full-scale sine of $1\,\text{MHz}$: $t_j < 19\,\text{ps}$
- **dielectric absorption**: held voltage creeps back towards the previous value

For a full-scale sine $\hat{V} = V_{\text{FS}}/2$ the jitter limit is $t_j < 1/(2\pi f\, 2^N)$. Jitter is the reason why fast converters need clean, low-noise clock sources. For our $5\,\text{Hz}$ system a jitter of a millisecond is harmless (exercise 8.6).

8.6 The Buffered Sample and Hold

The bare switch and capacitor only work with an ideal source and an ideal load. A real source has an internal resistance, for example the output resistance of a passive anti-aliasing filter, and a real load draws current, for example the input resistance of the converter or the reference ladder of a flash converter. The cure is a voltage follower on each side of the hold capacitor (chapter 6).

Follower – switch – capacitor – follower



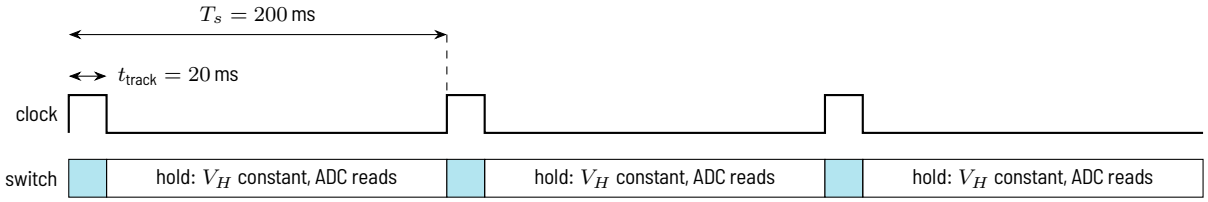
- **input buffer**: the charging current comes from the op-amp, not from the signal source
- **output buffer**: draws (almost) no current from C_H; only its input bias current adds to the leakage

Why both buffers are needed

- without buffers: source $R_i = 10\,\text{k}\Omega$, load $R_L = 100\,\text{k}\Omega$ (e.g. ADC input), $V_{\text{in}} = 3\,\text{V}$
  - track: $\tau = (R_i \parallel R_L)\, C_H = 9.1\,\text{ms}$, final value $3\,\text{V} \cdot \frac{100}{110} = 2.73\,\text{V}$
  - after $20\,\text{ms}$: $2.425\,\text{V}$
  - hold: discharge through R_L, $\tau = 100\,\text{ms}$: $0.401\,\text{V}$ after $180\,\text{ms}$
- with buffers: $3.000\,\text{V}$ during the whole period

▶ Falstad: `examples/samplehold/sh_buffered.txt` The unbuffered circuit fails in three ways at once: the source and the load form a voltage divider, the source resistance makes the charging slow, and the load discharges the capacitor in the hold phase. With buffers the source only sees the high input resistance of the first follower, and the capacitor only sees the high input resistance of the second follower. For the output buffer choose an op-amp with FET input (bias current in the pA range); its bias current is part of I_{leak}.

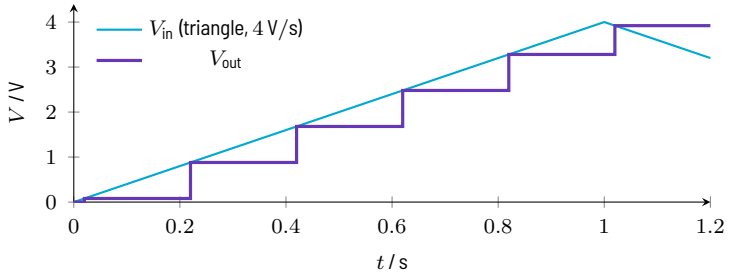
The sampling clock



- square wave with a short duty cycle $D = t_{\text{track}}/T_s$
- Falstad: rail with waveform *square*, $5\,\text{Hz}$, amplitude $2.5\,\text{V}$, offset $2.5\,\text{V}$, duty cycle 0.1
- in hardware: 555 timer or microcontroller output; 0/5 V logic level drives the switch
- the ADC result is read in the hold phase (e.g. in the middle or at the end)

A short track phase keeps the output constant for most of the period. The track phase must not be too short either: it must be longer than the acquisition time. With $D = 0.1$ at $5\,\text{Hz}$ we have $20\,\text{ms}$ for charging and $180\,\text{ms}$ of stable output.

Sampling a ramp



- buffered S&H, triangle $0\,\text{V} \ldots 4\,\text{V}$ at $0.5\,\text{Hz}$: steps of $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8\,\text{V}$
- Falstad: $0.08\,\text{V}$, $0.88\,\text{V}$, $1.68\,\text{V}$, $2.48\,\text{V}$, $3.28\,\text{V}$

▶ Falstad: `examples/samplehold/sh_staircase.txt` A step of $0.8\,\text{V}$ is larger than one LSB of the 3-bit converter ($0.5\,\text{V}$): consecutive samples skip codes. This is no error of the S&H stage. The input simply changes too fast for the sampling rate; such fast signals have to be removed by the anti-aliasing filter (see the sampling theorem in the next chapter).

Practical hints

- op-amps: rail-to-rail output for the full range $0\,\text{V} \ldots 4\,\text{V}$ on a single $5\,\text{V}$ supply, FET input for the output buffer
- guard ring around the hold node on the PCB: surface leakage
- integrated S&H amplifiers (LF398, AD783) contain both buffers and the switch; only C_H is external
- in most modern ADCs (SAR, sigma-delta) the S&H is built in: a switched capacitor array at the input
  - the driving op-amp must recharge this capacitor in the acquisition time: the same calculation as above

The LF398 is a classic monolithic S&H amplifier; its data sheet contains curves for acquisition time and droop rate as functions of C_H that are a good illustration of this chapter. Modern SAR converters in microcontrollers sample with an internal capacitor of a few picofarads; the source resistance must be low enough that this capacitor charges to $\frac{1}{2}$ LSB within the programmed sampling time, which is exactly the calculation $t_{\text{acq}} = \tau \ln 2^{N+1}$.

8.7 Exercises

Exercise 8.1: Choosing the hold capacitor for $f_s = 5\,\text{Hz}$

A buffered S&H stage samples at $f_s = 5\,\text{Hz}$ with a clock duty cycle of $10\,\%$. The charging path consists of $R_S = 1\,\text{k}\Omega$ and a switch with $R_{\text{on}} = 20\,\Omega$. The total leakage current at the hold node (switch, capacitor, buffer bias) is at most $I_{\text{leak}} = 10\,\text{nA}$. The stage drives a 3-bit converter with $V_{\text{FS}} = 4\,\text{V}$.

a) Calculate the track and hold times.

b) The capacitor must settle to $\frac{1}{2}$ LSB for a full-scale step within the track phase. Find the upper limit for C_H.

c) The droop during the hold phase must stay below $\frac{1}{10}$ LSB. Find the lower limit for C_H.

d) Choose C_H and calculate acquisition time and droop for your choice.

e) Check the charging curve in Falstad.

Solution 8.1

a) $T_s = 1/f_s = 200\,\text{ms}$, $t_{\text{track}} = 0.1 \cdot T_s = 20\,\text{ms}$, $t_{\text{hold}} = 180\,\text{ms}$.

b) LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$. Settling to $\frac{1}{2}$ LSB needs $t = \tau \ln 2^4 = 2.77\,\tau \le 20\,\text{ms}$, so $\tau \le 7.21\,\text{ms}$ and $C_H \le 7.21\,\text{ms}/1020\,\Omega = 7.1\,\mu\text{F}$.

c) $\Delta V_{\text{max}} = 50\,\text{mV}$: $C_H \ge I_{\text{leak}}\, t_{\text{hold}}/\Delta V_{\text{max}} = 10\,\text{nA} \cdot 0.18\,\text{s}/50\,\text{mV} = 36\,\text{nF}$.

d) The window is $36\,\text{nF}$ …$7.1\,\mu\text{F}$; the geometric middle is about $0.5\,\mu\text{F}$. We choose $C_H = 1\,\mu\text{F}$ (film capacitor). Then $\tau = 1.02\,\text{ms}$, $t_{\text{acq}} = 2.77\,\tau = 2.83\,\text{ms}$ (seven times faster than required) and $\Delta V_{\text{droop}} = 10\,\text{nA} \cdot 0.18\,\text{s}/1\,\mu\text{F} = 1.8\,\text{mV}$ (28 times smaller than allowed).

e) ▶ Falstad: `examples/samplehold/sh_acquisition.txt`
With a $4\,\text{V}$ input, `Vhold` shows $2.53\,\text{V}$ after $1.02\,\text{ms}$, $3.75\,\text{V}$ after $2.83\,\text{ms}$ and $4.00\,\text{V}$ at the end of the track phase. (Pause the simulation and use the scope, or reduce the simulation speed.)

Exercise 8.2: Droop and a too large capacitor

The S&H stage of exercise 8.1 has an (exaggerated) leakage current of $50\,\text{nA}$. The input is a constant $3\,\text{V}$.

a) Calculate the droop rate and the voltage at the end of the first hold phase for $C_H = 10\,\text{nF}$ and for $C_H = 1\,\mu\text{F}$.

b) With $C_H = 100\,\mu\text{F}$, which fraction of the remaining difference is charged in one track window? Calculate the held voltage after the first and after the fifth sample (leakage negligible).

c) Compare with Falstad.

Solution 8.2

a) $C_H = 10\,\text{nF}$: $dV/dt = 50\,\text{nA}/10\,\text{nF} = 5\,\text{V/s}$, droop in $180\,\text{ms}$: $0.9\,\text{V}$, the held voltage falls from $3.00\,\text{V}$ to $2.10\,\text{V}$. $C_H = 1\,\mu\text{F}$: $50\,\text{mV/s}$, droop $9\,\text{mV}$.

b) $\tau = 1.02\,\text{k}\Omega \cdot 100\,\mu\text{F} = 102\,\text{ms}$, fraction $a = 1 - e^{-20/102} = 0.178$. After n samples $V_n = 3\,\text{V}\,(1-(1-a)^n)$: $V_1 = 0.534\,\text{V}$, $V_5 = 3\,\text{V}\,(1-0.822^5) = 1.876\,\text{V}$.

c) ▶ Falstad: `examples/samplehold/sh_capsize.txt`
`V_10n` drops from $3.00\,\text{V}$ to $2.10\,\text{V}$ in every hold phase, `V_1u` loses $9\,\text{mV}$ ($3.00\,\text{V} \to 2.99\,\text{V}$), `V_100u` climbs in steps: $0.534\,\text{V}$ after the first, $1.876\,\text{V}$ after the fifth sample.

Exercise 8.3: Unbuffered sample and hold

A switch and $C_H = 1\,\mu\text{F}$ are connected directly between a source ($V_{\text{in}} = 3\,\text{V}$, internal resistance $R_i = 10\,\text{k}\Omega$) and a load $R_L = 100\,\text{k}\Omega$. Clock: $5\,\text{Hz}$, $10\,\%$. Neglect R_{on}.

a) Draw the equivalent circuit for the track phase and calculate its final value and time constant.

b) Calculate V_H at the end of the first track phase and at the end of the first hold phase.

c) How does the buffered circuit change the result? Check in Falstad.

Solution 8.3

a) Thevenin equivalent seen by C_H: $V_{\text{th}} = 3\,\text{V} \cdot \frac{100}{110} = 2.727\,\text{V}$, $R_{\text{th}} = R_i \parallel R_L = 9.09\,\text{k}\Omega$, $\tau = 9.09\,\text{ms}$.

b) End of track: $V_H = 2.727\,\text{V}\,(1 - e^{-20/9.09}) = 2.425\,\text{V}$. Hold: C_H discharges through R_L with $\tau = 100\,\text{ms}$: $V_H = 2.425\,\text{V} \cdot e^{-1.8} = 0.401\,\text{V}$.

c) The input follower removes the divider and the slow charging, the output follower removes the discharge: $V_{\text{out}} = 3.000\,\text{V}$ throughout.

▶ Falstad: `examples/samplehold/sh_buffered.txt`

`Vout_unbuf` shows $2.425\,\text{V}$ and $0.401\,\text{V}$, `Vout_buf` $3.000\,\text{V}$.

Exercise 8.4: Pedestal error

The gate of the switch has a coupling capacitance $C_{gd} = 10\,\text{pF}$ to the hold node. The clock falls from $5\,\text{V}$ to $0\,\text{V}$ when the switch opens.

a) Calculate the pedestal for $C_H = 1\,\text{nF}$ and $C_H = 100\,\text{nF}$.

b) Which C_H is needed for a pedestal below $\frac{1}{2}$ LSB of a 12-bit converter with $V_{\text{FS}} = 4\,\text{V}$?

c) Is the pedestal a problem for the 3-bit capstone with $C_H = 1\,\mu\text{F}$?

Solution 8.4

a) $\Delta V = -5\,\text{V} \cdot \frac{10\,\text{pF}}{1.01\,\text{nF}} = -49.5\,\text{mV}$; with $100\,\text{nF}$: $-0.50\,\text{mV}$.

▶ Falstad: `examples/samplehold/sh_pedestal.txt`: $2\,\text{V}$ input, held values $1.9505\,\text{V}$ and $1.9995\,\text{V}$.

b) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{13} = 0.49\,\text{mV}$: $C_H \geq 5\,\text{V} \cdot 10\,\text{pF}/0.49\,\text{mV} = 0.1\,\mu\text{F}$.

c) $\Delta V = 50\,\mu\text{V}$, i.e. 10^{-4} of an LSB of $0.5\,\text{V}$: no problem.

Exercise 8.5: Ramp input and step size

A buffered S&H stage ($f_s = 5\,\text{Hz}$) samples a triangle that rises from $0\,\text{V}$ to $4\,\text{V}$ in $1\,\text{s}$.

a) Calculate the height of the steps of the output staircase.

b) The sampling instants are at the end of the track windows, $t_k = 20\,\text{ms} + k \cdot 200\,\text{ms}$. List the first five held values.

c) What is the maximum slope of the input if consecutive samples may differ by at most one LSB of the 3-bit converter? What does this mean for a sine with an amplitude of $2\,\text{V}$?

Solution 8.5

a) $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8\,\text{V}$.

b) $V(t_k) = 4\,\text{V/s} \cdot t_k$: $0.08\,\text{V}$, $0.88\,\text{V}$, $1.68\,\text{V}$, $2.48\,\text{V}$, $3.28\,\text{V}$.

▶ Falstad: `examples/samplehold/sh_staircase.txt`: `Vout` shows the same values ($4\,\text{mV}$ lower, because C_H lags by $\tau \cdot 4\,\text{V/s}$).

c) $0.5\,\text{V}/0.2\,\text{s} = 2.5\,\text{V/s}$. Sine: $2\pi f \cdot 2\,\text{V} \leq 2.5\,\text{V/s}$ gives $f \leq 0.2\,\text{Hz}$. Faster signals are still sampled correctly as long as $f < f_s/2$ (next chapter), but the codes of neighbouring samples jump.

Exercise 8.6: Aperture jitter

The $5\,\text{Hz}$ clock of the capstone comes from a microcontroller whose software has a timing uncertainty of $t_j = 1\,\text{ms}$. The input is a sine with $\hat{V} = 2\,\text{V}$ and $f = 2.5\,\text{Hz}$ (the highest frequency the anti-aliasing filter passes).

a) Calculate the maximum sampling error caused by the jitter.

b) Compare with $\frac{1}{2}$ LSB of the 3-bit converter.

c) Which jitter would a 16-bit converter tolerate for the same signal ($V_{\text{FS}} = 4\,\text{V}$)?

Solution 8.6

a) $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j = 2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V} \cdot 1\,\text{ms} = 31\,\text{mV}$.

b) $\frac{1}{2}\,\text{LSB} = 250\,\text{mV}$: the jitter error is eight times smaller, harmless.

c) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{17} = 30.5\,\mu\text{V}$: $t_j < 30.5\,\mu\text{V}/(2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V}) = 0.97\,\mu\text{s}$. A software-timed clock is no longer good enough; a hardware timer is needed.

9 Analog-to-Digital Conversion

9.1 Sampling and Aliasing

An analog-to-digital converter (ADC) does two things: it samples the signal in time, and it quantizes each sample in amplitude. Both steps lose information, and both losses can be controlled by design. This section deals with sampling: how fast must we sample, and what happens if the signal contains frequencies that are too high for the sampling rate? The answer is the sampling theorem of Nyquist and Shannon [10]. Its practical consequence is a filter that every sampled system needs in front of the S&H stage: the anti-aliasing low-pass.

The sampling theorem

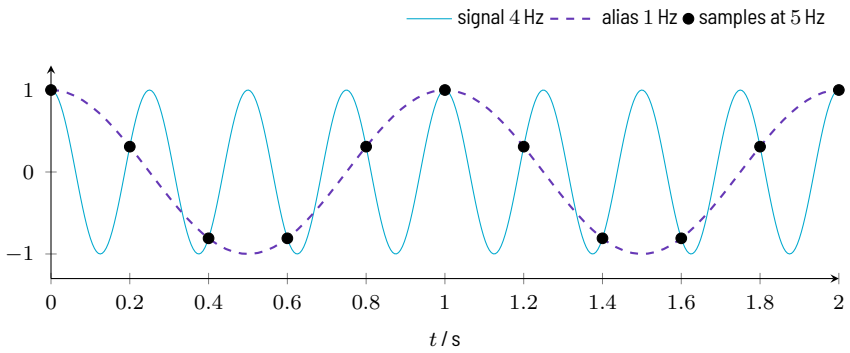
- a signal is sampled at the instants $t_k = k\,T_s$ with the sampling frequency $f_s = 1/T_s$

Sampling theorem (Nyquist–Shannon)
A signal that contains no frequencies at or above $f_s/2$ is completely determined by its samples taken at the rate f_s. $f_s/2$ is called the **Nyquist frequency**.

- the condition is $f_{\text{max}} < f_s/2$, i.e. more than two samples per period of the fastest component
- it applies to **every** component of the input, also to noise and interference
- capstone: $f_s = 5$ Hz $\Rightarrow$ only components below 2.5 Hz are allowed

The theorem says that the original signal can be reconstructed from its samples, provided that the band limit is respected. If the condition is violated, the samples are still taken, but they are the samples of a different, lower frequency. No processing after the sampler can tell the two cases apart.

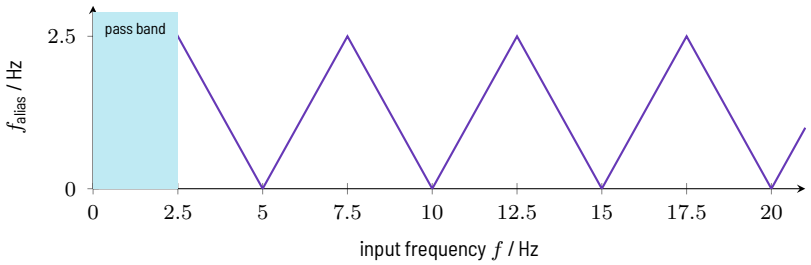
Aliasing



- a 4 Hz cosine sampled at 5 Hz gives the same samples as a 1 Hz cosine

Alias frequency
$f_{\text{alias}} = |f - k\,f_s|$, with the integer k that makes $f_{\text{alias}} \le f_s/2$

Folding



- every frequency is folded into the range $0 \ldots f_s/2$
- $f_s = 5$ Hz: 4 Hz $\to 1$ Hz; 12 Hz $\to 2$ Hz; 49 Hz $\to 1$ Hz; 50 Hz $\to 0$ Hz
- after sampling an alias cannot be distinguished from a real signal component

The 50 Hz problem

- mains interference (50 Hz) is coupled into every sensor cable
- capstone: $f_s = 5\,\text{Hz} \Rightarrow f_{\text{alias}} = |50\,\text{Hz} - 10 \cdot 5\,\text{Hz}| = 0\,\text{Hz}$
  - every sample sees the 50 Hz sine at the same phase
  - the result is a **constant DC offset** that biases every reading
  - its size depends on the (random) phase: anywhere between $-\hat{V}$ and $+\hat{V}$
  - nothing on the display looks wrong: the error is invisible
- with $f_s = 4.9\,\text{Hz}$ the alias is $|50\,\text{Hz} - 10 \cdot 4.9\,\text{Hz}| = 1\,\text{Hz}$: a slow beat
- **the anti-aliasing low-pass in front of the S&H stage is mandatory**

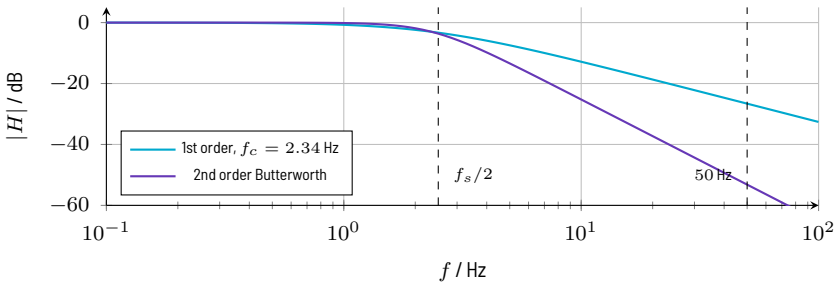
Because f_s is an integer fraction of the mains frequency, the interference is sampled at the same point of its period every time. The digital system sees a perfectly stable value, shifted by an unknown amount. With a slightly different sampling rate the same interference would show up as a slow oscillation of the readings, which is at least visible. The only remedy is to remove the interference before sampling.

Falstad: sampling 50 Hz hum

- input: 2 V DC (the wanted signal) plus 0.5 V at 50 Hz
- two S&H stages ($R = 1\,\text{k}\Omega$, $C_H = 100$ nF, track window 1 ms)
  - $f_s = 5$ Hz: every sample is 2.14 V: an offset of 0.14 V
  - $f_s = 4.9$ Hz: 2.14 V, 2.50 V, 2.14 V, 1.58 V, ...: a 1 Hz sine
- change the phase of the 50 Hz source: the offset at 5 Hz changes between 1.5 V and 2.5 V

▶ Falstad: `examples/adc/alias50.txt` The track window is only 1 ms here, so that the hold capacitor can follow the 50 Hz signal ($\tau = 0.1$ ms). The held value is the input at the end of the window, $t = 1$ ms: $2\,\text{V} + 0.5\,\text{V} \cdot \sin(2\pi \cdot 50\,\text{Hz} \cdot 1\,\text{ms} - 1.8°) = 2.14\,\text{V}$, where $1.8°$ is the phase lag of the RC charging path.

Anti-aliasing filter



- pass the wanted signal, attenuate everything above $f_s/2$ below the resolution
- at 50 Hz: 1st order -26.6 dB (factor 21), 2nd order -53 dB (factor 460)
- the filter must be **before** the S&H: after sampling the damage is done

Design card: Nyquist & Anti-aliasing

- sampling theorem: all input components must be below $f_s/2$ (Nyquist frequency)
- alias: $f_{\text{alias}} = |f - kf_s|$, nearest multiple kf_s; result lies in $0 \ldots f_s/2$
- capstone: $f_s = 5$ Hz, 50 Hz hum $\rightarrow |50 - 10 \cdot 5|$ Hz $= 0$ Hz: constant offset on every reading
- low-pass before the S&H: $f_c \approx f_s/2$ (capstone 2.5 Hz), e.g. RC: $f_c = 1/(2\pi RC)$
- required attenuation: interference amplitude $\times |H(f)| < \frac{1}{2}$ LSB (better $\frac{1}{4}$ LSB)
- 1st order: $|H| \approx f_c/f$ (-20 dB/decade), 2nd order: $(f_c/f)^2$ (-40 dB/decade)
- higher f_s (oversampling) relaxes the filter: more distance between signal band and $f_s/2$

9.2 Quantization

The second step is quantization: the continuous voltage range is divided into 2^N intervals of equal width, and each sample is replaced by the number of its interval. The width of the interval is the smallest voltage step the converter can resolve; it is called LSB (least significant bit), because a change of the input by one LSB changes the least significant bit of the output code [6, 2].

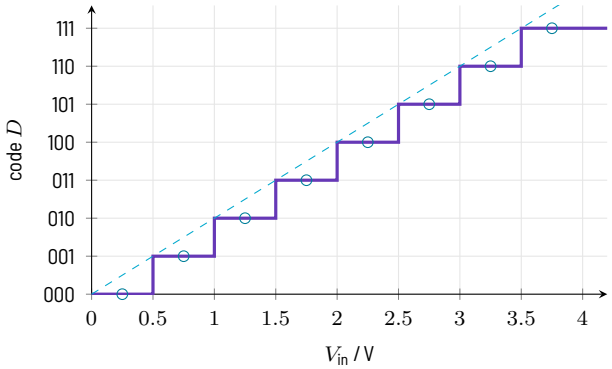
Resolution and LSB

LSB

$\text{LSB} = \dfrac{V_{\text{FS}}}{2^N}$ (V_{FS}: full-scale range, N: number of bits)

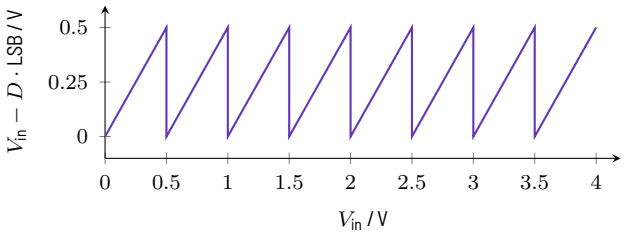
- capstone: $N = 3$, $V_{\text{FS}} = 4$ V: LSB $= 0.5$ V, 8 codes 000 …111
- 10 bit, 3.3 V (microcontroller): 3.2 mV; 16 bit, 5 V: 76 µV
- code for an input voltage (our flash converter):
  $D = \left\lfloor \dfrac{V_{\text{in}}}{\text{LSB}} \right\rfloor$, limited to $0 \ldots 2^N - 1$
- back to a voltage: $V_D = D \cdot$ LSB (lower edge) or $(D + \frac{1}{2})$ LSB (bin centre)

Transfer characteristic



- staircase: thresholds at $k \cdot$ LSB ($k = 1 \ldots 7$), the code changes at 0.5 V, 1.0 V, …, 3.5 V
- circles: bin centres 0.25 V, 0.75 V, …, 3.75 V: safe test points

Quantization error



- with thresholds at $k \cdot$ LSB: error $0 \ldots 1$ LSB (truncation), mean $\frac{1}{2}$ LSB
- shift all thresholds by $\frac{1}{2}$ LSB (ladder: $R/2$ at the bottom, $3R/2$ at the top): error $\pm\frac{1}{2}$ LSB
- or read the code as the bin centre $(D + \frac{1}{2})$ LSB: also $\pm\frac{1}{2}$ LSB

The two variants differ only by a constant offset of half an LSB. The capstone uses the simple ladder of eight equal resistors; its thresholds are at integer multiples of 0.5 V. A digital value D then means "the input is between $D \cdot 0.5$ V and $(D + 1) \cdot 0.5$ V".

Quantization noise and SNR

- for a busy signal the error behaves like noise, uniformly distributed over one LSB

  - rms value $V_{q,\text{rms}} = \dfrac{\text{LSB}}{\sqrt{12}}$
- full-scale sine: $V_{\text{rms}} = \frac{V_{\text{FS}}}{2\sqrt{2}} = \frac{2^N\,\text{LSB}}{2\sqrt{2}}$

Signal-to-noise ratio of an ideal N-bit converter

$$\text{SNR} = 20\log_{10}\frac{2^N\sqrt{12}}{2\sqrt{2}}\,\text{dB} = 6.02\,N + 1.76\,\text{dB}$$

- 3 bit: 19.8 dB; 8 bit: 49.9 dB; 12 bit: 74.0 dB; 16 bit: 98.1 dB
- every additional bit: 6 dB more (factor 2 in amplitude)

Falstad: quantization made visible

- 3-bit flash converter (next sections) followed by a 3-bit R-2R DAC
  - the DAC turns the code back into $V_q = D \cdot 0.5$ V
- input: triangle 0 V … 4 V, 0.25 Hz
- the scope shows V_{in} and the staircase V_q; the difference is always between 0 V and 0.5 V
- example readings: $V_{\text{in}} = 1.8$ V $\rightarrow V_q = 1.5$ V; $V_{\text{in}} = 3.98$ V $\rightarrow V_q = 3.5$ V

▶ Falstad: `examples/adc/quant_error.txt`

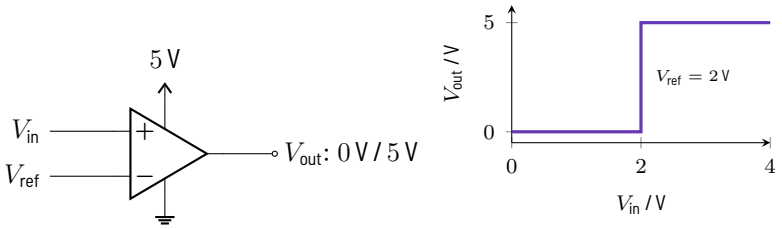
Design card: Reference Ladder & Quantization

- LSB $= V_{\text{FS}}/2^N$; capstone: $4\,\text{V}/8 = 0.5$ V
- ladder of 2^N equal resistors from $V_{\text{ref}} = V_{\text{FS}}$ to ground: taps $V_k = k \cdot V_{\text{ref}}/2^N$, $k = 1 \ldots 2^N - 1$
- capstone: 8×1 kΩ from 4 V: taps 0.5 V, 1.0 V, …, 3.5 V; $I = 0.5$ mA
- code $D = \lfloor V_{\text{in}}/\text{LSB} \rfloor$; quantization error $0 \ldots 1$ LSB; with $R/2$ at the bottom and $3R/2$ at the top: $\pm\frac{1}{2}$ LSB
- test points: bin centres $(D + \frac{1}{2})$ LSB = 0.25 V, 0.75 V, …, 3.75 V
- ladder current $\gg$ comparator input currents; tap resistance $\leq R_{\text{total}}/4 = 2$ kΩ
- reference: stiff voltage source (reference IC + buffer); its error scales all thresholds
- SNR of an ideal converter: $6.02\,N + 1.76$ dB

9.3 The Comparator

The comparator is the 1-bit converter: it decides whether its input is above or below a threshold. Every converter contains at least one. An op-amp without feedback is a comparator, because its huge open-loop gain drives the output into one of the rails as soon as the input difference is more than a few microvolts. Dedicated comparator ICs are faster and have logic-compatible outputs [1, 2].

Open-loop op-amp as comparator



- $V_{out} = V_{OH}$ if $V_{in} > V_{ref}$, otherwise V_{OL}
- single 5 V supply: output levels fit 5 V logic directly (rail-to-rail output)
- Falstad: op-amp with *max output* 5 V, *min output* 0 V

Real comparators

- op-amps are slow when used open loop: recovery from saturation, slew rate (µs)
- comparator ICs (LM393, LM339, TLV3501): fast, no frequency compensation, logic outputs
  - LM393/LM339: **open-collector** output, needs a pull-up resistor (e.g. 4.7 kΩ to 5 V)
- input offset voltage (mV): shifts the threshold; must be $\ll$ LSB
- input bias current: flows through the ladder resistors, small threshold shift
- never use a comparator IC as an op-amp: it has no compensation and oscillates with feedback

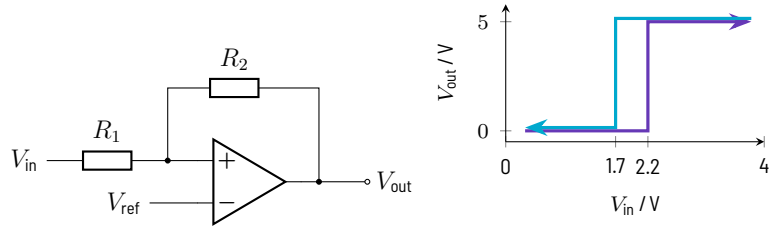
For the 3-bit capstone with an LSB of 0.5 V, any op-amp or comparator is good enough: offsets of a few millivolts are three orders of magnitude below the LSB. For an 8-bit flash converter with 4 V full scale (LSB $= 15.6$ mV) the offset of each comparator already matters.

Chatter at slow, noisy inputs

- a slowly changing input with noise crosses the threshold several times
- the output switches back and forth (**chatter**): a counter counts wrong, an LED flickers
- remedy: **hysteresis**: two thresholds, V_{T+} for rising and V_{T-} for falling inputs
  - positive feedback (Schmitt trigger)
  - the hysteresis $V_{T+} - V_{T-}$ must be larger than the peak-to-peak noise
- Falstad: triangle 0 V … 4 V at 1 Hz plus 0.2 V at 50 Hz
  - plain comparator: 10 edges in one period; Schmitt trigger: exactly 2

▶ Falstad: `examples/adc/comparator_hyst.txt`

Schmitt trigger (non-inverting)



- switching when $V_+ = V_{ref}$, with $V_+ = V_{in}\frac{R_2}{R_1+R_2} + V_{out}\frac{R_1}{R_1+R_2}$:
  $$V_T = V_{ref}\left(1 + \frac{R_1}{R_2}\right) - V_{out}\frac{R_1}{R_2}$$

- V_{T+} (output low): $V_{\text{ref}}(1 + R_1/R_2)$; V_{T-} (output high): $V_{T+} - V_{\text{OH}} R_1/R_2$
- $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $V_{\text{ref}} = 2\,\text{V}$: $V_{T+} = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$, hysteresis $0.5\,\text{V}$

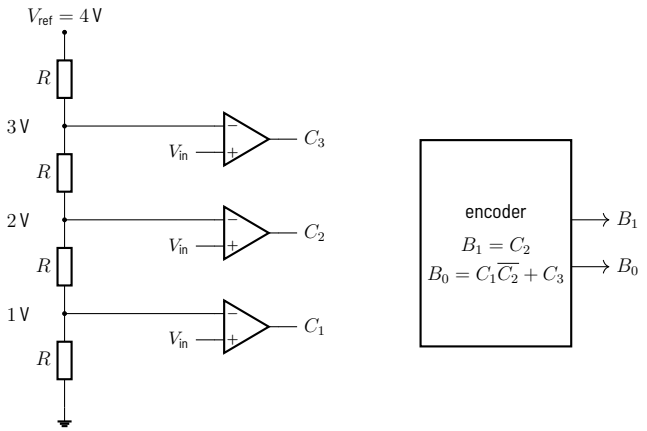
Design card: Comparator

- $V_{\text{out}} =$ high if $V_+ > V_-$, else low; op-amp open loop or comparator IC
- logic compatible: supply $0/5\,\text{V}$, rail-to-rail output (or open collector + pull-up $4.7\,\text{k}\Omega$)
- Falstad: op-amp, max output $5\,\text{V}$, min output $0\,\text{V}$; logic inputs switch at $2.5\,\text{V}$
- offset voltage $\ll$ LSB; input current $\times$ source resistance $\ll$ LSB
- hysteresis against chatter: R_1 from V_{in} to V_+, R_2 from V_{out} to V_+, V_{ref} at V_-
- $V_{T+} = V_{\text{ref}}(1+R_1/R_2)$, $V_{T-} = V_{T+} - V_{\text{OH}} R_1/R_2$, hysteresis $V_{\text{OH}} R_1/R_2 >$ noise (peak-to-peak)
- in a flash ADC behind an S&H: no hysteresis needed (the held input is constant), hysteresis would shift the thresholds

9.4 The Flash Converter

The flash (or parallel) converter is the most direct way to convert a voltage: compare it with all thresholds at the same time. An N-bit flash converter needs $2^N - 1$ comparators and a reference ladder that provides the thresholds. The comparator outputs form a thermometer code, which a small logic circuit turns into a binary number. The conversion takes only the delay of one comparator and a few gates, which makes flash converters the fastest of all architectures. The price is the exponential growth of the number of comparators: 7 for 3 bit, 255 for 8 bit.

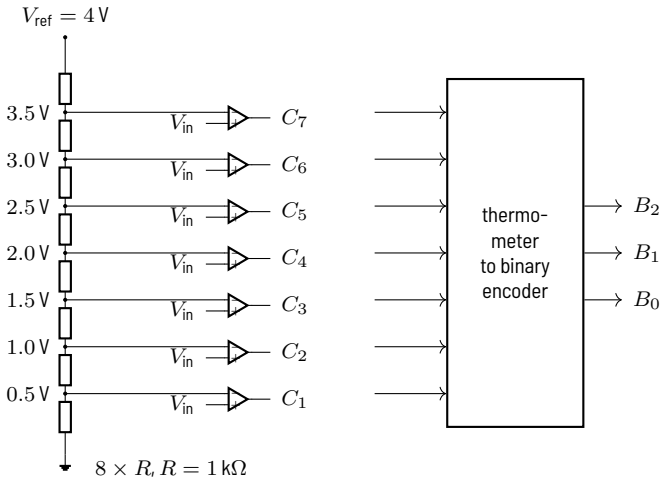
2-bit flash converter



- LSB $= 1$ V; thresholds 1 V, 2 V, 3 V; 3 comparators
- $C_k = 1$ if $V_{in} > k \cdot$ LSB

▶ Falstad: `examples/adc/flash2.txt`

3-bit flash converter (capstone)



- 7 comparators, $C_k = (V_{in} > k \cdot 0.5\,V)$; ladder current $4\,V / 8\,k\Omega = 0.5$ mA

Thermometer code

V_{in} / V	D	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.00 …0.50	0	0	0	0	0	0	0	0	0	0	0
0.50 …1.00	1	0	0	0	0	0	0	1	0	0	1
1.00 …1.50	2	0	0	0	0	0	1	1	0	1	0
1.50 …2.00	3	0	0	0	0	1	1	1	0	1	1
2.00 …2.50	4	0	0	0	1	1	1	1	1	0	0
2.50 …3.00	5	0	0	1	1	1	1	1	1	0	1
3.00 …3.50	6	0	1	1	1	1	1	1	1	1	0
3.50 …4.00	7	1	1	1	1	1	1	1	1	1	1

- like the column of a thermometer: all comparators below the input level are 1

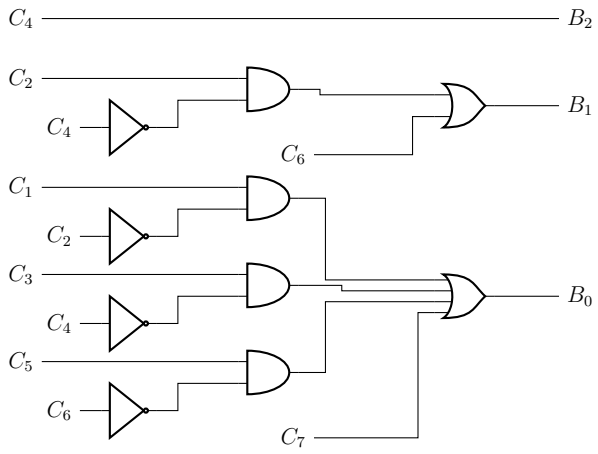
- the number of ones is the code D; only 2^N of the 2^{2^N-1} patterns are valid

Deriving the encoder

- the top of the column: $H_k = C_k\,\overline{C_{k+1}}$ is 1 only for $D = k$ (one-hot code, $C_8 := 0$)
- each output bit is the OR of the H_k whose code has this bit set:
  - $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$
  - $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_3} + C_3\overline{C_4} + C_6\overline{C_7} + C_7 = C_2\overline{C_4} + C_6$
  - $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- simplification uses the thermometer property: $C_{k+1} = 1 \Rightarrow C_k = 1$
  - $C_2\overline{C_3} + C_3\overline{C_4} = C_2\overline{C_4}$: "between level 2 and level 4"
  - B_2: four ones in a row means $D \geq 4$

The derivation can also be read directly from the table: B_2 is 1 for the lower half of the table, exactly where C_4 is 1. B_1 is 1 for the codes 2, 3 and 6, 7: the input is above level 2 but not above level 4, or it is above level 6. B_0 alternates with every code; it is 1 if the top of the thermometer column is at an odd position. The equations need three inverters, four 2-input AND gates, one 2-input OR gate and one 4-input OR gate.

Encoder logic



- 3 NOT, 4 AND, 2 OR gates: the same gates as in the Falstad model

Falstad: 3-bit flash converter with display

- input: slider 0 V … 4 V (label `Vin`); ladder 8×1 kΩ, 4 V
- comparator outputs and encoder signals are labeled nodes `C1`…`C7`, `nC2`, `a24`, …, `B0`…`B2`
- 7 LEDs as bar graph, 3 LEDs for $B_2B_1B_0$, each with $330\,\Omega$
- checked at the bin centres 0.25 V, 0.75 V, …, 3.75 V: codes 000, 001, …, 111
- default 2.6 V: five bar LEDs on, binary 101

▶ Falstad: `examples/adc/flash3.txt` Labeled nodes keep the drawing readable: every comparator output gets a label, and the gate inputs carry the same label names instead of long wires. Hover over a label to see its voltage. The tap voltages of the ladder are 0.5 V …3.5 V, the ladder current is 0.50 mA.

Priority encoder and bubble errors

- **bubble**: a wrong comparator decision inside the column, e.g. by offset or noise near a threshold

  - true code 3: 0000111; with bubble at C_2: 0000101
  - our encoder: $B_0 = C_1\overline{C_2} = 1$, $B_1 = 0$, $B_2 = 0 \Rightarrow$ 001: error of 2 LSB
- **priority encoder** (74HC148, 8 inputs $\rightarrow$ 3 bit): outputs the number of the **highest** active input
  - ignores everything below the top: a bubble does no harm
- bubble correction with 3-input gates: $H_k = C_{k-1}\, C_k\, \overline{C_{k+1}}$
- large flash converters: ones counter (Wallace tree) or Gray-code encoding

In the capstone the comparators decide on a held, constant voltage with an LSB of 0.5 V, so bubbles do not occur. In fast converters with hundreds of comparators they are a real problem, because the comparators near the input level see differences of only microvolts and decide at slightly different times.

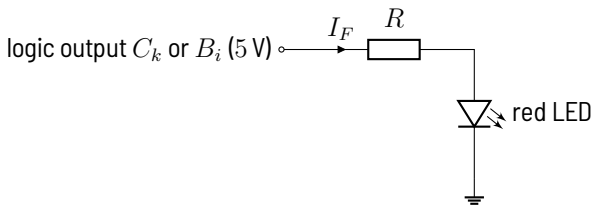
Design card: Thermometer-to-Binary Encoder

- thermometer code: $C_k = 1$ for all $k \le D$; the number of ones is the code D
- one-hot: $H_k = C_k\overline{C_{k+1}}$ (top of the column); $B_i =$ OR of all H_k with bit i of k set
- 3 bit (capstone): $B_2 = C_4 \quad B_1 = C_2\overline{C_4} + C_6 \quad B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- gates: 3 NOT ($\overline{C_2}, \overline{C_4}, \overline{C_6}$), 4 AND2, 1 OR2, 1 OR4
- check: sweep V_{in} over the bin centres 0.25 V … 3.75 V in steps of 0.5 V: 000 …111
- alternative: priority encoder (74HC148, active-low inputs and outputs), robust against bubbles

9.5 Displaying the Result

In the lab the digital value is shown on LEDs: a bar graph driven directly by the thermometer code, and three LEDs for the binary code. The bar graph is easy to read without any decoding; the binary LEDs show that the encoder works. Every LED is driven by a logic output (or comparator output) through its own current-limiting resistor.

LED with series resistor



Series resistor

$$R = \frac{V_{OH} - V_F}{I_F} = \frac{5\,\text{V} - 2\,\text{V}}{9\,\text{mA}} = 333\,\Omega \quad \rightarrow \quad \text{E12: } 330\,\Omega,\ I_F = 9.1\,\text{mA}$$

- red LED: $V_F \approx 1.8\,\text{V} \ldots 2.0\,\text{V}$; bright enough at $5\,\text{mA} \ldots 10\,\text{mA}$
- **one resistor per LED**: with a common resistor the current would depend on the number of LEDs that are on
- Falstad LED model: $V_F = 1.77\,\text{V}$, $I_F = 9.8\,\text{mA}$ with $330\,\Omega$

Current budget

- code 7: 7 bar LEDs + 3 binary LEDs on: $10 \times 9.1\,\text{mA} \approx 91\,\text{mA}$ from the $5\,\text{V}$ supply
- each output must deliver its LED current at a high level close to $5\,\text{V}$
  - in our circuit C_4 drives its bar LED, the B_2 LED and the gate inputs: $18\,\text{mA}$
  - 74HC gates: $\pm 25\,\text{mA}$ absolute maximum per output, V_{OH} drops under load
- better: buffer gates, low-current LEDs ($2\,\text{mA}$), or a driver IC (74HC245; ULN2003 sinks)
- LED to ground (output sources current) or LED to $5\,\text{V}$ (output sinks, active low): both are fine

The Falstad model has ideal outputs, so it does not show any of these loading effects. In hardware, look up the guaranteed output voltage at the required current in the data sheet. For a CMOS gate supplied with $5\,\text{V}$, V_{OH} is guaranteed above about $3.8\,\text{V}$ only up to a few milliamperes; at higher currents the LED current decreases accordingly.

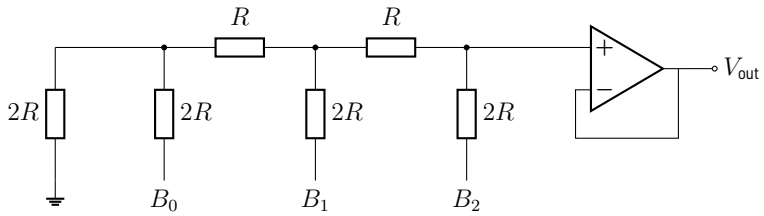
Design card: Falstad digital cheat-sheet

- gates: *Draw* → *Logic Gates, Input and Output*: AND, OR, NAND, NOR, XOR, Inverter; number of inputs and high level ($5\,\text{V}$) in the edit dialog (double click)
- logic levels: input high above $V_{high}/2 = 2.5\,\text{V}$; outputs are ideal $0\,\text{V}/5\,\text{V}$ sources
- comparator: op-amp with max output $5\,\text{V}$ and min output $0\,\text{V}$; inputs $-$ and $+$ as drawn
- *Logic Input* (click to toggle) and *Logic Output* (shows H/L) for quick tests
- *Labeled Node*: nodes with the same name are connected; use them for $C_1 \ldots C_7$, $B_0 \ldots B_2$; hover to read the voltage
- input voltage: DC voltage source or rail with a slider (edit dialog); clock: rail with square wave, frequency, duty cycle
- analog switch: closed if control $> 2.5\,\text{V}$, $R_{on} = 20\,\Omega$
- LED: always with a series resistor ($330\,\Omega$ at $5\,\text{V}$); lights above a few mA
- scope: right click on a node or element → *View in Scope*; set the time step to $0.1\,\text{ms}$ for $5\,\text{Hz}$ clocks

9.6 Other Converter Architectures

The flash converter is fast but expensive in comparators. Other architectures trade speed for resolution. Most of them contain a digital-to-analog converter (DAC) in a feedback loop, so we start with the simplest DAC, the R-2R ladder. Then we look at the successive-approximation converter found in every microcontroller, the dual-slope converter of digital multimeters and the sigma-delta converter used for audio and precision measurement [6, 7].

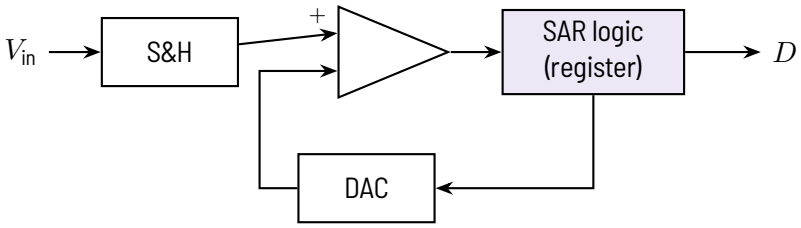
R-2R ladder DAC



- bit inputs: logic levels 0 V or V_{ref}; output resistance of the ladder: always R
- $V_{out} = V_{ref}\left(\frac{B_2}{2} + \frac{B_1}{4} + \frac{B_0}{8}\right) = V_{ref}\frac{D}{2^N}$
- only two resistor values, any number of bits: easy to integrate
- Falstad: $R = 10\,\text{k}\Omega$, 5 V logic, counter at 1 Hz/0.5 Hz/0.25 Hz: staircase in steps of 0.625 V, $D = 5$: 3.125 V

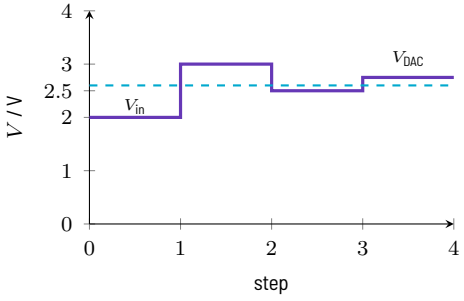
▶ Falstad: `examples/adc/r2r_dac.txt` Every node of the ladder sees a resistance of $2R$ to the left (the rest of the ladder) and $2R$ downwards (the bit input). Seen from each bit input, the current therefore splits in half at every node on its way to the output, so each bit has twice the weight of the next lower bit. A Thevenin conversion from left to right gives the formula. Loading the output with $4R$ instead of a follower scales it by $4R/(R + 4R) = 0.8$; this is used in the Falstad example `quant_error` to get exactly 0.5 V per code from 5 V logic levels.

Successive approximation (SAR)



- binary search with one comparator and a DAC: one bit per clock cycle, MSB first
- set the bit, compare V_{in} with V_{DAC}: keep the bit if $V_{in} \geq V_{DAC}$, otherwise clear it
- N steps for N bits; needs an S&H (built in); 8 …18 bit, up to some MS/s

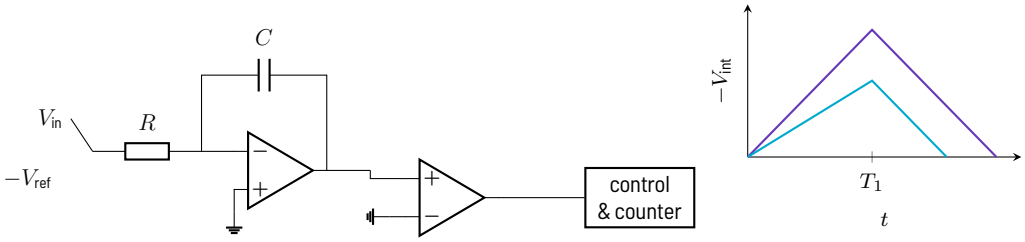
SAR: worked example



step	trial	V_{DAC}	$V_{in} \geq V_{DAC}$?
1	**1**000	2.00 V	yes: keep
2	1**1**00	3.00 V	no: clear
3	1**0**10	2.50 V	yes: keep
4	1**1**11	2.75 V	no: clear
	1010	2.50 V	$D = 10$

- 4 bit, $V_{FS} = 4$ V, LSB $= 0.25$ V, $V_{in} = 2.6$ V
- result 1010: V_{in} lies between 2.50 V and 2.75 V

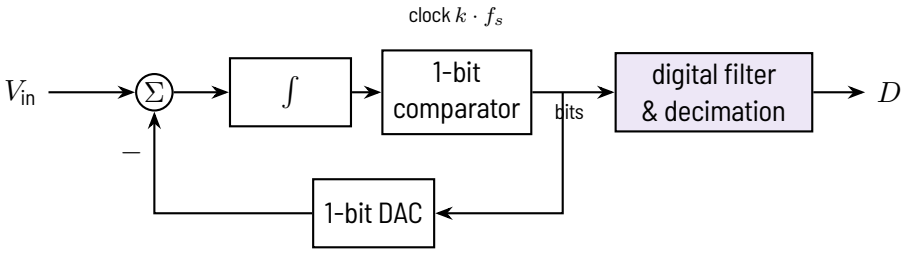
Dual-slope converter



- integrate V_{in} for a fixed time T_1, then integrate $-V_{ref}$ until the output is zero again (T_2)
- $V_{in}\,T_1 = V_{ref}\,T_2 \quad \Rightarrow \quad D = N_1 \dfrac{V_{in}}{V_{ref}}$: independent of R, C and the clock frequency
- $T_1 = 20$ ms (or 100 ms): 50 Hz interference integrates to zero

The dual-slope converter is slow (a few conversions per second) but very accurate, and it rejects mains interference by design: the integral of a 50 Hz sine over a whole number of periods is zero. That is why digital multimeters use it. It is the opposite strategy to our capstone, where the 50 Hz interference is removed by the anti-aliasing filter before sampling.

Sigma-delta converter (concept)



- oversampling with a 1-bit quantizer in a feedback loop: the density of ones is proportional to V_{in}
- the loop shapes the quantization noise to high frequencies; a digital low-pass removes it
- 16 …24 bit, audio and precision measurement (weighing scales, thermocouples)
- the digital filter is also the anti-aliasing filter: the analog filter can be simple

Comparison

	flash	SAR	dual-slope	sigma-delta
principle	$2^N - 1$ comparators	binary search	integration	oversampling
steps per sample	1	N	$2 \cdot 2^N$ clocks	many (oversampling)
resolution	4 …8 bit	8 …18 bit	12 …22 bit	16 …24 bit
speed	up to GS/s	kS/s …MS/s	1 …100 S/s	Hz …MHz bandwidth
cost, power	high (exponential)	low	low	medium
typical use	oscilloscopes, radio	microcontrollers	multimeters	audio, scales

- the capstone uses a 3-bit flash converter because every part of it is visible: ladder, comparators, logic

9.7 Exercises

Exercise 9.1: Aliasing

A system samples at $f_s = 5$ Hz.

a) Calculate the alias frequencies of input components at 2 Hz, 4 Hz, 12 Hz, 49 Hz and 50 Hz.

b) The sensor signal is 2 V DC with 0.5 V of 50 Hz hum. Describe the sampled values for $f_s = 5$ Hz and for $f_s = 4.9$ Hz.

c) An RC low-pass with $6.8\,\text{k}\Omega$ and $10\,\mu\text{F}$ is inserted before the S&H stage. Calculate f_c and the remaining hum amplitude. Compare with the LSB of the 3-bit converter (0.5 V).

d) Check part b) in Falstad.

Solution 9.1

a) 2 Hz: below $f_s/2$, no alias. 4 Hz: $|4-5| = 1$ Hz. 12 Hz: $|12-2\cdot5| = 2$ Hz. 49 Hz: $|49-10\cdot5| = 1$ Hz. 50 Hz: $|50 - 10\cdot 5| = 0$ Hz.

b) $f_s = 5$ Hz: the hum is sampled at the same phase every time; all samples are equal, $2\,\text{V} + 0.5\,\text{V}\sin\varphi_0$, i.e. a constant offset between -0.5 V and 0.5 V. $f_s = 4.9$ Hz: the phase advances by 0.204 periods per sample; the samples follow a 1 Hz sine between 1.5 V and 2.5 V.

c) $f_c = 1/(2\pi\cdot 6.8\,\text{k}\Omega\cdot 10\,\mu\text{F}) = 2.34$ Hz. $|H(50\,\text{Hz})| = 1/\sqrt{1+(50/2.34)^2} = 0.047$ (-26.6 dB): the hum is reduced to 23 mV, less than $\frac{1}{20}$ LSB. The offset error disappears from the readings.

d) ▶ Falstad: `examples/adc/alias50.txt`
`Vs_5Hz` holds 2.14 V at every sample (offset 0.14 V for the phase in the example); `Vs_4.9Hz` shows 2.14 V, 2.50 V, 2.14 V, 1.58 V, 1.62 V, 2.20 V, …

Exercise 9.2: Quantization

A 3-bit converter has a full scale of 4 V and thresholds at $k\cdot$ LSB.

a) Calculate the LSB and the codes for 0.3 V, 1.8 V, 2.0 V and 3.98 V.

b) Which voltage range does the code 101 represent? What is the largest quantization error?

c) Calculate the SNR of the ideal 3-bit converter. How many bits are needed for an SNR of at least 60 dB?

d) Observe the quantization in Falstad.

Solution 9.2

a) LSB $= 0.5$ V. $D = \lfloor V_\text{in}/0.5\,\text{V}\rfloor$: 0.3 V $\to$ 000, 1.8 V $\to$ 011, 2.0 V $\to$ 011 or 100 (exactly on the threshold, the comparator offset decides), 3.98 V $\to$ 111.

b) 101 = 5: 2.5 V … 3.0 V. The error $V_\text{in} - D\cdot$ LSB is between 0 and 0.5 V (1 LSB); relative to the bin centre 2.75 V it is ± 0.25 V.

c) SNR $= 6.02\cdot 3 + 1.76 = 19.8$ dB. $6.02\,N + 1.76 \geq 60 \Rightarrow N \geq 9.7$: 10 bits.

d) ▶ Falstad: `examples/adc/quant_error.txt`
At $V_\text{in} = 1.8$ V the DAC output is $V_q = 1.5$ V, at 3.98 V it is 3.5 V; $V_\text{in} - V_q$ stays between 0 V and 0.5 V.

Exercise 9.3: Designing the 3-bit flash converter

Design the converter of the capstone: full scale $4.00\,\text{V}$, 3 bit, comparators $C_k = (V_{\text{in}} > k \cdot 0.5\,\text{V})$.

a) Dimension the reference ladder with $1\,\text{k}\Omega$ resistors. Calculate the tap voltages, the ladder current and the power taken from the reference.

b) The comparators have an input bias current of $100\,\text{nA}$. Estimate the worst-case threshold shift. (Hint: Thevenin resistance of a tap.)

c) Write down the truth table from thermometer code to binary and derive the logic equations for B_2, B_1, B_0.

d) How must the ladder be changed for thresholds at $(k - \frac{1}{2}) \cdot \text{LSB}$?

e) Check the converter in Falstad at the bin centres.

Solution 9.3

a) 8 resistors of $1\,\text{k}\Omega$ from $4\,\text{V}$ to ground. Taps $V_k = k \cdot 4\,\text{V}/8 = 0.5\,\text{V}, 1.0\,\text{V}, \ldots, 3.5\,\text{V}$. $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$, $P = 2\,\text{mW}$.

b) The largest Thevenin resistance is at the middle tap: $4\,\text{k}\Omega \parallel 4\,\text{k}\Omega = 2\,\text{k}\Omega$. Shift: $100\,\text{nA} \cdot 2\,\text{k}\Omega = 0.2\,\text{mV}$, negligible compared with $0.5\,\text{V}$.

c) Truth table as on the slide *Thermometer code*. With $H_k = C_k\overline{C_{k+1}}$: $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$; $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_4} + C_6$; $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$.

d) Total resistance stays $8R$: bottom resistor $R/2 = 500\,\Omega$, six resistors R, top resistor $3R/2 = 1.5\,\text{k}\Omega$. The taps are then $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.25\,\text{V}$.

e) ▶ Falstad: `examples/adc/flash3.txt`
Taps $0.5\,\text{V}$ ...$3.5\,\text{V}$, ladder current $0.50\,\text{mA}$. Slider at $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.75\,\text{V}$: the bar graph shows 0 ...7 LEDs, the binary LEDs 000, 001, ..., 111.

Exercise 9.4: Bubble error

Because of noise, the comparators of the 3-bit flash converter deliver the pattern 0010111 (C_7 first), a bubble at C_4.

a) Which input range is plausible? Which code would be correct?

b) Which code does the encoder of the capstone produce?

c) Which code does a priority encoder produce?

Solution 9.4

a) $C_5 = 1$ but $C_4 = 0$ contradict each other. Either C_4 is wrong (input between $2.5\,\text{V}$ and $3.0\,\text{V}$, code 5) or C_5 is wrong (input between $1.5\,\text{V}$ and $2.0\,\text{V}$, code 3).

b) $B_2 = C_4 = 0$; $B_1 = C_2\overline{C_4} + C_6 = 1$; $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7 = 0 + 1 + 1 + 0 = 1$: code 011 = 3, one of the plausible values. (For the pattern 0000101 of the slide, however, it gives 001 instead of 011.)

c) The priority encoder looks at the highest active input, C_5: code 101 = 5, the other plausible value. For 0000101 it gives 011 = 3, the correct value; it never produces a code below the top of the column.

Exercise 9.5: LED display

The bar graph and the binary LEDs are driven by $5\,\text{V}$ logic outputs. Red LEDs with $V_F \approx 2\,\text{V}$ shall carry $5\,\text{mA} \ldots 10\,\text{mA}$.

a) Choose the series resistor from the E12 series and calculate the LED current.

b) Calculate the total LED current for the code 111.

c) In the Falstad model, C_4 drives its bar LED and also the B_2 LED. Which current does this output deliver?

d) Why must each LED have its own resistor?

Solution 9.5

a) $R = (5\,\text{V} - 2\,\text{V})/I_F$ gives $300\,\Omega \ldots 600\,\Omega$. Choose $330\,\Omega$: $I_F = 3\,\text{V}/330\,\Omega = 9.1\,\text{mA}$.

▶ Falstad: `examples/adc/flash3.txt`

The Falstad LED has $V_F = 1.77\,\text{V}$, so the current is $3.23\,\text{V}/330\,\Omega = 9.8\,\text{mA}$.

b) 10 LEDs: $10 \cdot 9.1\,\text{mA} = 91\,\text{mA}$.

c) $2 \cdot 9.1\,\text{mA} = 18\,\text{mA}$ (plus negligible gate input currents). That is near the limit of a CMOS logic output; a buffer or lower LED currents ($680\,\Omega$: $4.4\,\text{mA}$) are safer.

d) LEDs of different colours and samples have different V_F. With a common resistor the LED with the lowest V_F takes most of the current, and the current per LED changes with the number of LEDs that are on.

Exercise 9.6: Schmitt trigger, R-2R DAC and SAR

a) A non-inverting Schmitt trigger has $V_{\text{ref}} = 2\,\text{V}$, $R_1 = 10\,\text{k}\Omega$ and output levels $0\,\text{V}/5\,\text{V}$. Choose R_2 for a hysteresis of $0.5\,\text{V}$ and calculate both thresholds.

b) A 3-bit R-2R DAC is driven by $5\,\text{V}$ logic levels. Calculate the step size and the output for $D = 5$.

c) A 4-bit SAR converter ($V_{\text{FS}} = 4\,\text{V}$) converts $V_{\text{in}} = 1.3\,\text{V}$. Write down the trial codes, the DAC voltages and the result.

d) Check a) and b) in Falstad.

Solution 9.6

a) Hysteresis $V_{0\text{H}} R_1/R_2 = 0.5\,\text{V} \Rightarrow R_2 = 5\,\text{V} \cdot 10\,\text{k}\Omega/0.5\,\text{V} = 100\,\text{k}\Omega$. $V_{T+} = 2\,\text{V} \cdot 1.1 = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$.

b) LSB $= 5\,\text{V}/8 = 0.625\,\text{V}$; $D = 5$: $3.125\,\text{V}$.

c) LSB $= 0.25\,\text{V}$. 1000: $2.00\,\text{V}$, no $\rightarrow$ 0. 0100: $1.00\,\text{V}$, yes $\rightarrow$ 1. 0110: $1.50\,\text{V}$, no $\rightarrow$ 0. 0101: $1.25\,\text{V}$, yes $\rightarrow$ 1. Result 0101 = 5 ($1.25\,\text{V} \ldots 1.50\,\text{V}$).

d) ▶ Falstad: `examples/adc/comparator_hyst.txt`

The Schmitt trigger switches at $2.2\,\text{V}$ (rising) and $1.7\,\text{V}$ (falling) and gives exactly two edges per period; the plain comparator produces 10 edges.

▶ Falstad: `examples/adc/r2r_dac.txt`

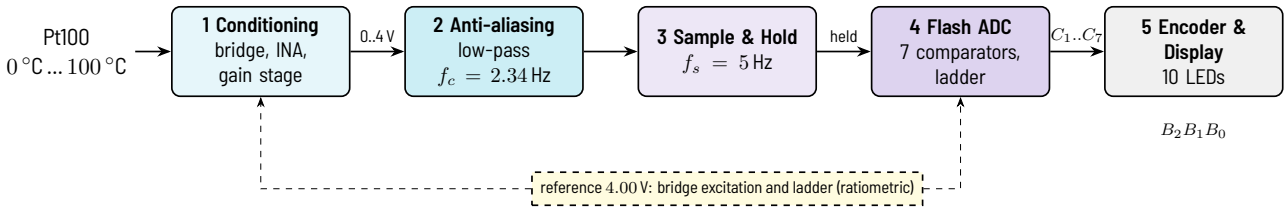
`Vout` steps through $0\,\text{V}$, $0.625\,\text{V}$, ..., $4.375\,\text{V}$; at $D = 5$ it shows $3.125\,\text{V}$.

10 From Sensor to Digital Value

10.1 The Measurement Chain

This chapter puts all building blocks of the lecture together. We design a simple digital thermometer: a resistive temperature sensor measures $0\,^{\circ}\mathrm{C} \ldots 100\,^{\circ}\mathrm{C}$, and the result is shown as a 3-bit code on LEDs. Every stage of the chain is a circuit from an earlier chapter: the bridge from chapter 5, the instrumentation amplifier from chapter 6, the low-pass from chapter 7, the sample and hold from chapter 8 and the flash converter from chapter 9. The LED with its series resistor even goes back to chapter 1.
The design is done twice: on paper, with all calculations, and in Falstad, where the complete chain runs and can be checked against the paper values. The chapter is also the worked solution of last year's capstone task (exercise 10.1).

Five stages from sensor to code



- requirement: $0\,°\text{C} \ldots 100\,°\text{C} \rightarrow 0\,\text{V} \ldots 4.00\,\text{V} \rightarrow$ code 000 …111
- $4\,\text{V}$ full scale makes the step size round: $\text{LSB} = 4\,\text{V}/2^3 = 0.5\,\text{V} \mathrel{\widehat{=}} 12.5\,°\text{C}$
- interference: 50 Hz, 5 mV in series with the sensor (overhead lamps)

Specification and design values

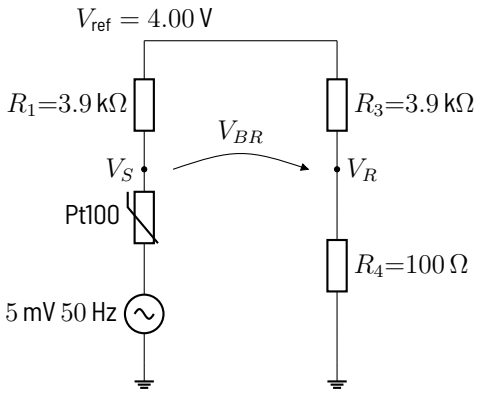
Quantity	Requirement	Design value
sensor	resistive, slider in Falstad	Pt100, $100\,\Omega \ldots 138.5\,\Omega$
sensor current	≤ 1 mA	1.000 mA … 0.990 mA
full scale	$0\,\text{V} \ldots 4.00\,\text{V}$	$G = 10 \cdot 10.76 = 107.6$
gain per stage	< 100	1, 10, 10.76
anti-aliasing	$f_c \approx 2.5$ Hz	$6.8\,\text{k}\Omega$, $10\,\mu\text{F}$: 2.34 Hz
sampling	$f_s = 5$ Hz	window 20 ms, $C_H = 1\,\mu\text{F}$
ADC	3 bit, LSB 0.5 V	$8 \times 1\,\text{k}\Omega$, 7 comparators
display	LED + resistor	$10\times$ (LED + $330\,\Omega$), 9.8 mA

▶ Falstad: `examples/capstone/capstone.txt`

10.2 Stage 1: Signal Conditioning

The sensor is a Pt100 platinum resistor with $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 100\,\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$, so it changes from $100\,\Omega$ at $0\,°\text{C}$ to $138.5\,\Omega$ at $100\,°\text{C}$. We feed the bridge from the same $4.00\,\text{V}$ reference that later supplies the ADC ladder. If the reference is $1\,\%$ too high, the bridge output and all thresholds of the converter are $1\,\%$ too high as well, and the code does not change: the measurement is *ratiometric*. The reference only needs to be stable while it is shared, not accurate.

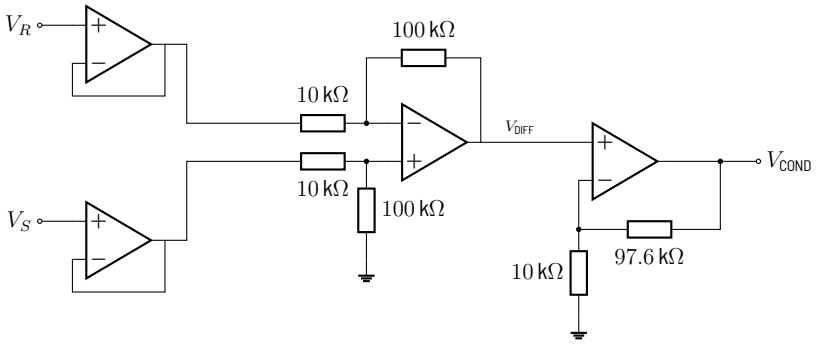
Bridge design



- sensor current $I_S = V_{\text{ref}}/(R_1 + R) = 4\,\text{V}/4000\,\Omega = 1.000\,\text{mA}$ at $0\,°\text{C}$ ($0.990\,\text{mA}$ at $100\,°\text{C}$)
- balanced at $0\,°\text{C}$: $R_4 = R(0\,°\text{C})$, so $V_{BR} = 0$
- $V_{BR}(T) = V_{\text{ref}}\left(\frac{R(T)}{R_1+R(T)} - \frac{R_4}{R_3+R_4}\right)$, $\quad V_{BR}(100\,°\text{C}) = 37.18\,\text{mV}$

The bridge output is not exactly linear in T, because the sensor current falls slightly with rising R. With $R_1 = 39\,R_0$ the error is small: at $50\,°\text{C}$ the output is $0.09\,\text{mV}$ higher than the straight line, $0.25\,\%$ of full scale, far below one LSB ($12.5\,\%$). A larger R_1 would reduce it further, but also the signal.

Instrumentation amplifier and gain stage



- buffers: input current zero, the bridge is not loaded
- difference amplifier: $V_{\text{DIFF}} = \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}(V_S - V_R) = 10\,V_{BR}$
- gain stage: $G_2 = 1 + \frac{97.6\,\text{k}\Omega}{10\,\text{k}\Omega} = 10.76$

Gain: how much, and in how many stages?

- required total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: $G_1 = 10$ (difference amplifier), $G_2 = 4.00/(10 \cdot 0.03718) = 10.758$
  - E96: $R_F = 97.6\,\text{k}\Omega \Rightarrow G_2 = 10.76$, full scale $4.0005\,\text{V}$
  - in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer, adjusted at $100\,°\text{C}$
- finite open-loop gain $A = 10^5$: relative gain error $\approx G/A$
  - one stage $G = 107.6$: $0.11\,\%$; two stages: $2 \times 0.01\,\%$
  - a single stage with $G > 1000$ would lose more than $1\,\%$: keep $G < 100$ per stage

Conditioning

$$V_{\mathrm{COND}} = G_1 G_2 \, V_{\mathrm{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right), \quad G_1 G_2 = 107.6$$

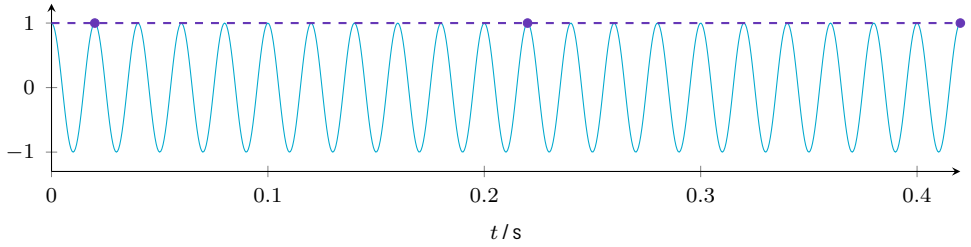
Splitting the gain over two stages has a second advantage: the zero point is set by the bridge (R_4) and the span by one resistor (R_F). The two adjustments do not influence each other, so the trimming procedure is simple: set the sensor to $100\,\Omega$ and check 0 V, then set it to $138.5\,\Omega$ and adjust R_F for 4.00 V.

10.3 Stage 2: Anti-Aliasing

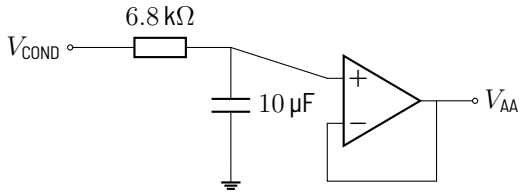
The temperature changes slowly, so five samples per second are plenty. The sampling theorem then limits the usable bandwidth to $f_s/2 = 2.5$ Hz. Everything above this frequency must be removed *before* the sampler, because after sampling it can no longer be distinguished from a real signal. The 50 Hz interference is the dangerous part: after the gain of 107.6 it has an amplitude of about half a volt, one full LSB.

Where does the 50 Hz go?

- sampling with f_s folds a frequency f to $f_{\text{alias}} = |f - k f_s|$ (nearest k)
- $f = 50$ Hz, $f_s = 5$ Hz: $|50 - 10 \cdot 5| = 0$ Hz
  - every sample sees the interference at the same phase
  - the result is a **constant offset**: nothing on the scope looks wrong
- amplitude at V_{COND}: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} \cdot 107.6 = 0.52\,\text{V}$ (about 1 LSB)



Anti-aliasing low-pass



- $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}} = 2.34\,\text{Hz} \le f_s/2$
- at 50 Hz: $|H| = 1/\sqrt{1 + (50/2.34)^2} = 0.047$ (-26.6 dB)
- residual offset: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$
- the follower decouples the RC network from the S&H input

For 3 bits a first-order filter is sufficient. For a 10-bit converter (LSB$/2 = 2$ mV) the residual 25 mV would be twelve times too large; a second- or third-order filter (chapter 7), a higher sampling rate followed by digital averaging, or a sampling rate that is *not* a sub-multiple of 50 Hz would then be needed.

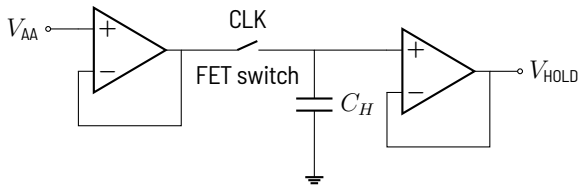
Design card: Nyquist & anti-aliasing (capstone values)

$f_s = 5$ Hz $\Rightarrow$ bandwidth ≤ 2.5 Hz; $f_{\text{alias}} = |f - k f_s|$; $f_c = 1/(2\pi RC) \le f_s/2$; residual $= \hat{v}\,/\sqrt{1 + (f/f_c)^2}$ must stay $<$ LSB$/2$.

10.4 Stage 3: Sample and Hold

The flash converter is fast and would not strictly need a sample and hold. We use one anyway, because it defines the sampling instant: the input of the converter changes only once per sampling period, so the LEDs show a stable code, and the aliasing experiment becomes visible. The circuit is the buffered S&H of chapter 8.

Buffered sample and hold



- clock 5 Hz, duty $10\,\%$: switch closed for 20 ms (track), open for 180 ms (hold)
- input buffer: charges C_H quickly; output buffer: no load on C_H

Choosing the hold capacitor

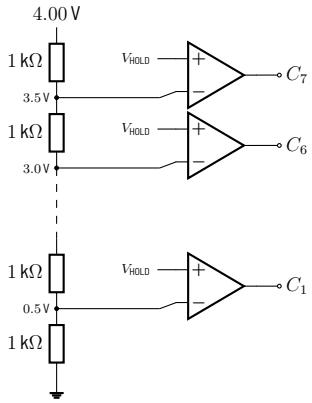
- **charging:** $\tau = R_{\text{on}} C_H = 20\,\Omega \cdot 1\,\mu\text{F} = 20\,\mu\text{s}$
  - settle to LSB$/2$ of 3 bits: $t = \tau \ln 2^4 = 2.8\,\tau = 55\,\mu\text{s} \ll 20$ ms
- **droop:** leakage $I_L = 1$ nA (switch + buffer input)
  - $\Delta V = I_L\, t_{\text{hold}} / C_H = 1\,\text{nA} \cdot 0.18\,\text{s} / 1\,\mu\text{F} = 0.18\,\text{mV}$
- both limits leave a wide range: 10 nF ...$100\,\mu$F would work
  - choose a film capacitor with low leakage and dielectric absorption: $C_H = 1\,\mu$F

In hardware a small resistor (e.g. $1\,\text{k}\Omega$, as in the examples of chapter 8) is usually put in series with the switch: it limits the charging current that the input buffer must deliver and isolates the buffer from the capacitive load. Then $\tau = 1.02$ ms and settling to LSB$/2$ takes 2.8 ms, still well inside the window. The limits are far apart because the signal is slow and the resolution low. They move together quickly in a fast, high-resolution system: a 12-bit converter sampling at 1 MHz has a track time of a few hundred nanoseconds and needs settling to 2^{-13}, which is why real S&H stages use picofarad capacitors and low-resistance switches.

10.5 Stages 4 and 5: Flash ADC, Encoder and Display

The held voltage is compared with seven thresholds at once. The thresholds come from a ladder of eight equal resistors across the 4.00 V reference. Each comparator answers one question, "is the input above $k \cdot 0.5$ V?", and together they form a thermometer code: all comparators below the input voltage output 1, all above output 0. An encoder turns the seven bits into the 3-bit binary number.

Reference ladder and comparators



- ladder current $4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; comparator inputs draw no current
- $C_k = 1$ if $V_{\text{HOLD}} > k \cdot 0.5\,\text{V}$; the number of ones is the level 0…7

Quantization and encoder

V_{HOLD} / V	level	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.0–0.5	0	0	0	0	0	0	0	0	0	0	0
0.5–1.0	1	0	0	0	0	0	0	1	0	0	1
1.0–1.5	2	0	0	0	0	0	1	1	0	1	0
1.5–2.0	3	0	0	0	0	1	1	1	0	1	1
2.0–2.5	4	0	0	0	1	1	1	1	1	0	0
2.5–3.0	5	0	0	1	1	1	1	1	1	0	1
3.0–3.5	6	0	1	1	1	1	1	1	1	1	0
3.5–4.0	7	1	1	1	1	1	1	1	1	1	1

- $B_2 = C_4$ (level ≥ 4)
- $B_1 = C_2\overline{C_4} + C_6$ (levels 2, 3, 6, 7)
- $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$ (odd levels: the edge of the thermometer is at an odd position)

Display

- 7 red LEDs from $C_1 \ldots C_7$: bar graph, a thermometer you can read without decoding
- 3 green LEDs from $B_2B_1B_0$: the binary result
- each LED with its own series resistor (chapter 1), driven by a 5 V logic output: $R = \frac{5\,\text{V} - V_F}{I_F} = \frac{5\,\text{V} - 1.8\,\text{V}}{10\,\text{mA}} = 320\,\Omega \rightarrow$ E12: $330\,\Omega$
  - Falstad: $I_F = 9.8\,\text{mA}$, $V_F = 1.78\,\text{V}$
- never one common resistor for several LEDs: the current would depend on how many are on

10.6 The Complete Chain in Falstad

The Falstad model contains all five stages in labelled boxes. The stages are connected with labeled nodes (`VCOND`, `VAA`, `VHOLD`, `C1`…`C7`, `B0`…`B2`); nodes with the same name are connected, which keeps the drawing readable. The sensor resistance is set with the slider "Sensor R". The first scope shows the filtered signal and the held staircase, the second the conditioned signal and the held value.

Using the model

- open ▶ Falstad: `examples/capstone/capstone.txt`
- move the slider from $100\,\Omega$ to $138.5\,\Omega$: the bar graph fills up, the binary LEDs count
- hover over a label to read its voltage; the held value changes every 200 ms
- after a slider step, wait about 0.5 s: the low-pass needs $5\tau = 5 \cdot 68$ ms

Verification table

For each test point the sensor resistance is computed from the conditioning equation so that V_{COND} lies in the centre of a code bin. The "Falstad" columns are the values of the simulation (with the 50 Hz interference switched on) just before the next sample.

T / °C	R / Ω	V_{COND} target	V_{HOLD} Falstad	code expected	code Falstad
6.2	102.38	0.25 V	0.251 V	000	000
18.6	107.16	0.75 V	0.751 V	001	001
43.5	116.75	1.75 V	1.749 V	011	011
56.0	121.56	2.25 V	2.250 V	100	100
81.1	131.22	3.25 V	3.249 V	110	110
93.7	136.07	3.75 V	3.750 V	111	111

- the temperatures differ slightly from the linear mapping $25\,°\mathrm{C/V}$ (6, 19, 44, 56, 81, $94\,°$C) because of the bridge nonlinearity

The aliasing experiment

- ▶ Falstad: `examples/capstone/capstone_nofilter.txt`: the S&H takes V_{COND} directly
- V_{COND} carries 0.52 V of 50 Hz; the S&H always samples at the same phase

	V_{COND} (DC)	V_{HOLD}	code
without filter	1.75 V	2.27 V	100 (wrong)
without filter	0.25 V	0.77 V	001 (wrong)
with filter	1.75 V	1.75 V	011

- without the filter the held value is a clean staircase, just 0.52 V too high: one full code too high
- the size of the offset depends on the phase between mains and sampling clock (here the worst case)

10.7 Exercises

Exercise 10.1: From sensor to digital value (capstone task 2025/26)

Build a complete data-acquisition chain in Falstad that turns a slowly changing sensor signal ($0\,°$C … $100\,°$C) into a 3-bit code. Label the five stages Conditioning | Anti-aliasing | Sample & Hold | Flash ADC | Encoder & Display.

a) Conditioning: resistive sensor in a Wheatstone bridge, sensor current ≤ 1 mA, sensor as a resistor with a slider; instrumentation-amplifier front end (two unity-gain buffers and a difference amplifier); gain trimmed to 0 V at the bottom and exactly 4.00 V at the top of the range; less than 100 per stage.

b) Anti-aliasing: sampling clock $f_s = 5$ Hz. Add the 50 Hz, 5 mV noise source in series with the sensor. Observe the held value without a filter, then add an active low-pass with $f_c \approx 2.5$ Hz and justify f_c from the Nyquist criterion.

c) Sample and hold at 5 Hz: follower, FET switch, hold capacitor, follower. Choose the hold capacitor.

d) Flash ADC: ladder $8 \times 1\,\text{k}\Omega$ from 4 V, seven comparators, thermometer code.

e) Encoder ($B_2 = C_4$, $B_1 = C_2\overline{C_4} + C_6$, $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$) and display: 7-LED bar graph and three binary LEDs, each LED with its own resistor.

f) Verification table for six inputs near the centres of different code bins, and one line on the aliasing experiment.

Solution 10.1

The design of this chapter is one possible solution; the complete model is
▶ Falstad: `examples/capstone/capstone.txt`.

a) Pt100 with $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$ at $V_{\text{ref}} = 4.00$ V: $I_S = 1.000$ mA, bridge output 0 mV … 37.18 mV. Buffers, difference amplifier $G_1 = 10$, non-inverting stage $G_2 = 1 + 97.6\,\text{k}\Omega/10\,\text{k}\Omega = 10.76$: $V_{\text{COND}} = 0$ V … 4.00 V.

b) Without filter the 50 Hz aliases to $|50 - 10 \cdot 5| = 0$ Hz; the held value is offset by up to 0.52 V (Falstad: 1.75 V becomes 2.27 V, code 100 instead of 011, ▶ Falstad: `examples/capstone/capstone_nofilter.txt`). Filter $R = 6.8\,\text{k}\Omega$, $C = 10\,\mu$F, $f_c = 2.34$ Hz $\leq f_s/2$, 50 Hz attenuated to $4.7\,\%$: 25 mV residual, below LSB$/2$.

c) $C_H = 1\,\mu$F: $\tau = 20\,\mu$s charges within the 20 ms window; droop 0.18 mV per hold period for 1 nA leakage.

d) Taps at $0.5, 1.0, \ldots, 3.5$ V; ladder current 0.5 mA.

e) 3 inverters, 4 AND gates, 2 OR gates (one with 4 inputs); 10 LEDs with $330\,\Omega$, 9.8 mA each.

f) See the verification table above. Aliasing line: "Without the anti-aliasing filter the held value showed a constant offset of 0.52 V (one code too high); with the 2.34 Hz low-pass the offset fell below 30 mV and all codes were correct."

Exercise 10.2: Ratiometric measurement

The 4.00 V reference of the capstone is in fact 4.10 V.

a) How large is V_{COND} at $100\,°$C?

b) Which code does the converter show at $100\,°$C and at $56\,°$C? Compare with a design in which the bridge is supplied from a separate, exact 4.00 V source.

Solution 10.2

a) The bridge output scales with the reference: $V_{\text{COND}} = 4.0005\,\text{V} \cdot 4.10/4.00 = 4.10$ V.

b) The thresholds scale as well ($k \cdot 0.5125$ V). The ratio $V_{\text{COND}}/V_{\text{ref}}$ is unchanged, so both codes are unchanged: 111 at $100\,°$C, 100 at $56\,°$C ($V_{\text{COND}} = 2.25\,\text{V} \cdot 1.025 = 2.306$ V, threshold of level 4: 2.05 V, of level 5: 2.5625 V). With an exact bridge supply and a 4.10 V ladder the thresholds would be $2.5\,\%$ too high while the signal is correct; a value just above a threshold (e.g. 2.02 V) would then be read one code too low. Ratiometric operation removes the reference from the error budget.

Exercise 10.3: Another sampling rate

The sampling clock of the capstone is changed to $f_s = 4$ Hz and the anti-aliasing filter is removed.

a) To which frequency does the 50 Hz interference alias?

b) Describe what the LEDs show for a constant temperature with $V_{\text{COND}} = 1.75$ V.

c) Is this better or worse than the 0 Hz alias at 5 Hz?

Solution 10.3

a) $|50 - 12 \cdot 4| = 2$ Hz, exactly the Nyquist frequency $f_s/2$: consecutive samples see the interference with a phase step of $2\pi \cdot 50/4 = 25\pi$, so its sign alternates from sample to sample.

b) The held value alternates between $1.75 + \hat{v}\sin\varphi$ and $1.75 - \hat{v}\sin\varphi$ ($\hat{v} = 0.52$ V): the LEDs blink at 2 Hz between two codes, e.g. 100 and 010 in the worst case, or stay at 011 if the phase happens to be $\varphi \approx 0$.

c) It is visible, so it is "better" than a silent offset, but the reading is still wrong. Only the low-pass in front of the sampler removes the error for every sampling rate.

Exercise 10.4: Four bits

The capstone shall be extended to 4 bits with the same 4.00 V full scale.

a) What is the new LSB, in volts and in °C? How many comparators and ladder resistors are needed?

b) Is the first-order anti-aliasing filter still sufficient?

c) Is the bridge nonlinearity still acceptable?

Solution 10.4

a) LSB $= 4\,\text{V}/16 = 0.25\,\text{V} \mathrel{\widehat{=}} 6.25\,°\text{C}$; $2^4 - 1 = 15$ comparators, 16 resistors (taps at $k \cdot 0.25$ V).

b) Residual 25 mV $<$ LSB$/2 = 125$ mV: yes, still a factor five of margin.

c) The maximum deviation from the straight line is about $0.09\,\text{mV} \cdot 107.6 = 10$ mV $\ll 125$ mV: yes.

Exercise 10.5: Hold capacitor at the limits

In the capstone S&H the analog switch has $R_{\text{on}} = 1\,\text{k}\Omega$ and the total leakage current is 10 nA. Determine the smallest and the largest usable hold capacitor if the droop during one hold period and the remaining charging error at the end of the window shall both stay below LSB$/4 = 125$ mV for a full-scale step.

Solution 10.5

Droop: $I_L\, t_{\text{hold}}/C_H < 0.125\,\text{V} \Rightarrow C_H > 10\,\text{nA} \cdot 0.18\,\text{s}/0.125\,\text{V} = 14.4$ nF. Charging: $4\,\text{V} \cdot e^{-t_w/\tau} < 0.125\,\text{V}$ $\Rightarrow t_w > \tau \ln 32 = 3.47\,\tau$, so $\tau < 20\,\text{ms}/3.47 = 5.8$ ms and $C_H < 5.8\,\text{ms}/1\,\text{k}\Omega = 5.8\,\mu\text{F}$. Any value between about 15 nF and 5.8 µF works; 1 µF is a good choice in the middle (on a logarithmic scale). In Falstad set the switch's on resistance to $1\,\text{k}\Omega$ and C_H to 10 µF in

▶ Falstad: `examples/capstone/capstone.txt`: after a slider step the held staircase needs several samples to reach the new value.

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